

Common AI Information and Capital Allocation ^{*}

Haoning Sun[†]

September 2026

Abstract

This paper studies how AI-generated financial information affects real capital allocation. Investors choose between private research and an AI signal with a shared model error, and firms learn from equilibrium asset prices before investing. Motivating evidence shows that institutional managers with greater common alignment with AI-extractable information subsequently hold more similar portfolios. In the model, investors may underestimate the overlap between AI information and prices. Under complete correlation neglect, cheaper AI raises aggregate TFP near a boundary where high costs keep AI use negligible. Improvements from very low model accuracy can instead lower TFP: endogenous reliance increases prices' exposure to common errors faster than informed trading reduces noise-trader contamination. Numerical equilibria exhibit a J-shaped TFP response to model accuracy and a hump-shaped response to lower costs, including under partial recognition of common errors. These responses arise from changes in actual capital allocation, with production technology held fixed. General acquisition costs preserve the local comparison under explicit conditions on information choices.

Keywords: artificial intelligence, information acquisition, portfolio similarity, price discovery, capital allocation.

JEL Classification: D82, G11, G23, O33.

^{*}I am deeply indebted to Feng Dong for invaluable advice, guidance, and support. I also thank Zhen Huo, Jinpu Li, Pengfei Wang, Yizhen Wang, Zhenyang Xu, Liyan Yang and Fei Zhou for comments and suggestions. All errors are mine.

[†]School of Economics and Management, Tsinghua University. Email: sunhn22@mails.tsinghua.edu.cn

1 Introduction

Artificial intelligence is changing how financial institutions acquire and process information. In a 2024 survey, the Bank of England and the Financial Conduct Authority found that 75% of responding financial firms used AI and that one-third of reported AI use cases relied on third-party implementations.¹ Common models can make research cheaper, but they can also expose investment decisions to shared errors. Then SEC Chair Gary Gensler warned that market participants could make similar decisions after receiving “the same signal from a base model or data aggregator”.² The Financial Stability Board similarly identifies common data and models as potential sources of correlated predictions and financial decisions (Financial Stability Board, 2024).³

The consequences of common information extend beyond investors’ portfolios. Asset prices aggregate information about firms’ prospects, and firms can use this information when making real investment decisions. Evidence that investment responds more strongly to prices containing more private information supports this learning channel (Chen et al., 2007); Bond et al. (2012) review its broader implications for the real effects of financial markets. If a common model error enters investors’ demands, market clearing can transmit it into the prices firms learn from. Because the model error is unrelated to the true fundamental, greater reliance on AI among investors could also make the price signal noisier for firms. The effect of AI on financial information aggregation can therefore affect real capital allocation even when AI does not directly change production technology.

Whether investors recognize the overlap in their information is central to this mechanism. Shared models and training data need not be transparent to their users: the Financial Stability Board notes that market participants can unknowingly rely on the same underlying data (Financial Stability Board, 2024). Evidence from professional AI use also suggests that information overlap may be difficult to assess. In an experiment with radiologists, Agarwal et al. (2026) find that participants fail to account fully for the dependence between their own assessments and AI predictions. Although this evidence comes from a different profession than financial investors, it motivates a specific question for financial markets: might an investor treat a price partly driven by a common AI signal as independent confirmation of that signal? Further, how would this behavior affect their trading behavior, the price informativeness, and the macroeconomy?

¹Bank of England and Financial Conduct Authority, *Artificial Intelligence in UK Financial Services – 2024*, See <https://www.bankofengland.co.uk/report/2024/artificial-intelligence-in-uk-financial-services-2024>.

²Gary Gensler, “Isaac Newton to AI,” remarks before the National Press Club, July 17, 2023: <https://www.sec.gov/newsroom/speeches-statements/gensler-isaac-newton-ai-remarks-07-17-2023>.

³Financial Stability Board, *The Financial Stability Implications of Artificial Intelligence*, November 14, 2024, Section 4.2.2: <https://www.fsb.org/uploads/P14112024.pdf>.

This paper studies how AI progress changes the information that financial markets convey to firms and, through that information, aggregate productivity. We find that a better AI signal is not always beneficial to aggregate allocative efficiency. A cheaper AI signal initially increases aggregate TFP, but the expansionary effect can reverse when costs are low enough and the AI’s crowd-out effect on private research is prominent. The effect of higher AI signal accuracy also depends on how technological progress improves aggregate information acquisition and how it increases exposure to common AI error. At the adoption margin, the latter effect could dominate, implying that a marginal increase in AI model accuracy could instead reduce the price signal’s informativeness and the aggregate TFP level. As a result, the common reliance on AI information could affect the real macroeconomic allocation efficiency through the market prices, generating a non-monotonic effect on the aggregate TFP level.

We first document a pattern consistent with common reliance on AI-related information. Following the revealed-reliance approach of [Sheng et al. \(2026\)](#), we measure managers’ alignment with signals extracted from earnings calls by a large language model. Manager pairs with greater common alignment subsequently hold more similar portfolios, controlling for current similarity. This relationship is stronger after ChatGPT’s public release and in settings where language models appear to have a clearer information advantage. These predictive associations motivate a shared information source.

Inspired by the empirical motivations, we then build an equilibrium model with endogenous information acquisition in a production economy. Investors choose the quantity and composition of information between private research and AI before trading claims on firms’ revenues. Private research errors are investor-specific, whereas AI model errors are shared across investors researching a given firm. Investors also learn from prices, but may underestimate the covariance between their AI information and the price signal. With financial claims in fixed net supply, common shifts in desired holdings enter equilibrium prices. Firms observe these prices, update their beliefs, and choose capital. The model thus connects a common source of investment information to real allocation through endogenous price formation.

Under complete correlation neglect, we obtain analytical results near two boundaries at which AI use vanishes. At fixed model accuracy, a marginal reduction in a sufficiently high acquisition cost expands total information even though it displaces some private research. The resulting increase in informed trading reduces noise-trader contamination at first order. Exposure to the common error contributes only at second order because its price loading starts near zero. Information gains therefore dominate, and cheaper AI raises price informativeness and aggregate TFP sufficiently close to this boundary.

The balance differs when negligible AI use reflects low model accuracy. When the model accuracy is low enough, investors acquire nearly all the effective precision available from a very inaccurate model, even though the AI share of their information remains close to zero.

A marginal accuracy improvement then expands effective AI information almost one-for-one. The induced common-error exposure and the reduction in noise-trader contamination both affect price informativeness at first order. TFP initially falls when the former effect dominates. The result concerns endogenous reliance on an improving information source: a decline in its underlying error variance need not reduce the error transmitted through prices.

Numerical analysis traces these responses beyond the local boundaries. Improvements in accuracy can generate a J-shaped TFP curve, while reductions in acquisition costs can generate a hump. The nonmonotonicities also occur under partial recognition of common errors, although greater recognition changes both the strength of informed trading and prices' exposure to AI. Extensions to more general acquisition costs identify the conditions on marginal costs and interactions between research sources that preserve the local results.

To sum up, this paper shows that reliance on common AI models as information providers could affect the real economy through financial markets and price informativeness, and that the productivity consequences of an information technology depend on how its use changes the information markets aggregate. Investors can acquire more precise signals while firms receive less informative prices. Distinguishing cheaper access from greater model accuracy, and individual precision from the information conveyed by prices, helps explain why AI progress need not translate monotonically into better capital allocation.

Related literature. This paper relates to research on AI and data technologies in financial decision-making. [Coleman et al. \(2022\)](#) compare robo-analyst and traditional analyst recommendations, while [Cao et al. \(2024\)](#) study the gains from combining machine and human stock analysis. [Sheng et al. \(2026\)](#) examine generative AI in asset management using a revealed-reliance measure, which we adapt to study common information exposure across managers.⁴ Data technologies can also change the allocation of financing. [Bonelli \(2026\)](#) finds that venture capitalists adopting these technologies improve their screening of startups resembling historical observations but shift investment toward those firms and become less likely to finance innovative startups that achieve rare, major successes. Our paper studies how shared information affects dependence across investors and the prices that guide firms' investment. It connects individual information acquisition to real allocation through the aggregation of common AI signals in financial markets.

The portfolio evidence also relates to institutional herding and common holdings. [Lakonishok et al. \(1992\)](#), [Wermers \(1999\)](#), and [Sias \(2004\)](#) study correlated institutional trading, and [Pool et al. \(2015\)](#) connect portfolio similarity to social interaction among fund managers. Com-

⁴Related work studies machine learning and language models in return prediction and portfolio construction ([Gu et al., 2020](#); [Lopez-Lira and Tang, 2026](#); [Cong et al., 2026](#)), as well as the portfolio effects of robo-advice ([D'Acunto et al., 2019](#); [Rossi and Utkus, 2024](#)).

mon ownership is associated with return comovement ([Antón and Polk, 2014](#)) and liquidity commonality ([Koch et al., 2016](#)); [Girardi et al. \(2021\)](#) examine the relationship between portfolio similarity and asset liquidation.⁵ We document that common alignment with AI-extractable information predicts subsequent portfolio convergence, and use a model that is aligned with this phenomenon to study the consequences for price discovery and capital allocation.

Theoretically, the paper builds on endogenous information acquisition in financial markets ([Grossman and Stiglitz, 1980](#); [Verrecchia, 1982](#); [Van Nieuwerburgh and Veldkamp, 2010](#)). [Veldkamp \(2006\)](#) shows how information-production costs encourage investors to acquire common signals, generating comovement in asset prices. [Hellwig and Veldkamp \(2009\)](#) explain how strategic motives in decisions carry over to information choices, while [Myatt and Wallace \(2012\)](#) study the allocation of attention across information sources. [Goldstein and Yang \(2015\)](#) examine the role of information diversity in trading and acquisition incentives, and [Colombo et al. \(2014\)](#) analyze the welfare consequences of endogenous information acquisition. Particularly close to our question, [Farboodi and Veldkamp \(2020\)](#) study how advances in data processing change investors’ learning about asset fundamentals and other traders’ demands. We consider a choice between information sources about the same fundamental that differ in the extent to which their errors are shared. AI progress changes both total effective precision and its composition between private research and AI. This distinction allows us to compare cheaper access with greater model accuracy and to characterize when the induced common-error exposure outweighs the reduction in noise-trader contamination, reducing the aggregate capital allocation efficiency.

In our model, investors could have biased beliefs about model error commonality, which is related to the literature on correlation neglect and repeated information. [DeMarzo et al. \(2003\)](#) and [Eyster and Rabin \(2010\)](#) study how failures to account for the origins of information distort learning, and [Enke and Zimmermann \(2019\)](#) provide experimental evidence on the treatment of correlated signals. In financial settings, [Fedyk and Hodson \(2023\)](#) document difficulties in recognizing recombined old news, while [Guo and Wachter \(2026\)](#) relate stock-market return patterns to investors’ incomplete recognition of repetitive earnings information. These studies motivate imperfect recognition of information overlap in our setting. Investors learn from prices and correctly assess the marginal precision of their own AI signals, but may underestimate their interdependence. We study how this belief affects the information they choose to acquire, the common-error component of equilibrium prices, and the investment decisions firms make using those prices.

The macroeconomic analysis contributes to research on the real effects of financial information and information-driven capital misallocation. [Chen et al. \(2007\)](#) provide evidence consis-

⁵Related work studies crowded mutual fund portfolios ([Gonzalez et al., 2024](#)) and the implications of common exposures for systemic liquidation risk ([Wagner, 2011](#); [Greenwood et al., 2015](#)).

tent with firms learning from stock prices when investing, and [Bond et al. \(2012\)](#) review the channels through which financial markets affect real decisions. [David et al. \(2016\)](#) quantify the role of imperfect information in capital misallocation, and [Benhabib et al. \(2016\)](#) and [Benhabib et al. \(2019\)](#) study the feedback between information acquisition, uncertainty, and economic activity.⁶ We introduce endogenous reliance on an AI signal with a shared error into this environment. Firms learn from the price signal and invest optimally, but the information available to them depends on investors' acquisition choices and beliefs. The resulting allocation effects distinguish the precision an individual investor acquires from the information financial markets convey to firms.

Finally, our findings relate to the productivity paradox highlighted by [Solow \(1987\)](#), in which rapid technological progress coexists with weak aggregate productivity growth. Research on general purpose technologies emphasizes the complementary innovations needed to realize their productivity gains ([Bresnahan and Trajtenberg, 1995](#)). [David \(1990\)](#) traces the delayed gains from electrification to slow diffusion and the reorganization of production, while [Greenwood and Yorukoglu \(1997\)](#) show how learning to use new technologies can temporarily slow productivity growth. In the context of AI, [Brynjolfsson et al. \(2019\)](#) emphasize implementation lags and complementary investment. [Brynjolfsson et al. \(2021\)](#) formalize a productivity J-curve in which measured productivity growth is initially depressed by unmeasured intangible investment and rises as its returns materialize. We provide an additional explanation through financial information and capital allocation. Improvements in model accuracy can increase reliance on AI, raising the common-error component of prices and reducing their informativeness. Firms then base investment on less informative prices, so aggregate TFP can fall even as total information acquisition rises. This effect is especially prominent when the investors underestimate the correlation between AI signals. Further accuracy gains can restore price informativeness and TFP, generating a J-shaped relationship between AI model accuracy and equilibrium TFP. This comparison across equilibria isolates an informational channel through capital allocation that complements explanations based on implementation lags and intangible investment.

The remainder of the paper is organized as follows. Section 2 presents the portfolio evidence and the economic motivation for the model. Section 3 introduces the environment, and Section 4 characterizes information acquisition, price formation, and capital allocation. Section 5 studies the effects of AI progress analytically and numerically. Section 6 considers general information costs, and Section 7 concludes.

⁶More broadly, imperfect information shapes expectations and macroeconomic dynamics ([Angeletos and Huo, 2021](#); [Schaal and Taschereau-Dumouchel, 2025](#)).

2 Evidence and Economic Motivation

This section examines whether common alignment with AI-extractable information predicts subsequent portfolio convergence and whether this relationship is stronger when language models have an information advantage. Because institutional adoption is not directly observed, we interpret the estimates as predictive associations. We then discuss the external evidence and economic mechanisms motivating investors’ beliefs and firms’ use of financial prices.

2.1 Data and Measurement

Institutional holdings and sample coverage. We obtain quarterly institutional holdings from Form 13F filings and construct a manager–stock–quarter panel. Position values are converted into portfolio weights, and consecutive reports are used to measure portfolio adjustments. In each quarter, we retain the 300 managers with the largest reported portfolios and form every unordered pair of managers. The resulting universe contains up to $300 \times 299/2 = 44,850$ manager pairs per quarter. The top 300 managers collectively account for more than 85% of portfolio value in the source Form 13F panel.

The earnings-call sample consists primarily of S&P 500 constituents and related large-cap U.S. firms for which transcript text, firm identifiers, and the required controls can be matched. These scored firms account for more than 65% of the portfolio value held by the top 300 managers. Combining the manager and firm coverage, the manager–firm observations underlying the AI-reliance measure represent more than 55% of total portfolio value in the full Form 13F holdings panel. The transcript sample is concentrated among large public firms, but it covers an economically important segment of institutional portfolios in which correlated adjustments could matter at the aggregate level.

LLM-based firm signals. We follow [Sheng et al. \(2026\)](#) and use earnings-call transcripts to construct fourteen directional firm-quarter signals. The prompt asks GPT-4o-mini to evaluate the implications of each call for U.S. and global economic conditions, industry conditions, and eleven firm-level outcomes covering financial prospects, earnings, revenue, investment, labor costs, employment, product demand, production, output prices, input prices, and the cost of capital. We exclude the firm name, ticker, industry, and call date from the prompt and explicitly instruct the model to rely only on management’s statements in the excerpt rather than outside knowledge. Each score lies between -1 and 1 ; zero is assigned when the transcript does not support a directional assessment. Long transcripts are divided sequentially into chunks of approximately 12,000 words. GPT-4o-mini has a 128,000-token context window, and this conservative chunk size leaves room for the complete instructions and structured JSON response

while avoiding context truncation. We aggregate chunk-level outputs to the transcript level and then average across transcripts for the same firm-quarter. The fourteen scores enter the portfolio-adjustment regressions separately rather than through a composite index.

Manager-level revealed AI reliance. Following the empirical logic of [Sheng et al. \(2026\)](#), we construct a revealed measure of alignment with AI-extractable information from the incremental explanatory power of LLM-extracted information for a manager’s portfolio decisions. For every manager i and quarter t , we estimate two cross-sectional regressions across securities s :

$$\Delta w_{i,s,t} = \alpha_{i,t} + \gamma'_{i,t} X_{s,t-1} + \varepsilon_{i,s,t}, \quad (2.1)$$

$$\Delta w_{i,s,t} = \alpha_{i,t} + \gamma'_{i,t} X_{s,t-1} + \sum_{m=1}^{14} \beta^m_{i,t} GPT^m_{s,t-1} + \varepsilon_{i,s,t}, \quad (2.2)$$

where $\Delta w_{i,s,t} = w_{i,s,t} - w_{i,s,t-1}$ is the baseline portfolio-weight change. The public-information vector $X_{s,t-1}$ contains log market capitalization, book-to-market, return on assets, the stock return, and the change in the I/B/E/S recommendation consensus. All five controls and all GPT signals are dated $t - 1$, so the information set precedes the portfolio adjustment.

Based on the two regressions, we define manager i ’s revealed AI-reliance measure as:

$$Reliance_{i,t} = R^2_{full,i,t} - R^2_{base,i,t}. \quad (2.3)$$

This construction follows the revealed-public-information approach of [Kacperczyk and Seru \(2007\)](#) and its recent application to generative AI by [Sheng et al. \(2026\)](#). [Sheng et al. \(2026\)](#) further show that a closely related holdings-based measure aligns with survey evidence on managers’ reported use of generative AI. In the present setting, the measure captures a manager’s revealed behavioral alignment with information that a language model extracts from corporate disclosures. It is therefore an information-alignment measure rather than a provider-specific indicator of model adoption. Because the full regression nests the baseline specification, the raw increase in in-sample R^2 is mechanically nonnegative. Appendix D considers partial- R^2 , adjusted- R^2 , and adjusted partial- R^2 measures that either normalize the incremental explanatory power of the GPT scores or penalize the additional regressors.

At the pair level, we define common AI reliance as:

$$Common_Reliance_{i,k,t} = Reliance_{i,t} \times Reliance_{k,t}. \quad (2.4)$$

We also define $AverageReliance_{i,k,t} = (Reliance_{i,t} + Reliance_{k,t})/2$. Conditional on average re-

liance, the product has a particularly useful interpretation:

$$Reliance_{i,t} \times Reliance_{k,t} = AverageReliance_{i,k,t}^2 - \frac{(Reliance_{i,t} - Reliance_{k,t})^2}{4}. \quad (2.5)$$

Thus, holding the average fixed, a larger product means a smaller gap between the two managers' reliance and captures greater common alignment with the same AI-extractable information. Finally, we define a pair-level indicator, $Joint_GPT_Alignment_{i,k,t}$, which equals one when the joint F -tests on the fourteen GPT coefficients are significant at the 5% level for both managers. This indicator provides an extensive-margin measure of joint GPT alignment.

Portfolio similarity. We use two methods to measure the similarity between two asset managers' portfolio structures. We follow [Wahal and Wang \(2011\)](#), [Cici et al. \(2017\)](#) and [Gonzalez et al. \(2024\)](#) to use portfolio overlap as the main measure:

$$Overlap_{i,k,t} = \sum_s \min\{w_{i,s,t}, w_{k,s,t}\}, \quad (2.6)$$

where $w_{i,s,t}$ denotes the portfolio share of asset s by manager i at time t . The portfolio overlap measure captures the fraction of portfolio holdings that can be directly matched between two asset managers. This measure ranges from zero to one. A value of zero indicates that the two managers have no common holdings, whereas a value of one indicates identical portfolios. Economically, this measure represents the share of portfolio mass that can be matched security by security and therefore provides a direct measure of the magnitude of their common asset exposure.

We also use cosine similarity as an alternative measure ([Hoberg et al., 2018](#); [Girardi et al., 2021](#)):

$$Cosine_{i,k,t} = \frac{\sum_s w_{i,s,t} w_{k,s,t}}{\sqrt{\sum_s w_{i,s,t}^2} \sqrt{\sum_s w_{k,s,t}^2}}. \quad (2.7)$$

Cosine similarity measures the extent to which the two portfolios are oriented in the same *direction* in the asset-weight space. Relative to portfolio overlap, it places greater weight on securities that constitute large positions in both portfolios and is therefore more sensitive to similarity in managers' dominant exposures. By contrast, the overlap measure has a more direct interpretation as the total amount of common portfolio exposure. Thus, while both measures increase when two managers hold similar portfolios, portfolio overlap primarily captures the magnitude of shared holdings, whereas cosine similarity emphasizes relative allocation patterns across securities. The main empirical patterns are similar across the two measures.

2.2 Common AI Reliance and Future Portfolio Similarity

We first examine whether common AI reliance predicts incremental portfolio convergence. Period- t reliance is estimated from holdings in $t - 1$ and t , while the outcome is portfolio similarity in $t + 1$. Controlling for period- t similarity absorbs the substantial persistence in portfolio composition. The construction of the reliance measure also imposes no directional restriction: given the same GPT-generated information, managers may increase or decrease their holdings of a stock, and neither the direction nor the magnitude of their adjustments needs to coincide. The specification therefore asks whether pairs with greater common reliance converge further from their current level of similarity. We estimate:

$$\begin{aligned} \text{Similarity}_{i,k,t+1} = & \beta \times \text{Common_Reliance}_{i,k,t} + \lambda \times \text{Average_Reliance}_{i,k,t} \\ & + \rho \times \text{Similarity}_{i,k,t} + \theta' Z_{i,k,t} + \varphi_{i,k} + \psi_t + \epsilon_{i,k,t}. \end{aligned} \quad (2.8)$$

where $\varphi_{i,k}$ and ψ_t are manager-pair and quarter fixed effects. The vector $Z_{i,k,t}$ contains the control variables, including pair average and absolute difference in log portfolio value and number of holdings (Wahal and Wang, 2011). Standard errors are clustered by manager pair.⁷

The timing and controls serve distinct purposes. Using $t + 1$ similarity places the outcome after the portfolio adjustment used to estimate reliance. Current similarity absorbs the considerable persistence in institutional portfolios, so β measures whether common reliance predicts additional convergence from the pair's existing position. Average reliance separates commonality from the overall level of AI exposure: conditional on the average, the product is larger when the two managers have more similar reliance estimates. Pair fixed effects remove time-invariant matching patterns between institutions, and quarter fixed effects absorb aggregate shifts in portfolio similarity and the information environment.

Table 1 presents four specifications for each similarity measure. Panel A is the baseline and controls for the pair's average AI reliance. Conditional on this average, pairs with greater common AI reliance have significantly higher next-quarter overlap and cosine similarity. The same pattern appears when Panel B controls separately for the two managers' reliance and when Panel C uses the joint-alignment indicator. Panel D adds squared average reliance to allow for a nonlinear relationship between the overall level of reliance and future similarity. The common-reliance coefficient remains positive for both measures and statistically significant for overlap. The current-similarity coefficients, approximately 0.80 for overlap and 0.68 for cosine similarity, show that portfolio composition is highly persistent. The reliance coefficients therefore capture differences in next-quarter similarity around this persistent baseline.

The continuous estimates describe modest one-quarter changes in portfolio similarity. Ap-

⁷Appendix D reports a two-manager clustering sensitivity exercise. The continuous baseline coefficients remain statistically significant, although some indicator specifications are estimated less precisely.

Table 1: Common AI Reliance and Future Portfolio Similarity

	(1) Future overlap	(2) Future cosine similarity
<i>Panel A: Controlling for average AI reliance</i>		
Common AI reliance	0.0271*** (6.27)	0.0627*** (6.80)
Average AI reliance	-0.0097*** (-11.29)	0.0052** (2.23)
Current similarity	0.7957*** (371.02)	0.6760*** (223.62)
<i>Panel B: Controlling for individual AI reliance</i>		
Common AI reliance	0.0272*** (6.29)	0.0628*** (6.81)
AI reliance, manager <i>i</i>	-0.0057*** (-10.46)	0.0018 (1.18)
AI reliance, manager <i>k</i>	-0.0041*** (-7.62)	0.0033** (2.30)
Current similarity	0.7956*** (371.03)	0.6760*** (223.62)
<i>Panel C: Joint GPT alignment</i>		
Joint GPT alignment	0.0020*** (9.86)	0.0105*** (13.47)
Average AI reliance	-0.0076*** (-10.86)	0.0059*** (3.19)
Current similarity	0.7956*** (371.06)	0.6760*** (223.84)
<i>Panel D: Controlling for nonlinear average AI reliance</i>		
Common AI reliance	0.0118** (2.34)	0.0180 (1.51)
Average AI reliance	-0.0173*** (-9.70)	-0.0170*** (-3.41)
Squared average AI reliance	0.0291*** (5.29)	0.0850*** (5.56)
Current similarity	0.7956*** (370.80)	0.6760*** (223.57)
Pair controls	Yes	Yes
Pair FE	Yes	Yes
Quarter FE	Yes	Yes
SE cluster	Pair	Pair
Observations	890,394	890,394

Notes: Each panel represents a separate regression. Robust *t*-statistics are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

pendix Table D.1 reports a standard deviation of approximately 0.011 for common AI reliance. A one-standard-deviation increase in common reliance predicts increases of 0.0003 in future portfolio overlap and 0.0007 in future cosine similarity. These estimates correspond to approximately 0.2% and 0.3% of a standard deviation in the respective similarity measures. The indicator estimates imply larger differences. In a pair-quarter in which both managers exhibit significant GPT alignment, future overlap is 0.0020 higher and future cosine similarity is 0.0105 higher, equivalent to approximately 1.2% and 4.1% of a standard deviation, respectively. Because the regressions condition on current similarity and pair fixed effects, these magnitudes measure incremental within-pair convergence over the next quarter.

Validation and heterogeneity. We next examine whether the common-reliance measure behaves as expected in settings where language models are plausibly more useful for processing corporate disclosures. These exercises provide construct validation beyond the baseline predictive relationship. The first exercise uses the broader accessibility and salience of generative AI after the public release of ChatGPT. If the measure captures economically meaningful reliance on AI-extractable information, its association with subsequent portfolio convergence should be stronger in the post-release period. We define $PostGPT_t$ to equal one from 2023Q1 onward. Because ChatGPT was released on November 30, 2022, 2022Q4 is treated as the last pre-period. Quarter fixed effects absorb the level of the post indicator, while its interaction with common reliance measures the change in the predictive relationship. We estimate:

$$\begin{aligned} Similarity_{i,k,t+1} = & \beta \times Common_Reliance_{i,k,t} + \lambda \times Common_Reliance_{i,k,t} \times PostGPT_t \\ & + \mathcal{L}_{i,k,t} + \rho \times Similarity_{i,k,t} + \theta' Z_{i,k,t} + \varphi_{i,k} + \psi_t + \epsilon_{i,k,t}, \end{aligned} \quad (2.9)$$

where $\mathcal{L}_{i,k,t}$ contains average reliance and its interaction with $PostGPT_t$. A positive λ means that the relationship between common AI reliance and future portfolio similarity is stronger in the post-GPT period, consistent with the broader availability of general-purpose AI tools. We also interact average reliance with the post-GPT indicator so that the interaction of interest continues to capture commonality conditional on the pair's average reliance.

The second exercise studies three dimensions of the firm information environment in which language models may have a greater comparative advantage:

First, GPT signal strength, which is defined as the firm-quarter mean absolute value of the fourteen transcript scores:

$$GPT_strength_{s,t} = \frac{1}{14} \sum_k |GPT_score_{k,s,t}|. \quad (2.10)$$

$GPT_strength_{s,t}$ is larger when the model extracts a more decisive directional signal. We expect

the post-GPT association between common reliance and portfolio convergence to be stronger for manager pairs jointly exposed to these firms.

Second, earnings-call length. We define firm-quarter earnings-call length as the mean transcript word count, which captures the amount of unstructured information that investors must process. Language models are plausibly more useful when disclosures are longer, so we expect a stronger post-GPT association for manager pairs jointly exposed to firms with longer calls.

Third, human analyst coverage. We define analyst coverage as the number of I/B/E/S forecasts, which proxies for the supply of human-produced information. AI-extractable information may carry greater incremental value for firms with fewer analyst forecasts.

Each firm characteristic is standardized within GICS sector and quarter, and then aggregated to the asset-pair level according to their portfolio structures. For manager pair (i, k) , the constructed moderator variable for firm characteristic X is defined as:

$$M_{i,k,t}^X = \frac{\sum_s \min\{\bar{w}_{i,s,t}^{hist}, \bar{w}_{k,s,t}^{hist}\} Z_{s,t}^X}{\sum_s \min\{\bar{w}_{i,s,t}^{hist}, \bar{w}_{k,s,t}^{hist}\}}, \quad (2.11)$$

where $\bar{w}_{i,s,t}^{hist}$ is manager i 's average portfolio weight on asset s over $t - 1$ through $t - 4$, and $Z_{s,t}^X$ is the sector-quarter standardized z-score for firm characteristic X . Historical overlap weights make the moderator reflect the information environment of firms to which the pair was jointly exposed before the current portfolio adjustment. A higher $M_{i,k,t}^X$ implies that the asset manager pair (i, k) is jointly more exposed to firms with higher levels of X .

For each moderator, we estimate a saturated triple-interaction model:

$$\begin{aligned} Similarity_{i,k,t+1} = & \beta_1 \times Common_Reliance_{i,k,t} + \beta_2 \times Common_Reliance_{i,k,t} \times PostGPT_t \\ & + \beta_3 \times Common_Reliance_{i,k,t} \times M_{i,k,t}^X \\ & + \lambda \times Common_Reliance_{i,k,t} \times M_{i,k,t}^X \times PostGPT_t \\ & + \mathcal{L}_{i,k,t} + \rho Similarity_{i,k,t} + \theta' Z_{i,k,t} + \varphi_{i,k} + \psi_t + \epsilon_{i,k,t}, \end{aligned} \quad (2.12)$$

where $\mathcal{L}_{i,k,t}$ contains average reliance and its corresponding interactions, the moderator, $M^X \times PostGPT$, and all remaining lower-order terms. In this regression, β_1 is the pre-GPT common-reliance slope at $M^X = 0$, β_2 is the post-GPT shift at that reference value, and λ measures how the shift varies with the information environment. We expect λ to be positive for earnings-call length and GPT signal strength, and negative for analyst coverage. These signs would indicate that the post-GPT association is stronger when manager pairs focus on firms for which machine processing is plausibly more useful.

Table 2 reports the results. Column (1) shows that the estimated common-reliance slope is larger and statistically significant in the post-GPT period for both similarity measures.

Columns (2) to (4) relate this shift to the information environment. The triple interactions for GPT signal strength and call length are positive for both outcomes: the post-GPT association is strongest when the model delivers a more decisive signal or when the underlying text is more costly to process. For analyst coverage, the triple interaction is negative, consistent with AI-extractable information having greater incremental value when human information production is limited. Taken together, these patterns provide construct validation for the revealed-reliance measure. The evidence is strongest for the post-GPT shift, GPT signal strength, and call length. The analyst-coverage result has the predicted sign for both outcomes but is less precisely estimated for portfolio overlap. Two-manager clustering widens the confidence intervals around the post-GPT and moderator interactions, as Appendix D reports.

Appendix D examines alternative fit measures, definitions of portfolio adjustment, manager-time controls, joint-alignment thresholds, and covariance estimators. The common-reliance coefficient for portfolio overlap remains positive across these specifications, although penalizing the additional GPT regressors reduces its precision. The cosine result remains positive under the alternative fit and portfolio-adjustment measures but is more sensitive to manager-by-quarter fixed effects. Under those fixed effects, the joint-alignment indicator remains positive and significant for both outcomes. Two-manager clustering increases standard errors but leaves the continuous baseline coefficients statistically significant. Figures D.1 and D.2 further show that the main post-GPT and GPT-signal-strength patterns are not driven by any single predictor quarter.

Portfolio similarity also predicts common selling and downside comovement, as reported in Appendix D.1. These additional associations describe the consequences of common positions without identifying whether the information behind those positions is accurate.

2.3 Implications for the Model

The evidence above identifies a predictive relationship between common alignment with AI-extractable information and subsequent portfolio similarity. This relationship is consistent with investors responding to a shared information source, but it does not identify the source of that common response. Investors may learn the same fundamentals, make the same errors, or follow related strategies for other reasons. We therefore use the evidence to motivate common information in the model, while keeping fundamental information and model error separate. Appendix A shows how both components can generate similar desired holdings at given prices. In our baseline model that follows, market clearing determines how a common shift in demand enters prices when the supply of financial claims is fixed.

Table 2: Validation and Heterogeneity

	(1) Post-GPT	(2) GPT strength	(3) Call length	(4) Analyst coverage
<i>Panel A: Future portfolio overlap</i>				
Common reliance	0.0192*** (4.10)	0.0125*** (2.60)	0.0148*** (2.95)	0.0040 (0.52)
Common reliance $\times$ Post-GPT	0.0202*** (2.72)	0.0252*** (3.31)	0.0207*** (2.77)	0.0574*** (4.04)
Common reliance $\times M^X \times$ Post-GPT	–	0.0807*** (4.73)	0.0773*** (3.44)	-0.0135 (-1.29)
Current similarity	0.7948*** (370.75)	0.7902*** (358.51)	0.7907*** (360.76)	0.7872*** (349.73)
<i>Panel B: Future cosine similarity</i>				
Common reliance	0.0180* (1.81)	-0.0020 (-0.20)	0.0121 (1.16)	-0.0745*** (-4.42)
Common reliance $\times$ Post-GPT	0.1575*** (8.81)	0.1762*** (9.65)	0.1704*** (9.57)	0.4174*** (11.36)
Common reliance $\times M^X \times$ Post-GPT	–	0.2626*** (6.62)	0.2386*** (4.47)	-0.1610*** (-6.13)
Current similarity	0.6736*** (222.89)	0.6700*** (220.07)	0.6705*** (220.68)	0.6644*** (213.14)
All lower-order interactions	Yes	Yes	Yes	Yes
Pair controls	Yes	Yes	Yes	Yes
Pair FE	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes
SE cluster	Pair	Pair	Pair	Pair
Observations	890,394	859,357	859,357	859,357

Notes: Each panel and column is a separate regression. Column (1) has no firm moderator. Columns (2)–(4) include all lower-order terms implied by the triple interaction and the corresponding interactions with average reliance. Robust t -statistics are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Recognition of shared information. Although the model error is common, the literature and survey both imply that the investors may not fully understand this. The Bank of England and Financial Conduct Authority survey reports that 46% of respondents using or planning to use AI had only a partial understanding of the technologies they employed ([Bank of England and Financial Conduct Authority, 2024](#)). In about one-third of AI use cases, a third-party provider provides the service, which may in turn rely on a common public model. The survey report also raises concerns about model concentration and correlated outcomes. The Financial Stability Board points more specifically to limited transparency about training data and the possibility of unknowingly using common information sources ([Financial Stability Board, 2024](#)). These observations imply that limited understanding of the commonality of AI model error by the investors should be taken seriously into account.

Experimental evidence helps distinguish this hypothesis from a general tendency to overtrust AI. [Agarwal et al. \(2026\)](#) find that radiologists both underweight AI advice and fail to account fully for its dependence on their own information. Thus, placing too little weight on AI on average can coexist with giving too much weight to overlapping evidence. The financial analogue in our model is an investor who treats an AI signal and a price partly driven by that signal as more independent than they are. Evidence on correlation neglect ([Enke and Zimmermann, 2019](#)) and on financial markets' response to recombined old news ([Fedyk and Hodson, 2023](#)) provides related support for difficulties in tracing information overlap. We model this difficulty as a perceived correlation parameter in the model discussed later.

Prices and real investment. The final link is firms' use of financial information. [Chen et al. \(2007\)](#) find that investment is more sensitive to stock prices when those prices contain more private information, consistent with managers learning from markets. The broader literature reviewed by [Bond et al. \(2012\)](#) studies how such learning connects financial prices to real decisions. We adopt this channel to examine the consequences of common AI information for capital allocation. Firms in the model understand the true precision of prices and invest optimally. Any deterioration in allocation arises from the information available to them, not from an additional mistake in their investment rule.

These observations motivate a model with common AI information, potentially imperfect recognition of its overlap with prices, and firms that learn from those prices. The portfolio evidence bears directly on the first element; the other two draw on external evidence and established economic mechanisms. The productivity response developed below is a theoretical implication of their interaction.

3 Model

The model studies how investors' information choices affect firms through financial prices. Its production and price-learning environment builds on [Benhabib et al. \(2019\)](#). The main departure is the choice between private research and AI information whose model error is shared across investors. Investors may underestimate this common component, while firms understand the true information content of prices.

3.1 Economic Environment and Timing

There are N intermediate-good firms, indexed by $j = 1, \dots, N$. Define

$$\delta \equiv \frac{1}{\theta}, \quad \chi \equiv 1 - \frac{1}{\theta}, \quad \theta > 1. \quad (3.1)$$

A competitive final-good producer combines intermediate goods according to:

$$Y_N = \left[\frac{1}{N} \sum_{j=1}^N \mathcal{E}_j^\delta Y_j^\chi \right]^{1/\chi}, \quad (3.2)$$

where $\mathcal{E}_j$ is a firm-specific product-demand shifter. The associated inverse demand curve is:

$$P_j = Y_N^\delta \mathcal{E}_j^\delta Y_j^{-\delta}. \quad (3.3)$$

Firm j produces with:

$$Y_j = Z A_j K_j^\eta, \quad 0 < \eta < 1, \quad (3.4)$$

where Z is an aggregate productivity component known before trading. Capital fully depreciates, and its price is normalized to one. Firm fundamental shocks $\{A_j, \mathcal{E}_j\}$ follow independent log-normal distributions:

$$\log A_j = \mu_a + a_j, \quad \log \mathcal{E}_j = \mu_d + d_j, \quad (3.5)$$

where:

$$a_j \sim \mathcal{N}(0, \tau_a^{-1}), \quad d_j \sim \mathcal{N}(0, \tau_d^{-1}). \quad (3.6)$$

The fundamental shocks are independent across firms. Firms and investors share these priors. Signal errors and noise-trader demands are independent of fundamentals and independent across firms; the AI error introduced below is common across investors within a firm.

Firms and investors observe a public signal of firm productivity:

$$s_j = a_j + e_j, \quad e_j \sim \mathcal{N}(0, \tau_s^{-1}), \quad (3.7)$$

Define

$$X_j \equiv \mathcal{E}_j^\delta A_j^\chi, \quad g \equiv \eta\chi \in (0, 1), \quad \mu_L \equiv \chi\mu_a + \delta\mu_d. \quad (3.8)$$

Then the firm's operational revenue is

$$V_j = P_j Y_j = Y_N^\delta Z^\chi X_j K_j^\delta. \quad (3.9)$$

Following [Benhabib et al. \(2019\)](#), the financial claim pays log revenue. Lowercase letters denote the logarithms of the corresponding capital, output, and aggregate productivity variables:

$$v_j \equiv \log V_j = \delta y_N + \chi z + \mu_L + \chi a_j + \delta d_j + g k_j. \quad (3.10)$$

Trading a log-revenue claim is a tractability device: its payoff is conditionally Gaussian, so portfolio demand and the equilibrium price are linear. The claim can be interpreted as a firm-value derivative rather than literal equity. The claim is in zero net supply; real capital remains an endogenous choice of the firm.

There is a unit mass of atomistic asset managers, indexed by $i \in [0, 1]$. We solve the economy in the limit as the number of firms tends to infinity, conditional on known Z . Aggregate output is then known before trading and investment. Finite- N expressions below provide allocation accounting under the limiting investment rule.

At date 0, managers choose how much private and AI information to acquire. At date 1, firm fundamentals, signal errors, and noise-trader demands are realized. Managers observe their signals and trade firm-value claims. Firms then observe their public productivity signals and the resulting prices and choose capital. Managers anticipate the firms' investment rule when trading. At date 2, firms produce and financial claims pay off.

3.2 Information Acquisition and Correlation Beliefs

Managers observe the firm's productivity signal s_j and can acquire private and AI signals about its demand shock d_j :

$$\begin{aligned} s_{ij}^P &= d_j + u_{ij} + \epsilon_{ij}^P, & u_{ij} &\sim \mathcal{N}(0, \tau_u^{-1}), \\ s_{ij}^A &= d_j + b_j + \epsilon_{ij}^A, & b_j &\sim \mathcal{N}(0, \tau_b^{-1}). \end{aligned} \quad (3.11)$$

The investor-specific errors satisfy $\epsilon_{ij}^P \sim \mathcal{N}(0, (x_{ij}^P)^{-1})$ and $\epsilon_{ij}^A \sim \mathcal{N}(0, (x_{ij}^A)^{-1})$. Managers pay to increase the raw precisions x_{ij}^P and x_{ij}^A . Private and acquisition errors are mutually independent and independent across managers. They are also independent of b_j and noise-trader demand, while b_j is shared by all investors researching firm j . The effective precisions and their quantity–composition representation are:

$$\begin{aligned} h_{ij}^P &= \frac{\tau_u x_{ij}^P}{\tau_u + x_{ij}^P}, & h_{ij}^A &= \frac{\tau_b x_{ij}^A}{\tau_b + x_{ij}^A}, \\ h_{ij} &= h_{ij}^P + h_{ij}^A, & \alpha_{ij} &= \frac{h_{ij}^A}{h_{ij}}. \end{aligned} \quad (3.12)$$

Thus h measures total effective precision and α the share supplied by AI. More acquisition reduces investor-specific noise, but effective precision remains bounded by the source’s underlying error: $h_{ij}^P < \tau_u$ and $h_{ij}^A < \tau_b$. When suppressing the firm index, we also write $h_{Pi} = h_i^P$ and $h_{Ai} = h_i^A$.

The baseline has separable power costs of raw precision, with positive cost parameters and exponent $\nu > 1$, as in the precision-cost approach of [Colombo et al. \(2014\)](#). In effective-precision coordinates, the per-firm cost is:

$$C^I(h_i, \alpha_i) = \frac{c_A}{\nu} \left(\frac{\tau_b \alpha_i h_i}{\tau_b - \alpha_i h_i} \right)^\nu + \frac{c_P}{\nu} \left(\frac{\tau_u (1 - \alpha_i) h_i}{\tau_u - (1 - \alpha_i) h_i} \right)^\nu, \quad \nu > 1. \quad (3.13)$$

Section 6 relaxes this specification. The distinction between quantity and composition will also be useful there.

Managers also extract information from the market price q_j . Since the production function is log-linear and the firm-value claim traded is the log of firm revenue, we conjecture that:

$$q_j = \beta_0 + \beta_s s_j + \beta_q \tilde{q}_j, \quad \tilde{q}_j = d_j + \psi b_j + \zeta n_j, \quad n_j \sim \mathcal{N}(0, \tau_n^{-1}), \quad (3.14)$$

where $\tilde{q}_j$ is the standardized price signal and n_j is noise-trader demand. Observing s_j and the equilibrium price coefficients allows investors to recover $\tilde{q}_j$ from q_j . They therefore combine three signals about demand, $(s_{ij}^P, s_{ij}^A, \tilde{q}_j)$. Accounting for a shared error helps distinguish demand information from the component repeated in AI and prices.

Correlation beliefs. The AI model error b_j is fully common, while managers may nevertheless perceive only a fraction $\kappa \in [0, 1]$ of its variance as common. Their subjective Gaussian representations of the AI signal and the normalized price signal are:

$$s_{ij}^A = d_j + \sqrt{\kappa} b_j^c + \sqrt{1 - \kappa} b_{ij}^I + \epsilon_{ij}^A, \quad (3.15)$$

$$\tilde{q}_j = d_j + \psi \sqrt{\kappa} b_j^c + \zeta n_j, \quad (3.16)$$

where b_j^c is the perceived AI common error, and b_{ij}^I is the perceived idiosyncratic shock. They are subjectively independent, and both have precision τ_b . This representation preserves the manager's perceived marginal variance of the AI signal, while changing only the perceived separation between the common and idiosyncratic parts. Investors use the equilibrium price coefficients but misspecify the cross-sectional structure of AI errors: they perceive a fraction κ of model-error variance as common. Consequently, they also understate the variance of the price error when $\kappa < 1$. At $\kappa = 1$, beliefs are correct. At $\kappa = 0$, a manager believes that the fixed AI error is idiosyncratic and therefore washes out of the price. The manager still learns from the price, but incorrectly treats it as an independent confirmation of the AI signal.

The parameter κ formalizes the difficulty of recognizing overlapping information discussed in Section 2.3. It does not change the perceived marginal accuracy of the investor's own AI signal, and it is not a general measure of trust in AI. A low value captures the possibility that shared data or models make seemingly distinct information sources less independent than their users believe. We treat it as a sensitivity parameter, rather than estimate it from the portfolio evidence.

3.3 Decisions and Equilibrium

Managers have mean-variance preferences with risk-aversion coefficient $\rho > 0$. Suppressing the firm index where there is no ambiguity, they anticipate a log-linear investment rule:

$$k = k_0 + k_s s + k_q \tilde{q}, \quad (3.17)$$

where (k_0, k_s, k_q) are common constants among firms. We will verify this log-linear investment decision later. Although investment occurs immediately after price formation, managers know this rule when trading. Hence, conditional on (s, q) , subsequent investment is known and affects expected payoff but not conditional payoff variance.

Managers allocate wealth between a risk-free asset and firm-value claims. The risk-free return is normalized to zero, and m_i denotes manager i 's position in a given claim.⁸ Let $\mathcal{I}_i$ contain the public productivity signal, the two acquired signals, and the price. Conditional on this information, manager i evaluates position m_i according to:

$$\mathcal{V}_i(m_i | \mathcal{I}_i) = m_i (\mathbb{E}_i^\kappa[v | \mathcal{I}_i] - q) - \frac{\rho}{2} m_i^2 \text{Var}_i^\kappa(v | \mathcal{I}_i). \quad (3.18)$$

⁸Following [Van Nieuwerburgh and Veldkamp \(2010\)](#) and [Benhabib et al. \(2019\)](#), we assume that the asset managers can hold unconstrained positions for the risk-free asset, so the claim position is not restricted by a binding portfolio budget constraint.

Before signals are realized, the information choice solves:

$$\max_{h_i, \alpha_i} \left\{ \mathbb{E}_{0,i}^K \left[\max_{m_i} \mathcal{V}_i(m_i | \mathcal{I}_i) \right] - C^I(h_i, \alpha_i) \right\}. \quad (3.19)$$

This criterion first evaluates the mean-variance trading problem conditional on each information realization and then averages its optimized value.

Firms take aggregate output as given and maximize expected revenue net of capital cost, $\mathbb{E}[V_j | s_j, q_j] - K_j$. They use the true distribution of the price signal. This separates the effect of investors' beliefs on information aggregation from the firms' optimal use of the information they receive.

The firm-value claim is in zero net supply, and noise traders hold n_j . Market clearing for each firm implies:

$$\int_0^1 m_{ij}^* di + n_j = 0. \quad (3.20)$$

A symmetric equilibrium consists of common information choices, individual portfolio rules, price coefficients, and firm investment rules. Managers' information and portfolio choices are optimal under their subjective beliefs, firms' investment choices are optimal under the true signal distribution, and financial markets clear under realized signals. Each manager takes the aggregate price and investment rules as given when choosing information. We focus on interior, regular equilibria satisfying the individual second-order conditions.

4 Information Aggregation and Capital Allocation

We solve backward from portfolio choice to information acquisition and price formation, then characterize the investment decisions supported by those prices. The key distinction is between investors' perceived information and the true precision of the signal available to firms.

4.1 Investor Learning and Portfolio Choice

Since all the shocks are independent across firms, we suppress the subscript j for simplicity. Define:

$$v_{Ai} \equiv (\alpha_i h_i)^{-1}, \quad v_{Pi} \equiv ((1 - \alpha_i) h_i)^{-1}, \quad v_q^\kappa \equiv \frac{\kappa \psi^2}{\tau_b} + \frac{\zeta^2}{\tau_n}, \quad c_{Aq}^\kappa \equiv \frac{\kappa \psi}{\tau_b}. \quad (4.1)$$

Conditional on d , the perceived error covariance matrix of $(s_i^P, s_i^A, \tilde{q})'$ is:

$$R_i^\kappa = \begin{pmatrix} h_{Pi}^{-1} & 0 & 0 \\ 0 & v_{Ai} & c_{Aq}^\kappa \\ 0 & c_{Aq}^\kappa & v_q^\kappa \end{pmatrix}. \quad (4.2)$$

Define $D_i^\kappa = v_{Ai}v_q^\kappa - (c_{Aq}^\kappa)^2 > 0$. The manager's subjective posterior distribution of demand d satisfies:

Lemma 1 (Managers' subjective posterior) *The subjective posterior of d is*

$$\hat{d}_i^\kappa = \ell_{Pi}^\kappa s_i^P + \ell_{Ai}^\kappa s_i^A + \ell_{qi}^\kappa \tilde{q}, \quad (4.3)$$

where:

$$V_{di}^\kappa \equiv \text{Var}_i^\kappa(d \mid s_i^P, s_i^A, \tilde{q}) = \left[\tau_d + h_{Pi} + \frac{v_{Ai} + v_q^\kappa - 2c_{Aq}^\kappa}{D_i^\kappa} \right]^{-1}, \quad (4.4)$$

and

$$\ell_{Pi}^\kappa = V_{di}^\kappa h_{Pi}, \quad \ell_{Ai}^\kappa = V_{di}^\kappa \frac{v_q^\kappa - c_{Aq}^\kappa}{D_i^\kappa}, \quad \ell_{qi}^\kappa = V_{di}^\kappa \frac{v_{Ai} - c_{Aq}^\kappa}{D_i^\kappa} \quad (4.5)$$

Proof. See Appendix B.1. ■

The public signal separately determines the posterior of productivity:

$$\hat{a}_j = \omega_s s_j, \quad \omega_s \equiv \frac{\tau_s}{\tau_a + \tau_s}, \quad \text{Var}(a_j \mid s_j) = \frac{1}{\tau_a + \tau_s}. \quad (4.6)$$

Thus, the conditional payoff variance satisfies:

$$\Omega_i^\kappa \equiv \text{Var}_i^\kappa(v \mid \mathcal{I}_i) = \frac{\chi^2}{\tau_a + \tau_s} + \delta^2 V_{di}^\kappa. \quad (4.7)$$

Investment does not contribute to conditional payoff variance because managers know the rule $k(s, \tilde{q})$. The following lemma characterizes each asset manager's optimal portfolio choice:

Lemma 2 *The optimal holding is*

$$m_i^* = \frac{\hat{v}_i^\kappa - q}{\rho \Omega_i^\kappa}, \quad \hat{v}_i^\kappa \equiv \mathbb{E}_i^\kappa[v \mid \mathcal{I}_i]. \quad (4.8)$$

The optimized conditional value is

$$\max_{m_i} \mathcal{V}_i(m_i \mid \mathcal{I}_i) = \frac{(\hat{v}_i^\kappa - q)^2}{2\rho \Omega_i^\kappa}. \quad (4.9)$$

Proof. See Appendix B.2. ■

4.2 Information Acquisition and Price Formation

Before signals arrive, managers choose (h_i, α_i) to maximize the expected optimized trading value net of acquisition costs. An atomistic deviator changes its own posterior coefficients $(\ell_{Pi}^\kappa, \ell_{Ai}^\kappa, \ell_{qi}^\kappa)$ when changing (h_i, α_i) , but takes the aggregate price rule, the firm's public investment rule, and aggregate demand as given. In equilibrium, the expected optimized conditional value is determined by $\mathbb{E}_{0,i}^\kappa(\hat{v}_i^\kappa - q)^2$, which is characterized by the following proposition:

Proposition 1 Let $\pi_{Ai} \equiv x_i^A = \tau_b h_{Ai} / (\tau_b - h_{Ai})$ denote raw AI precision, and define

$$H_i^\kappa = \frac{(\gamma_{Pi} + \gamma_{Ai} + \gamma_{qi})^2}{\tau_d} + \frac{\kappa(\gamma_{Ai} + \psi\gamma_{qi})^2}{\tau_b} + \frac{(1 - \kappa)\gamma_{Ai}^2}{\tau_b} + \frac{\gamma_{Pi}^2}{h_{Pi}} + \frac{\gamma_{Ai}^2}{\pi_{Ai}} + \frac{(\zeta\gamma_{qi})^2}{\tau_n}, \quad (4.10)$$

where:

$$\gamma_{Pi} = \delta\ell_{Pi}^\kappa, \quad \gamma_{Ai} = \delta\ell_{Ai}^\kappa, \quad \gamma_{qi} = \delta(\ell_{qi}^\kappa - \bar{\ell}_T^\kappa), \quad \bar{\ell}_T^\kappa \equiv \int \ell_{Pi}^\kappa di + \int \ell_{Ai}^\kappa di + \int \ell_{qi}^\kappa di, \quad (4.11)$$

Then the manager's subjective gross ex ante information value is

$$\mathcal{G}_i^\kappa(h_i, \alpha_i) = \mathbb{E}_{0,i}^\kappa \left[\frac{(\hat{v}_i^\kappa - q)^2}{2\rho\Omega_i^\kappa} \right] = \frac{H_i^\kappa}{2\rho\Omega_i^\kappa}. \quad (4.12)$$

Proof. See Appendix B.3. ■

The numerator H_i^κ measures the ex ante variation in perceived expected excess returns, or the trading opportunities generated by information. The denominator Ω_i^κ measures residual payoff risk after the signals are observed. Information can be valuable both because it changes the state-contingent expected return and because it lowers residual risk.

For an interior information choice, the first-order conditions are:

$$\frac{\partial \mathcal{G}_i^\kappa}{\partial h_i} = C_h^I, \quad \frac{\partial \mathcal{G}_i^\kappa}{\partial \alpha_i} = C_\alpha^I. \quad (4.13)$$

When differentiating (4.12), the deviator allows its own V_{di}^κ , posterior coefficients, H_i^κ , and Ω_i^κ to change, while holding $(\psi, \zeta, \bar{\ell}_T^\kappa)$ fixed.

In equilibrium, idiosyncratic signal errors wash out, whereas the common AI error does not:

$$\int_0^1 s_{ij}^P di = d_j, \quad \int_0^1 s_{ij}^A di = d_j + b_j. \quad (4.14)$$

Considering a symmetric equilibrium, where all investors make the same information acquisition decisions, the true signal realizations give:

$$\int_0^1 \widehat{d}_{ij}^\kappa di = (\ell_P^\kappa + \ell_A^\kappa + \ell_q^\kappa) d_j + (\ell_A^\kappa + \psi \ell_q^\kappa) b_j + \zeta \ell_q^\kappa n_j. \quad (4.15)$$

Combining the conjectured investment decisions, managers' portfolio holding decisions, and the market clearing condition, we can solve for the equilibrium price q_j :

Proposition 2 (Financial-information equilibrium) *Suppose the symmetric information choices are interior, the covariance matrices are positive definite, and the individual information objective is locally concave. In equilibrium, the price coefficients are:*

$$\beta_q = \delta(\ell_P^\kappa + \ell_A^\kappa + \ell_q^\kappa) + gk_q, \quad (4.16)$$

$$\beta_s = \chi\omega_s + gk_s, \quad (4.17)$$

$$\beta_0 = \delta y_N + \chi z + \mu_L + gk_0. \quad (4.18)$$

Proof. See Appendix B.4. ■

The symmetric equilibrium solves the following four equations:

$$\begin{aligned} \psi &= \frac{\ell_A^\kappa}{\ell_P^\kappa + \ell_A^\kappa}, & \zeta &= \frac{\rho\Omega^\kappa}{\delta(\ell_P^\kappa + \ell_A^\kappa)}, \\ \mathcal{G}_{h_P}^\kappa &= \mathcal{C}_{h_P}^I, & \mathcal{G}_{h_A}^\kappa &= \mathcal{C}_{h_A}^I. \end{aligned} \quad (4.19)$$

The equilibrium is a subjective-beliefs equilibrium. Managers optimize using their perceived covariance matrix, but prices clear under the true realization of the common error. When managers underestimate commonness, they treat overlapping information in AI and prices as more independent than it is. Two limiting cases clarify the mechanism. At $\kappa = 1$, the covariance matrix in (4.2) is the correct covariance matrix and investors correctly account for the dependence among all three demand signals. At $\kappa = 0$,

$$\ell_P^0 = V_d^0 h_P, \quad \ell_A^0 = V_d^0 h_A, \quad \psi = \frac{h_A}{h_P + h_A} = \alpha. \quad (4.20)$$

Thus complete correlation neglect delivers the reduced-form benchmark in which the true common-error loading of the price equals the AI information share. Price learning itself is not shut down: ℓ_q^0 generally remains positive. What disappears is the covariance correction connecting the price error to the manager's own AI error.

4.3 Firm Investment and Aggregate Productivity

Firm j observes (s_j, q_j) and, using the equilibrium price coefficients, recovers the normalized price:

$$\tilde{q}_j = \frac{q_j - \beta_0 - \beta_s s_j}{\beta_q} = d_j + \psi b_j + \zeta n_j. \quad (4.21)$$

Firms know the true distribution of the price signal. This assumption isolates the effect of managers' correlation neglect on price formation; it does not require firms to make the same mistake. Define the true price-error variance and precision by:

$$S_q^T \equiv \frac{\psi^2}{\tau_b} + \frac{\zeta^2}{\tau_n}, \quad \tau_q^T \equiv (S_q^T)^{-1}. \quad (4.22)$$

Given the productivity and price signals, the firm's posterior means of their productivity $\hat{a}_j^F$ and demand shocks $\hat{d}_j^F$ are respectively given by:

$$\hat{a}_j^F = \omega_s s_j, \quad \omega_s = \frac{\tau_s}{\tau_a + \tau_s}, \quad (4.23)$$

$$\hat{d}_j^F = \omega_q^T \tilde{q}_j, \quad \omega_q^T = \frac{\tau_q^T}{\tau_d + \tau_q^T}. \quad (4.24)$$

Because $\log X_j = \mu_L + \chi a_j + \delta d_j$, the conditional expectation of revenue productivity is:

$$M_j \equiv \mathbb{E}[X_j \mid s_j, \tilde{q}_j] = \exp \left\{ \mu_L + \chi \omega_s s_j + \delta \omega_q^T \tilde{q}_j + \frac{1}{2} \Sigma^F \right\}, \quad (4.25)$$

where

$$\Sigma^F \equiv \frac{\chi^2}{\tau_a + \tau_s} + \frac{\delta^2}{\tau_d + \tau_q^T}, \quad (4.26)$$

denotes the conditional variance of the log revenue productivity $\log X_j$. Given the posterior distribution of the fundamentals, the firms take the aggregate output as given, and choose their own investment, which turns out to follow a log-linear rule that verifies our conjecture:

Proposition 3 (Firm investment rules) *Let:*

$$\Theta \equiv \frac{1}{1-g} > 1. \quad (4.27)$$

Conditional on known aggregate output, the atomistic investment rule is:

$$K_j = \left[g Y_N^\delta Z^\chi M_j \right]^\Theta. \quad (4.28)$$

Thus, the coefficients in the log-linear investment rule (3.17) are given by:

$$k_s = \Theta \chi \omega_s, \quad (4.29)$$

$$k_q = \Theta \delta \omega_q^T, \quad (4.30)$$

$$k_0 = \Theta \left[\log g + \delta y_N + \chi z + \mu_L + \frac{1}{2} \Sigma^F \right]. \quad (4.31)$$

Proof. See Appendix B.5. ■

A realized common AI error affects investment according to:

$$\frac{\partial \log K_j}{\partial b_j} = k_q \psi > 0. \quad (4.32)$$

The firm understands that the price is noisy and optimally attenuates its response through ω_q^T . It nevertheless cannot distinguish d_j from b_j in a single price realization. Managers' correlation neglect therefore matters because it changes the true equilibrium coefficients (ψ, ζ) and hence the precision of the demand signal observed by the firm.

Aggregate TFP. Let $K_N = N^{-1} \sum_j K_j$ denote aggregate investment. The individual investment rule implies $K_j = K_N M_j^\Theta / [N^{-1} \sum_\ell M_\ell^\Theta]$, so aggregate output can be written as $Y_N = Z \mathcal{A}_N^R K_N^\eta$, where $\mathcal{A}_N^R$ can be regarded as the aggregate TFP. Appendix B.6 derives the realized finite-sample index $\mathcal{A}_N^R$, which depends on the realized firm-level fundamentals. As in Benhabib et al. (2019), the equilibrium interpretation applies as $N \rightarrow \infty$, when each firm is atomistic. Conditional on known Z , the law of large numbers replaces cross-sectional averages by their expectations, so limiting aggregate TFP is nonstochastic.

Proposition 4 (Limiting aggregate TFP) *Suppose the primitive firm-level shocks are independent and identically distributed across j and have finite relevant moments. Then as the number of firms $N \rightarrow \infty$, the aggregate TFP becomes:*

$$\mathcal{A}_N^R \xrightarrow{a.s.} \mathcal{A} = \left\{ \mathbb{E}[M^\Theta] \right\}^{1/\chi - \eta}. \quad (4.33)$$

Define:

$$\Sigma_0 \equiv \frac{\chi^2}{\tau_a} + \frac{\delta^2}{\tau_d}, \quad (4.34)$$

Then aggregate TFP can be expressed as:

$$\log \mathcal{A} = \left(\frac{1}{\chi} - \eta \right) \left[\Theta \mu_L + \frac{\Theta^2}{2} \Sigma_0 - \frac{\Theta(\Theta - 1)}{2} \Sigma^F \right]. \quad (4.35)$$

Proof. See Appendix B.7. ■

Although b_j is independent across firms and therefore washes out of the cross-sectional average in levels, its variance remains in each firm's price signal. It consequently affects the deterministic limiting TFP through Σ^F . As for the aggregate capital, the unit capital cost and firms' first-order conditions imply $K = gY$ in the limit with $N \rightarrow \infty$. Combining this identity with $Y = ZAK^\eta$ gives:

$$K = (gZA)^{1/(1-\eta)}, \quad Y = g^{\eta/(1-\eta)}(ZA)^{1/(1-\eta)}. \quad (4.36)$$

The allocation-efficiency object $\mathcal{A}$ separates changes in the distribution of investment from changes in aggregate capital. Holding production parameters fixed, output and TFP move in the same direction, with the output response amplified by endogenous aggregate investment.

Price informativeness and aggregate TFP. The effect of the financial equilibrium on real allocation is summarized by the true error variance of the standardized price signal,

$$S_q^T(\kappa; c_A, \tau_b) = \underbrace{\frac{\psi(\kappa; c_A, \tau_b)^2}{\tau_b}}_{\text{common-AI component}} + \underbrace{\frac{\zeta(\kappa; c_A, \tau_b)^2}{\tau_n}}_{\text{noise-trader component}}. \quad (4.37)$$

AI reliance $\alpha = h_A/(h_P + h_A)$ matters because it changes the equilibrium price loadings, but it is not by itself a sufficient statistic for allocation. A rise in AI reliance can increase ψ while a simultaneous increase in total information reduces ζ . Firms care about the sum of the two true error components. Since:

$$\frac{1}{\tau_d + \tau_q^T} = \frac{S_q^T}{1 + \tau_d S_q^T}, \quad (4.38)$$

the price error variance determines the equilibrium aggregate TFP. Only the second term of Σ^F depends on S_q^T . Equations (4.26) and (4.38) imply:

$$\frac{\partial \Sigma^F}{\partial S_q^T} = \frac{\delta^2}{(1 + \tau_d S_q^T)^2}.$$

Differentiating (4.35) then gives:

$$\frac{\partial \log \mathcal{A}}{\partial S_q^T} = -\frac{(1/\chi - \eta)\Theta(\Theta - 1)\delta^2}{2(1 + \tau_d S_q^T)^2} < 0. \quad (4.39)$$

Thus a parameter change improves allocation if and only if it reduces the true error variance of the price signal, holding production parameters fixed. AI progress affects both components

through information acquisition. More informed trading can reduce the price response to noise demand, while greater reliance on a shared signal can increase exposure to its common error. The equilibrium determines which effect dominates. Firms optimally discount a less precise price, but the resulting increase in posterior uncertainty still lowers allocation efficiency.

At fixed model accuracy, a change in AI acquisition costs affects price noise only through the equilibrium loadings:

$$\frac{dS_q^T}{dc_A} = \frac{2\psi}{\tau_b} \frac{d\psi}{dc_A} + \frac{2\zeta}{\tau_n} \frac{d\zeta}{dc_A}. \quad (4.40)$$

A cost reduction raises TFP if its effects on information acquisition reduce the total price-error variance. There is no direct change in the variance of the underlying AI error.

Higher model accuracy also changes the underlying variance of the common AI error, adding a direct information benefit:

$$\frac{dS_q^T}{d\tau_b} = \frac{2\psi}{\tau_b} \frac{d\psi}{d\tau_b} - \frac{\psi^2}{\tau_b^2} + \frac{2\zeta}{\tau_n} \frac{d\zeta}{d\tau_b}. \quad (4.41)$$

The middle term is the direct benefit of a less volatile common error. The other two terms capture endogenous changes in price exposure and noise. The next section uses the equilibrium system to compare these responses near the no-AI boundary and then studies their global shape numerically.

5 AI Progress and Aggregate Productivity

This section compares cheaper access to AI information with improvements in model accuracy. We first simplify the equilibrium under complete correlation neglect and establish the local productivity effects. Numerical analysis then traces the J-shaped response to accuracy and the contrasting response to lower costs beyond the no-AI boundary, allowing for different beliefs.

5.1 Information Acquisition under Correlation Neglect

Setting $\kappa = 0$ in the equilibrium system isolates the consequences of treating shared information as independent. Investors still learn from prices and correctly assess the marginal precision of their own AI signals. They omit the covariance correction linking those signals to prices. This benchmark yields a tractable characterization of information quantity and composition; it is not a general lower bound on TFP or an assertion that complete neglect describes all investors.

Define the precision of the price signal perceived by managers as:

$$r \equiv \frac{\tau_n}{\zeta^2}, \quad (5.1)$$

and let:

$$\Omega_a \equiv \frac{\chi^2}{\tau_a + \tau_s}. \quad (5.2)$$

The following lemma reduces the general equilibrium system at $\kappa = 0$:

Lemma 3 (Information equilibrium under complete correlation neglect) *Suppose $\kappa = 0$ and maintain the conditions for an interior symmetric financial-information equilibrium. The equilibrium conditions reduce to:*

$$\psi = \alpha, \quad (5.3)$$

$$r = \frac{\tau_n \delta^2 h^2}{\rho^2 [\Omega_a (\tau_d + h + r) + \delta^2]^2}, \quad (5.4)$$

$$C_h^I(h, \alpha) = m(h), \quad (5.5)$$

$$C_\alpha^I(h, \alpha) = 0. \quad (5.6)$$

Here, $r = \tau_n / \zeta^2$ is the perceived precision of the price signal. For each $h > 0$, the price equation has a unique positive solution $r(h)$, with $r'(h) > 0$. The function $m(h) > 0$ is the manager's marginal gross return to total information, evaluated at the symmetric equilibrium price rule; its expression is given in (B.14), and $m'(h) < 0$.

Proof. See Appendix B.8. ■

The lemma separates the choice of total information from the cost-minimizing composition at a given quantity. Under complete correlation neglect, private and AI information have the same marginal gross value per unit of effective precision. The AI share therefore minimizes the cost of obtaining h , while total precision equates marginal cost and marginal value. Prices remain endogenous: higher total information raises the perceived price precision $r(h)$ and changes the equilibrium marginal return. With $\kappa > 0$, the investor instead accounts for some of the overlap between AI and prices, so composition also affects gross information value.

Although composition does not affect the manager's gross information value at a fixed h and price rule, it affects the information that prices convey to firms. Since $\psi = \alpha$, the true price error variance becomes

$$S_q^T = \frac{\alpha^2}{\tau_b} + s(h), \quad s(h) \equiv \frac{1}{r(h)}, \quad s'(h) < 0. \quad (5.7)$$

Total information reduces the noise-trader component of the price error, whereas a larger AI

share increases its common-error component at a given τ_b . The latter component remains in actual prices even though managers omit it from their perceived price variance. Equation (4.39) maps changes in these two components into changes in endogenous aggregate TFP.

Under the cost specification in equation (3.13), the marginal costs expressed in effective precisions are:

$$\text{MC}_P(h_P) \equiv c_P \frac{\tau_u^{\nu+1} h_P^{\nu-1}}{(\tau_u - h_P)^{\nu+1}}, \quad (5.8)$$

$$\text{MC}_A(h_A; c_A, \tau_b) \equiv c_A \frac{\tau_b^{\nu+1} h_A^{\nu-1}}{(\tau_b - h_A)^{\nu+1}}. \quad (5.9)$$

Thus, an interior symmetric information equilibrium at $\kappa = 0$ satisfies:

$$m(h) = \text{MC}_P(h_P) = \text{MC}_A(h_A; c_A, \tau_b), \quad h = h_P + h_A. \quad (5.10)$$

5.2 The Initial Productivity Effects of AI Progress

We consider two routes to negligible AI use: prohibitively high acquisition costs at fixed accuracy, and vanishing model accuracy at fixed acquisition costs. Both approach $\alpha = 0$, but they have different first-order implications for the price-error variance. The following proposition characterizes those implications.

Proposition 5 (Local TFP effects of AI progress) *Suppose $\kappa = 0$ and $\nu > 1$. Assume that the no-AI limiting equilibrium is interior and regular and that the relevant symmetric equilibrium branches are differentiable and satisfy the local second-order conditions. Let h_0 denote the precision in the no-AI equilibrium which solves $m(h_0) = \text{MC}_P(h_0)$, and define:*

$$\begin{aligned} m_0 &\equiv m(h_0), \quad \lambda_0 \equiv \frac{\text{MC}'_P(h_0)}{\text{MC}'_P(h_0) - m'(h_0)} \in (0, 1), \quad S_0 \equiv s(h_0), \\ \Gamma_0 &\equiv \frac{(1/\chi - \eta)\Theta(\Theta - 1)\delta^2}{2(1 + \tau_d S_0)^2} > 0. \end{aligned}$$

Then the following results hold near the two zero-AI-reliance boundaries.

(i) Fix $\tau_b > 0$ and let $c_A \rightarrow \infty$. Then $\alpha \rightarrow 0$ and:

$$h - h_0 = \lambda_0 h_A + o(h_A), \quad \frac{d \log \mathcal{A}}{dc_A} < 0 \quad \text{for sufficiently large } c_A. \quad (5.11)$$

Hence, a marginal decline in c_A necessarily raises endogenous aggregate TFP when AI reliance is sufficiently close to zero along the high-cost boundary.

(ii) Fix $c_A \in (0, \infty)$ and let $\tau_b \downarrow 0$. Then $\alpha \rightarrow 0$ and:

$$\frac{h_A}{\tau_b} \rightarrow 1, \quad h - h_0 = \lambda_0 \tau_b + o(\tau_b), \quad \alpha = \frac{\tau_b}{h_0} + o(\tau_b). \quad (5.12)$$

The right-hand derivative of endogenous aggregate TFP at the zero-accuracy boundary is:

$$\left. \frac{d \log \mathcal{A}}{d \tau_b} \right|_{0^+} = \Gamma_0 \left\{ [-s'(h_0)] \lambda_0 - \frac{1}{h_0^2} \right\}. \quad (5.13)$$

A marginal increase in τ_b initially increases endogenous aggregate TFP if and only if:

$$\frac{1}{h_0^2} < -s'(h_0) \lambda_0. \quad (5.14)$$

Proof. See Appendix B.9. ■

The proposition distinguishes two reasons why AI use may initially be low: acquiring AI information is costly, or the underlying model is inaccurate. These constraints have different implications for the information that financial markets transmit to firms. Near the no-AI equilibrium, the first-order condition implies:

$$h - h_0 = \lambda_0 h_A + o(h_A), \quad \alpha = \frac{h_A}{h_0} + o(h_A). \quad (5.15)$$

Each additional unit of effective AI information displaces $1 - \lambda_0$ units of private information, so total precision rises by λ_0 units. This local substitution applies to both forms of AI progress. Substituting into the price-error variance gives:

$$S_q^T - S_0 = h_A \left[s'(h_0) \lambda_0 + \frac{1}{h_0^2} \frac{h_A}{\tau_b} \right] + o(h_A). \quad (5.16)$$

The first term captures the reduction in noise-trader error as total information precision increases, since $s'(h_0) < 0$, while the second captures exposure to the common AI error that reduces price informativeness and aggregate TFP. h_A/τ_b denotes the share of the AI signal's error variance attributable to the common model error.

When acquisition costs are high and model accuracy is fixed, $h_A/\tau_b \rightarrow 0$. The accuracy of AI information acquired is far from its limit, τ_b . In this case, investors acquire little AI information, and their signal errors are dominated by the idiosyncratic acquisition noise. Cheaper access thus increases total information and reduces the noise-trader component of prices, while the increase in common error exposure is negligible, so the initial effect on aggregate TFP is positive.

When model accuracy is very low, the information-choice condition instead implies $h_A/\tau_b \rightarrow 1$. With a fixed acquisition-cost parameter and $\nu > 1$, the marginal cost of raw precision approaches zero as acquisition vanishes. When model error is highly volatile, even a small research input can reduce acquisition noise relative to that error and raise effective signal precision at low cost. Investors therefore have an incentive to reduce acquisition noise, and they obtain nearly all the effective precision available from the model, even though both h_A and the AI share α approach zero. Greater accuracy then raises effective AI information almost one-for-one with τ_b . Although the variance of the common error falls, increased reliance makes its contribution to price errors rise. TFP falls when this increase outweighs the reduction in noise-trader error.

Both forms of progress raise investors' perceived posterior precision. Under correlation neglect, however, this improvement can coexist with higher common AI reliance and lower effective price-signal precision. Firms account for price noise when investing, but cannot infer demand fundamentals as accurately. This information loss could worsen capital allocation and reduce aggregate productivity, and Proposition 5 identifies these effects at the adoption margin.

Correlation beliefs. Investors' beliefs about common errors affect both information acquisition and price use. At an interior symmetric equilibrium with $0 < \psi < 1$, Bayesian updating and the information first-order conditions imply:

$$\frac{\mathcal{G}_{h_A}^\kappa}{\mathcal{G}_{h_P}^\kappa} = \frac{\text{MC}_A(h_A)}{\text{MC}_P(h_P)} = \left(\frac{v_q^\kappa - c_{Aq}^\kappa}{v_q^\kappa - (c_{Aq}^\kappa)^2 h_A} \right)^2 < 1, \quad 0 < \kappa \leq 1. \quad (5.17)$$

Effective AI precision has lower perceived marginal value because the price already contains information that overlaps with the AI signal. With strictly convex acquisition costs, investors therefore choose a smaller AI share than would minimize acquisition costs at the same total precision h . The comparison across beliefs also depends on how h adjusts. At the complete neglect margin $\kappa = 0$, a marginal change in the belief system κ would affect the AI information

share and price loadings according to:⁹

$$\frac{d\alpha}{d\kappa}|_{\kappa=0} = -\frac{2m(h)h\alpha r/\tau_b + C_{\alpha h}^I dh/d\kappa}{C_{\alpha\alpha}^I}, \quad \frac{d\psi}{d\kappa}|_{\kappa=0} = \frac{d\alpha}{d\kappa}|_{\kappa=0} - \frac{\alpha^2(1-\alpha)r}{\tau_b}. \quad (5.18)$$

Holding total information fixed, the direct effect of greater recognition on the AI share is negative. The overall response also depends on the adjustment in h , as the first line of (5.18) shows. The second line separates that acquisition response from a further decline in the common-error price loading: investors who recognize overlap discount the component repeated in AI and prices. Greater recognition can therefore moderate the common-error exposure associated with AI progress, but it also changes informed trading and the price response to noise demand. Since firms use a signal with error variance $\psi^2/\tau_b + \zeta^2/\tau_n$, lower common-error exposure need not imply higher TFP. The numerical exercises examine this balance without imposing a monotone ordering across beliefs.

5.3 Numerical Analysis

Proposition 5 characterizes the effects of AI progress near the no-AI equilibrium. We now examine how these effects change as AI use expands. The exercises vary model accuracy and acquisition costs separately, and trace their effects on the quantity and composition of information, price informativeness, and aggregate productivity and output. We solve the symmetric equilibrium in the large- N limit and compare different beliefs about the common AI error.

Parameterization. We first parameterize an economy without AI as a common reference for the exercises. Following Benhabib et al. (2019), we set the elasticity of substitution between intermediate goods to $\theta = 6$, the production exponent to $\eta = 0.8$, and the risk-aversion coefficient to $\rho = 8$.¹⁰ To discipline prior and signal uncertainty, we use the U.S. estimates in David et

⁹Let u denote the derivative of $\mathcal{G}_{h_p}^\kappa$ with respect to κ when (h_p, h_A) remain fixed and $(\psi, \zeta, \bar{\ell}_T^\kappa)$ adjust with the price equilibrium. The individual marginal value itself is computed holding the price rule fixed. Differentiating the information first-order conditions gives:

$$\frac{dh}{d\kappa} = \frac{u - \frac{MC'_P}{MC'_P + MC'_A} \frac{2m(h)\alpha r}{\tau_b}}{\frac{MC'_P MC'_A}{MC'_P + MC'_A} - m'(h)}.$$

Primes on marginal costs denote derivatives with respect to their own effective precisions. The denominator is positive because both marginal costs are increasing and $m'(h) < 0$. The numerator compares the change in private information value with the decline in AI's relative value, so total information can rise or fall.

¹⁰Benhabib et al. (2019)'s choice of η draws on Gopinath et al. (2017). The latter estimate revenue elasticities whose sum ranges from 0.75 to 0.91, providing evidence on curvature rather than a direct estimate of our production exponent.

al. (2016) where the standard deviations of fundamental innovations and firms' private-signal errors are 0.45 and 0.39, respectively. We use these magnitudes as a guide to uncertainty in our static environment, rather than as direct estimates of its priors. Taking inverse variances gives $\tau_a = 0.45^{-2} = 4.9383$ and $\tau_s = 0.39^{-2} = 6.5746$. The resulting posterior standard deviation of log productivity is approximately 0.295. We hold τ_s fixed as financial information changes. We set the prior precision of demand shocks to $\tau_d = 0.04$, using the value in Benhabib et al. (2019) as a literature benchmark.

For noisy trading volatility, we set $\tau_n = 1000$, corresponding to a noise-demand standard deviation of approximately 0.0316, using the quarterly noise-trading intensity in Peress and Schmidt (2021) as an order-of-magnitude reference.¹¹ For the information acquisition cost, we use quadratic costs, $\nu = 2$, and set the private-signal precision ceiling to $\tau_u = 0.02$, as illustrative choices: private research remains noisy even as acquisition effort increases. Conditional on these values, we choose $c_P = 1.12605$ so that the equilibrium price-signal precision without AI, $1/S_q^T$, matches the price-signal precision of 0.0655 in the higher-information equilibrium reported by Benhabib et al. (2019). The resulting private-information precision is $h_P = h = 0.01358 < \tau_u$, which implies that the private research reaches around 68% of the maximum private signal precision, τ_u .

In the numerical exercise, we compare four cases with $\kappa = 0$, $\kappa = 0.25$, $\kappa = 0.75$, and $\kappa = 1$, representing complete correlation neglect, partial recognition, and correct beliefs about the common error. These values illustrate how beliefs affect the transmission of AI progress; they are not estimates of investors' recognition of shared errors. The AI cost c_A and accuracy τ_b are specified within each exercise. We report TFP as an index with its value in the economy without AI normalized to 100; that reference economy is the same across beliefs because it contains no AI signal.

Holding production parameters fixed, output moves one-for-one with TFP in sign, with $\Delta \log Y = \Delta \log \mathcal{A}/(1 - \eta)$. We therefore show four panels for each exercise: TFP, the common-error component of price noise, total information, and the AI share. The no-AI economy provides a common normalization, not a calibration of current AI adoption.

Changes in AI accuracy. We first increase model accuracy τ_b while holding the acquisition-cost parameters equal, $c_A = c_P$. Figure 1 shows a J-shaped TFP response under complete correlation neglect: productivity initially declines, before recovering and rising above the no-AI benchmark.

Higher model accuracy improves the precision of a given amount of AI research and raises

¹¹At the quarterly frequency, Peress and Schmidt (2021) reports bounds of 2.2% to 4.3% for noise-trading intensity using price-improved TAQ trades from 2010 to 2015. These bounds concern market-level trading intensity constructed from turnover. They are not a direct estimate of the variance of our normalized claim demand; the mapping is illustrative and depends on the demand normalization.

the return to additional acquisition. Investors therefore acquire more AI information and partly substitute away from private research, increasing both total information h and the AI share α in Panels C and D. The expansion strengthens informed trading relative to noise demand and reduces the noise-trader component of price error. However, greater AI reliance also raises the equilibrium loading ψ . Starting from low accuracy, this loading increases sufficiently rapidly that the common-error component ψ^2/τ_b rises in Panel B, despite the decline in the underlying error variance $1/\tau_b$. A more accurate model thus exposes prices to more common error because investors use its information more extensively. Under complete correlation neglect, this increase initially outweighs the reduction in noise-trader contamination, worsening firms' information about demand and lowering TFP.

As accuracy improves further, the reduction in noise-trader contamination overtakes the increase in the common component, and TFP starts to recover. The recovery therefore begins before the common-error component itself falls. At higher accuracy, acquisition costs and the remaining investor-specific noise increasingly limit the gains from further improvements in the underlying model. The response of AI reliance moderates, and the decline in model-error variance eventually dominates the further increase in ψ . The common component then falls as well, reinforcing the recovery in capital allocation, TFP, and output. The J-shaped response reflects this changing balance between the expansion of AI reliance and the improvement in the information on which investors rely.

Along these equilibrium paths, greater recognition moderates the initial expansion in AI reliance. At $\kappa = 0.25$, TFP exhibits a shallow initial decline, whereas it increases throughout the plotted range at $\kappa = 0.75$ and $\kappa = 1$. Even under correct beliefs, the common-error component initially rises, but the reduction in noise-trader contamination is sufficient to improve price informativeness. At higher accuracy, the TFP ordering across beliefs can reverse: greater reliance on AI then conveys more demand information through prices. These comparisons illustrate why recognition of common errors has no uniform implication for allocative productivity.

These results also connect to the productivity J-curve literature. [Brynjolfsson et al. \(2021\)](#) show how complementary intangible investment and its incomplete measurement can depress measured productivity growth during the early adoption of a general-purpose technology, before the returns to these investments become visible. Our model identifies a complementary mechanism operating through information aggregation and capital allocation. When investors underestimate the AI model's common errors, improvements from low AI accuracy induce greater reliance on the shared signal. Although investors acquire more precise information, the resulting increase in common-error exposure initially outweighs the reduction in price contamination. Prices become less informative about demand, and firms allocate capital less efficiently. The productivity loss therefore reflects a deterioration in actual allocation, even as investors actively use the improving technology. Further gains in accuracy eventually improve price in-

formativeness and restore the productivity gains from AI. The horizontal axis here is model accuracy, not time. The exercise therefore describes a comparative-static J-shaped response in actual allocative productivity, rather than a transition path or a measurement shortfall. It shows how the information generated by a new technology can affect productivity through the prices guiding investment.

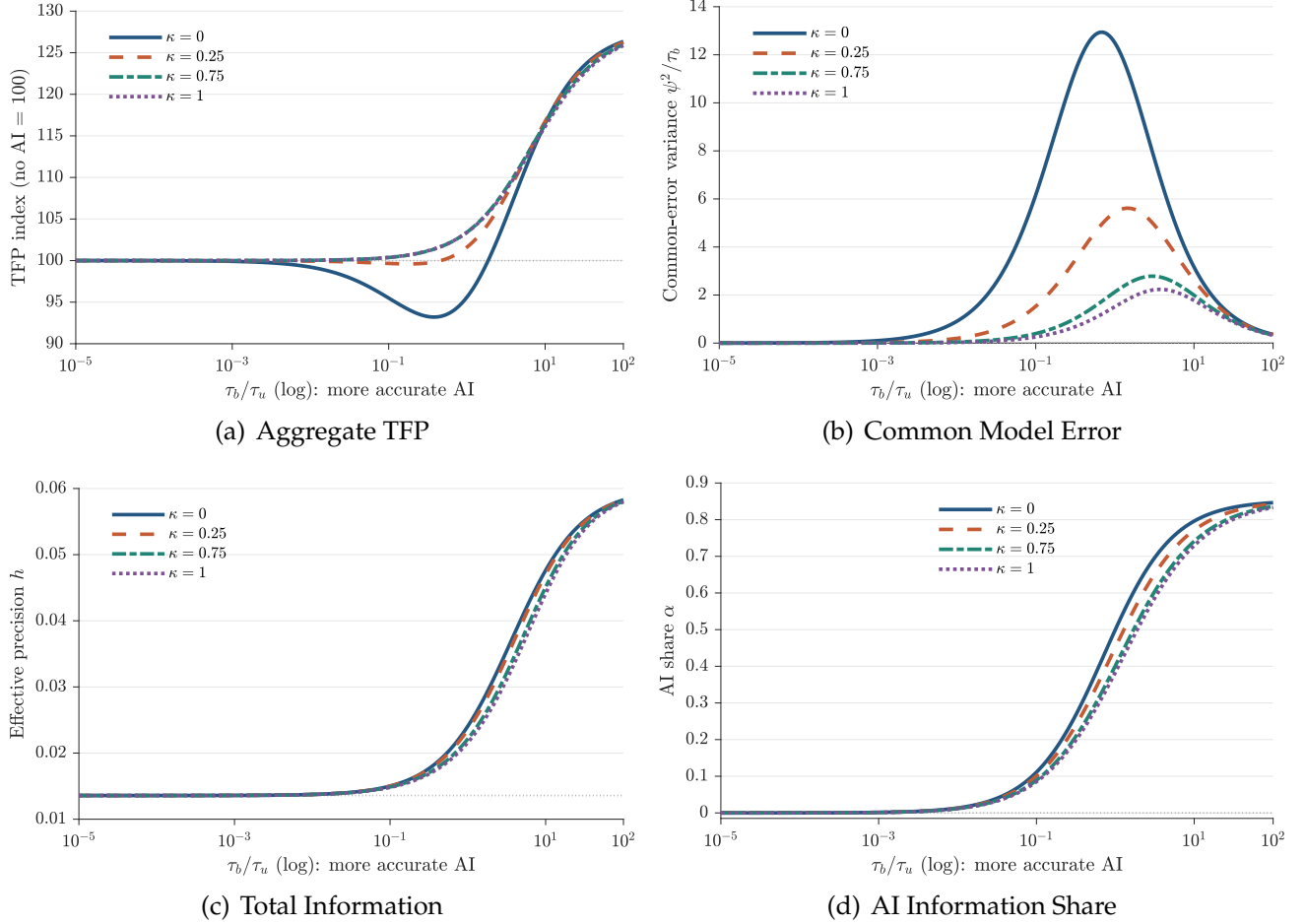


Figure 1: Equilibrium Effect of AI Accuracy Changes

Notes: This figure shows the equilibrium effect as AI accuracy τ_b changes. The acquisition cost is held at $c_A = c_P$. Moving right corresponds to an increase in τ_b ; the horizontal axis shows τ_b/τ_u on a logarithmic scale. The four curves correspond to $\kappa = 0, 0.25, 0.75, 1$. Gray dotted lines show the economy without AI. All other structural parameters are fixed across curves.

Changes in AI acquisition costs. We next reduce AI acquisition costs c_A while holding model accuracy at $\tau_b = \tau_u$. Figure 2 shows the equilibrium TFP index $\mathcal{A}$, the common-error component of price-signal noise ψ^2/τ_b , and the information structure (h, α) . Panel A shows that the initial productivity gain in Proposition 5 can reverse as AI becomes cheaper. Under complete correlation neglect, TFP first rises and then falls below its level in the economy without AI.

Lower acquisition costs make additional AI information cheaper at the margin, inducing investors to expand AI research. Since AI and private signals convey information about the same demand fundamental, this expansion partly substitutes for private research. Along the equilibrium path, investors acquire less private information, while both total information h and the AI share α increase, as shown in Panels C and D. The distinction between the two sources matters for information aggregation: additional AI acquisition reduces investor-specific signal noise but leaves the underlying common error unchanged. Whereas private signal errors average out across investors, the common AI error shifts their portfolio demands in the same direction. As investors place greater weight on AI information, market clearing therefore gives this error a larger loading in the price signal. Under complete correlation neglect, this connection is particularly direct: $\psi = \alpha$. With τ_b fixed, the rising AI share thus increases ψ^2/τ_b , as shown in Panel B.

At the same time, greater information acquisition strengthens informed trading relative to noise-trader demand, reducing the noise-trader component ζ^2/τ_n along all four paths. Initially, this reduction outweighs the increase in the common-error component, so prices become more informative and firms allocate capital more closely to demand fundamentals. As AI reliance grows, however, the balance between these effects changes. Under complete correlation neglect, $\psi = \alpha$, so the common-error component is quadratic in the AI share. A given increase in α therefore adds more common price error when investors already rely heavily on AI.

As acquisition costs decrease, further gains in total information also shrink. With model accuracy fixed, effective AI precision is bounded by τ_b : additional acquisition removes investor-specific noise but cannot eliminate the common model error. The accompanying decline in private research further limits the net expansion in h . Consequently, further cost reductions yield progressively smaller reductions in noise-trader contamination, while the continued shift toward AI raises the common-error loading. Under complete correlation neglect, the latter effect eventually dominates along the equilibrium path. Prices then become less informative even though investors continue to acquire more information. Firms optimally discount the less precise price signal, but their remaining uncertainty about demand rises, worsening capital allocation and lowering TFP. Output follows TFP, with the response amplified by the adjustment in aggregate investment.

Recognizing the common error moderates the shift toward AI along these paths and preserves a greater role for private research. The hump remains at $\kappa = 0.25$. At $\kappa = 0.75$, the later decline is small and TFP remains above the no-AI benchmark, while at $\kappa = 1$ it increases throughout the plotted range. The ordering across beliefs also changes with costs. When AI use is limited, the larger information expansion under greater neglect can improve price discovery. When AI becomes cheap, the associated common-error exposure can instead reduce TFP. Cheaper access therefore illustrates how expanding investors' information can eventually

impair the information that prices convey to firms.

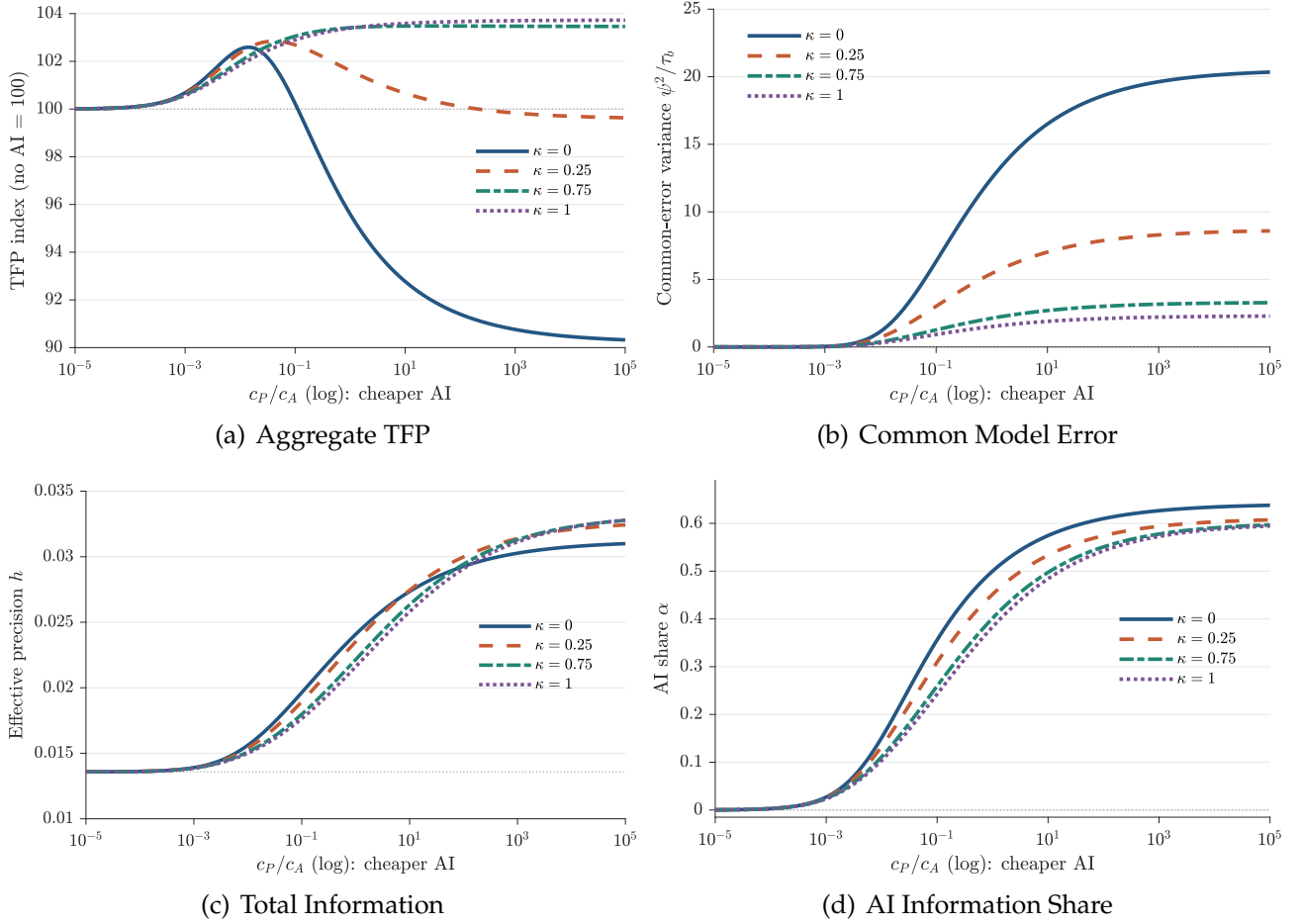


Figure 2: Equilibrium Effect of AI Cost Changes

Notes: This figure shows the equilibrium effect as AI information cost c_A changes. AI accuracy is held at $\tau_b = \tau_u$. Moving right corresponds to a decline in c_A ; the horizontal axis shows c_P/c_A on a logarithmic scale. The four curves correspond to $\kappa = 0, 0.25, 0.75, 1$.

6 General Information Acquisition Costs

This section considers the robustness of our main results under general information acquisition costs. Conditional on information choices, the portfolio and price equations do not require a particular acquisition cost. Costs determine how those choices respond to technological progress. We first extend the local results to general separable cost functions, then consider the entropy-based cost structure under certain conditions. Throughout, $\kappa = 0$, and the relevant equilibrium branches satisfy individual second-order conditions and the stated regularity conditions.

6.1 General Separable Acquisition Costs

We first retain the precision technology in (3.12) and allow general separable acquisition costs:

$$C = c_A f(x^A) + g(x^P, c_P), \quad x^A = \frac{\tau_b h_A}{\tau_b - h_A}, \quad x^P = \frac{\tau_u h_P}{\tau_u - h_P}. \quad (6.1)$$

This formulation follows the literature that specifies costs of attention or signal precision. Colombo et al. (2014) assume an increasing, convex private-precision cost with zero initial marginal cost. Models with several information sources also specify costs directly over the vector of attention inputs.¹² Equation (6.1) makes the source costs separable; the next subsection allows composition and total information to interact in costs.

Define the private cost in effective-precision coordinates by

$$G(p) = g\left(\frac{\tau_u p}{\tau_u - p}, c_P\right). \quad (6.2)$$

Let $h_0 \in (0, \tau_u)$ satisfy $m(h_0) = G'(h_0)$, where m is the marginal information value in Lemma 3. For this cost specification, let $m_0 = m(h_0)$ and define:

$$\lambda_0 = \frac{G''(h_0)}{G''(h_0) - m'(h_0)}, \quad \Gamma_0 = - \left. \frac{\partial \log \mathcal{A}}{\partial S_q^T} \right|_{S_q^T = s(h_0)} > 0. \quad (6.3)$$

Proposition 6 Suppose $f(0) = 0$, $f'(x) > 0$ for $x > 0$, and f is convex, continuously differentiable at zero, and twice continuously differentiable at positive inputs. When $f'_+(0) > 0$, additionally assume $\lim_{\pi \downarrow 0} \pi f''(\pi) = 0$. Suppose G is twice continuously differentiable near h_0 , with $G''(h_0) > 0$. Then:

- (i) Fix $\tau_b > 0$. On the interior equilibrium branch, a marginal reduction in c_A raises TFP whenever h_A is sufficiently close to zero:

$$\frac{d \log \mathcal{A}}{dc_A} < 0 \quad \text{for sufficiently small } h_A > 0. \quad (6.4)$$

The no-AI boundary is approached as $c_A \rightarrow \infty$ if $f'_+(0) = 0$, and as $c_A \uparrow m_0 / f'_+(0)$ from below if $f'_+(0) > 0$.

- (ii) Fix $c_A > 0$ satisfying $c_A f'_+(0) < m_0$. As $\tau_b \downarrow 0$:

$$\frac{d \log \mathcal{A}}{d\tau_b} \Big|_{\tau_b \rightarrow 0} = \Gamma_0 \left(1 - \sqrt{\frac{c_A f'_+(0)}{m_0}} \right) \times \left[[-s'(h_0)] \lambda_0 - \frac{1}{h_0^2} \left(1 - \sqrt{\frac{c_A f'_+(0)}{m_0}} \right) \right]. \quad (6.5)$$

¹²For example, Pavan (2025) uses a linear cost of total attention in the main text and allows a general increasing convex cost in the appendix. The information-source structure follows Myatt and Wallace (2012).

Proof. See Appendix C.1. ■

Proposition 6 shows which features of information costs matter for the two forms of AI progress. Under correlation neglect, the AI acquisition condition can be written as:

$$\left(1 - \frac{h_A}{\tau_b}\right)^2 = \frac{c_A f'(\pi_A)}{m(h)}. \quad (6.6)$$

The fraction of available AI precision that investors acquire therefore depends on the marginal cost of acquisition relative to the marginal value of effective information. With accuracy fixed, high costs keep raw acquisition small, so $h_A/\tau_b \rightarrow 0$. The resulting gain in total information reduces noise-trader contamination at first order, while exposure to the common error increases only at second order, leading to a marginal TFP gain. A positive initial marginal cost changes the location of the adoption boundary: above the finite threshold $m_0/f'_+(0)$, investors do not use AI, and cost reductions have no effect until they cross that threshold.

When accuracy approaches zero at a fixed acquisition cost, raw acquisition also vanishes. If $f'_+(0) = 0$, the marginal cost of acquisition reaches zero and investors choose to obtain nearly all the precision the model can provide. Thus $h_A/\tau_b \rightarrow 1$, even though the AI share α approaches zero. Equation (6.5) then recovers the boundary formula in Proposition 5. The power exponent determines how quickly acquisition approaches this limit, not the limit itself. If, instead, the marginal cost at the adoption margin is positive, $f'_+(0) > 0$, the investors would acquire information such that the aggregate AI information precision is a fraction of its frontier:

$$\frac{h_A}{\tau_b} \big|_{\tau_b \rightarrow 0} = 1 - \sqrt{\frac{c_A f'_+(0)}{m_0}}, \quad (6.7)$$

which remains positive if $c_A f'_+(0) < m_0$. This weakens both the gain in total information and the increase in common-error exposure. As a result, the marginal TFP effect of an increase in τ_b changes to (6.5). However, as long as $c_A f'_+(0) < m_0$, the sign of this marginal effect could be either positive or negative, as the expansion in common error exposure dominates when:

$$1 - \sqrt{\frac{c_A f'_+(0)}{m_0}} > -h_0^2 s'(h_0) \lambda_0. \quad (6.8)$$

6.2 Entropy-Based Non-Separable Information Costs

Information costs can also be expressed in terms of the entropy reduction from acquired signals (Van Nieuwerburgh and Veldkamp, 2010). For the two direct Gaussian signals, the entropy

reduction is:

$$\mathcal{K} = I(d; s^A, s^P) = \frac{1}{2} \log \left(1 + \frac{h}{\tau_d} \right), \quad h = H(\mathcal{K}) = \tau_d(e^{2\mathcal{K}} - 1). \quad (6.9)$$

The composition variable α retains its definition as the share of effective precision supplied by AI. At a given $\mathcal{K}$, different values of α may require different research inputs and generate different exposure to the common error. Inspired by this observation, we set the information acquisition cost as:

$$\mathcal{C}(\mathcal{K}, \alpha; c_A, c_P, \tau_b). \quad (6.10)$$

This representation accommodates entropy-based costs with source-dependent acquisition technology. It also includes (6.1): substitute $h_A = \alpha H(\mathcal{K})$ and $h_P = (1 - \alpha)H(\mathcal{K})$ into the raw-precision formulas. We retain the original feasibility bounds and consider interior equilibria. Assume that $\mathcal{C}$ is twice continuously differentiable at interior choices and that $\mathcal{C}(\mathcal{K}, 0; c_A, c_P, \tau_b) = \mathcal{C}_0(\mathcal{K}; c_P)$, so AI technology does not change private-research costs when AI is unused. For this cost, let $G(p) = \mathcal{C}_0(\frac{1}{2} \log(1 + p/\tau_d); c_P)$ and let $h_0 \in (0, \tau_u)$ solve $G'(h_0) = m(h_0)$. Assume G is twice continuously differentiable near h_0 , with $G''(h_0) - m'(h_0) > 0$. The constants $m_0 = m(h_0)$, λ_0 , and Γ_0 are evaluated at this no-AI equilibrium.

In the complete neglect benchmark, $\kappa = 0$, the information acquisition first-order conditions in a symmetric equilibrium are given by:

$$\mathcal{C}_\alpha = 0, \quad \mathcal{C}_\mathcal{K} = H'(\mathcal{K})m(H(\mathcal{K})). \quad (6.11)$$

This specification permits cost interactions across sources: $\mathcal{C}_{h_A h_P}$ can be nonzero. An additional unit of AI precision may raise or lower the marginal cost of private research. In quantity–composition coordinates, the response also depends on how total information changes the preferred mix. For $\vartheta \in \{c_A, \tau_b\}$, differentiating the composition condition gives:

$$\alpha_\vartheta = -\frac{\mathcal{C}_{\alpha\vartheta}}{\mathcal{C}_{\alpha\alpha}} - \frac{\mathcal{C}_{\mathcal{K}\alpha}}{\mathcal{C}_{\alpha\alpha}} \mathcal{K}_\vartheta. \quad (6.12)$$

The first term is the change in the preferred AI share at a fixed information quantity. The second is the further adjustment in composition as total information changes. The quantity response accounts for this adjustment as well as the equilibrium change in the marginal value of information.¹³

We first consider the equilibrium response to a change in AI information cost c_A . A cost

¹³Let $B(\mathcal{K}) = H'(\mathcal{K})m(H(\mathcal{K}))$. We restrict attention to equilibrium branches on which $\mathcal{C}_{\alpha\alpha} > 0$ and the denominator below is positive. These are equilibrium regularity conditions in addition to individual optimality. The

reduction can make AI more attractive relative to private research and reduce the cost of acquiring additional information. The interaction between quantity and composition determines how much of the response is an expansion in information rather than a replacement of private research. Define:

$$\begin{aligned} U_c &= \mathcal{C}_{\alpha\alpha}\mathcal{C}_{\mathcal{K}c_A} - \mathcal{C}_{\mathcal{K}\alpha}\mathcal{C}_{\alpha c_A}, \\ V_c &= [\mathcal{C}_{\mathcal{K}\mathcal{K}} - B'(\mathcal{K})]\mathcal{C}_{\alpha c_A} - \mathcal{C}_{\mathcal{K}\alpha}\mathcal{C}_{\mathcal{K}c_A}. \end{aligned}$$

Then (6.12) implies that $\alpha_{c_A} < 0$, or a marginal decrease in AI information cost increases the share of AI information, if and only if $V_c > 0$. Given $V_c > 0$, a cost reduction also expands total information near the no-AI boundary if:

$$\liminf_{h_A \downarrow 0} \frac{h_{c_A}}{\alpha_{c_A}} = \liminf_{h_A \downarrow 0} \frac{H'(\mathcal{K})U_c}{V_c} > 0, \quad (6.13)$$

where the limit follows this cost path.¹⁴ Under these two conditions, the increase in total information does not vanish relative to the increase in the AI share. At fixed τ_b , the marginal common-error contribution is proportional to α and vanishes at the no-AI boundary. The positive lower bound in (6.13) ensures that the reduction in noise-trader contamination remains first order relative to the change in composition. It therefore dominates sufficiently close to that boundary, and a marginal cost reduction raises TFP.

We next consider higher AI accuracy. The following lemma links the fraction of available precision acquired and the expansion of total information to the behavior of marginal costs near the boundary:

Lemma 4 (Information acquisition at low AI accuracy) *Fix c_A, c_P and consider an interior equilibrium branch with $h \rightarrow h_0$ as $\tau_b \downarrow 0$. Suppose $\mathcal{C}_{\alpha\alpha} > 0$ on the feasible composition interval for total information near h_0 . Assume the following conditions:*

(i) *For each $q \in (0, 1)$, the following limit exists:*

$$F(q) = \lim_{\substack{\tau_b \downarrow 0 \\ H(\mathcal{K}) \rightarrow h_0}} \mathcal{C}_\alpha \left(\mathcal{K}, \frac{q\tau_b}{H(\mathcal{K})}; c_A, c_P, \tau_b \right) \quad (6.14)$$

There is a $q_0 \in (0, 1]$ such that $F(q) < 0$ for $q < q_0$ and $F(q) > 0$ for $q > q_0$.

response of aggregate information acquisition intensity is:

$$\mathcal{K}_\theta = -\frac{\mathcal{C}_{\mathcal{K}\theta} - \mathcal{C}_{\mathcal{K}\alpha}\mathcal{C}_{\alpha\theta}/\mathcal{C}_{\alpha\alpha}}{\mathcal{C}_{\mathcal{K}\mathcal{K}} - B'(\mathcal{K}) - \mathcal{C}_{\mathcal{K}\alpha}^2/\mathcal{C}_{\alpha\alpha}}.$$

¹⁴The response of $\mathcal{K}$ and α are explicitly given in Appendix C.2.

(ii) In effective-precision coordinates, C_{h_P} is continuous at $h_A = 0$ for each $\tau_b > 0$, and:

$$\lim_{\substack{\tau_b \downarrow 0, h_P \rightarrow h_0 \\ 0 < h_A < \tau_b}} C_{h_P}(h_A, h_P; c_A, c_P, \tau_b) = \gamma_0 \in \mathbb{R}. \quad (6.15)$$

Then:

$$\frac{h_A}{\tau_b} \rightarrow q_0, \quad h = h_0 + L_0 \tau_b + o(\tau_b), \quad L_0 = q_0 \frac{G''(h_0) - \gamma_0}{G''(h_0) - m'(h_0)}. \quad (6.16)$$

Proof. See Appendix C.3. ■

Condition (i) in Lemma 4 compares feasible allocations that use a fixed fraction $q \in (0, 1)$ of available AI precision as accuracy vanishes. Its sign conditions imply that equilibrium utilization converges to q_0 . They also cover $q_0 = 1$: when $F(q) < 0$ for every $q < 1$, investors approach the precision frontier. This recovers (6.7) under separable costs. Condition (ii) measures how AI information acquisition changes the marginal cost of private research. A positive γ_0 implies that higher AI information acquisition increases the marginal cost of private research, while a negative γ_0 implies that AI information acquisition complements private research. The finite joint limit rules out an arbitrarily large change in private marginal cost per unit of AI precision as the AI frontier shrinks. It does not require the interaction to vanish. Total information rises at first order with accuracy if and only if $\gamma_0 < G''(h_0)$: the increase in private marginal cost induced by AI must be smaller than the curvature of private research cost.

These acquisition responses determine the marginal TFP effects:

Proposition 7 *Under the maintained equilibrium regularity conditions, the local TFP effects are:*

(i) Fix $\tau_b > 0$. On an interior cost path approaching $(h, \alpha) = (h_0, 0)$ and satisfying $V_c > 0$ and (6.13), a marginal reduction in c_A raises TFP sufficiently close to the boundary:

$$\frac{d \log \mathcal{A}}{dc_A} < 0. \quad (6.17)$$

(ii) Under the conditions of Lemma 4, the right derivative of TFP at zero accuracy is:

$$\left. \frac{d \log \mathcal{A}}{d\tau_b} \right|_{0+} = \Gamma_0 \left\{ [-s'(h_0)] L_0 - \frac{q_0^2}{h_0^2} \right\}, \quad (6.18)$$

Proof. See Appendix C.4. ■

Condition (6.13), Lemma 4, and Proposition 7 characterize the marginal effect of AI technology growth on the aggregate TFP index. Under general cost functions, the marginal response of information acquisition depends on how acquisition from one source affects the marginal cost

of the other. Under the stated conditions, a marginal cost reduction raises TFP sufficiently close to the no-AI boundary, while a marginal accuracy increase can reduce allocation efficiency at the adoption margin.

Relation to separable costs. For separable costs, $C_{h_A h_P} = 0$ and hence $L_0 = \lambda_0 q_0$. The relative marginal-cost limit in (6.14), divided by h_0 , is $c_A f'_+(0)/(1-q)^2 - m_0$. Under $c_A f'_+(0) < m_0$, it gives $q_0 = 1 - \sqrt{c_A f'_+(0)/m_0}$. When $f'_+(0) = 0$, this limit is negative at every $q < 1$, so $q_0 = 1$ and $L_0 = \lambda_0$. Equation (6.18) therefore recovers the boundary results in Proposition 6.

These extensions above identify the acquisition incentives behind the local productivity effects of AI progress. Marginal acquisition costs determine how much of the available AI precision is obtained, while cost interactions between information sources govern the accompanying adjustment in private research. Under correlation neglect and the stated cost conditions, cheaper access at fixed accuracy raises TFP near the no-AI boundary because the expansion in total information reduces noise-trader contamination more than it increases common-error exposure. Improvements from very low accuracy can instead increase common-error exposure enough to outweigh the information gain. This comparison depends on how private research adjusts: rising marginal costs as AI acquisition expands strengthen displacement and weaken growth in total information, whereas cost complementarities support private research and strengthen that growth. The local comparison continues to depend on how investors' acquisition choices balance the reduction in noise-trader contamination against the common error transmitted through prices to firms' investment decisions.

7 Conclusion

This paper studies how common reliance on artificial intelligence affects the information that financial markets convey to firms and, through that information, the allocation of capital. The portfolio evidence shows that manager pairs with greater common alignment with AI-extractable information subsequently hold more similar portfolios. Motivated by this association, this paper then builds a model that connects investors' choices over the quantity and composition of information to price formation and firms' investment. Shared AI information helps investors learn about fundamentals, but also exposes their decisions to common model errors that survive aggregation and enter financial prices.

The analysis distinguishes two forms of AI progress: cheaper access to information and greater model accuracy. When investors underestimate the correlation of AI signals they obtain, a marginal cost reduction raises TFP when high acquisition costs initially keep AI use close to zero. Additional information strengthens informed trading, and the reduction in noise-

trader contamination dominates the increase in common-error exposure. Improvements from very low model accuracy can instead reduce TFP. Investors acquire more precise AI information, but their increased reliance on the shared signal can make prices less informative about fundamentals. Thus, economies with similarly limited AI use can have different marginal productivity effects, depending on whether costs or accuracy constrain acquisition. Numerical exercise illustrate how these local effects can extend to a hump-shaped relationship between cheaper access and TFP, and a J-shaped relationship between model accuracy and TFP.

These results distinguish the information an investor acquires from the information that markets aggregate. Improvements in individual signals need not translate into more informative prices when investors adjust their reliance on shared sources. Firms study from the market price and invest optimally, yet their allocation decisions deteriorate when those prices reveal less about demand. With production technology held fixed, the resulting productivity changes reflect actual capital allocation. Recognition of common errors changes both information acquisition and price formation, without a uniform effect on TFP. The general-cost extensions identify the acquisition incentives under which the local comparison between cheaper access and greater accuracy continues to hold. Evaluating the productivity gains from financial AI therefore requires attention to both individual signal quality and the information that prices convey.

Further empirical work could connect these mechanisms more directly to institutional AI use. Observations of the models investors employ, the overlap in their information, and their beliefs about shared errors would help assess the magnitude of the effects on prices and real investment. The J-shaped relationship in this paper compares equilibria at different levels of model accuracy. A dynamic extension could examine how this information channel interacts with capital adjustment and investment in complementary technologies. These questions extend the paper's central finding: the productivity consequences of AI depend on how investors use its information and how financial markets transmit that information to firms.

References

- Agarwal, Nikhil, Alex Moehring, Pranav Rajpurkar, and Tobias Salz, "Combining Human Expertise with Artificial Intelligence: Experimental Evidence from Radiology," 2026. Working paper, revised August 12, 2026.
- Angeletos, George-Marios and Zhen Huo, "Myopia and Anchoring," *American Economic Review*, April 2021, 111 (4), 1166–1200.
- Antón, Miguel and Christopher Polk, "Connected Stocks," *Journal of Finance*, 2014, 69 (3), 1099–1127.
- Bank of England and Financial Conduct Authority, "Artificial Intelligence in UK Financial Services: 2024," Technical Report, Bank of England and Financial Conduct Authority November 2024.
- Benhabib, Jess, Xuewen Liu, and Pengfei Wang, "Endogenous information acquisition and countercyclical uncertainty," *Journal of Economic Theory*, 2016, 165, 601–642.
- , —, and —, "Financial Markets, the Real Economy, and Self-Fulfilling Uncertainties," *Journal of Finance*, 2019, 74 (3), 1503–1557.
- Bond, Philip, Alex Edmans, and Itay Goldstein, "The Real Effects of Financial Markets," *Annual Review of Financial Economics*, 2012, 4, 339–360.
- Bonelli, Maxime, "Data-Driven Investors," *The Review of Financial Studies*, 07 2026, 39 (7), 1909–1969.
- Bresnahan, Timothy F. and Manuel Trajtenberg, "General Purpose Technologies: 'Engines of Growth'?", *Journal of Econometrics*, 1995, 65 (1), 83–108.
- Brynjolfsson, Erik, Daniel Rock, and Chad Syverson, "Artificial Intelligence and the Modern Productivity Paradox: A Clash of Expectations and Statistics," in Ajay Agrawal, Joshua Gans, and Avi Goldfarb, eds., *The Economics of Artificial Intelligence: An Agenda*, University of Chicago Press, 2019, pp. 23–57.
- , —, and —, "The Productivity J-Curve: How Intangibles Complement General Purpose Technologies," *American Economic Journal: Macroeconomics*, 2021, 13 (1), 333–372.
- Cameron, A. Colin, Jonah B. Gelbach, and Douglas L. Miller, "Robust Inference with Multiway Clustering," *Journal of Business and Economic Statistics*, 2011, 29 (2), 238–249.
- Cao, Sean, Wei Jiang, Junbo Wang, and Baozhong Yang, "From Man vs. Machine to Man + Machine: The Art and AI of Stock Analyses," *Journal of Financial Economics*, 2024, 160, 103910.
- Chen, Qi, Itay Goldstein, and Wei Jiang, "Price Informativeness and Investment Sensitivity to Stock Price," *The Review of Financial Studies*, 2007, 20 (3), 619–650.
- Cici, Gjergji, Stefan Jaspersen, and Alexander Kempf, "Speed of Information Diffusion within Fund Families," *Review of Asset Pricing Studies*, 2017, 7 (1), 144–170.
- Coleman, Braiden, Kenneth Merkley, and Joseph Pacelli, "Human Versus Machine: A Com-

- parison of Robo-Analyst and Traditional Research Analyst Investment Recommendations,” *The Accounting Review*, 2022, 97 (5), 221–244.
- Colombo, Luca, Gianluca Femminis, and Alessandro Pavan**, “Information Acquisition and Welfare,” *Review of Economic Studies*, 2014, 81 (4), 1438–1483.
- Cong, Lin William, Ke Tang, and Jingyuan Wang**, “AlphaPortfolio: Goal-Oriented Investment Management Through Deep Reinforcement Learning,” Working Paper 35195, National Bureau of Economic Research May 2026.
- D’Acunto, Francesco, Nagpurnanand Prabhala, and Alberto G. Rossi**, “The Promises and Pitfalls of Robo-Advising,” *Review of Financial Studies*, 2019, 32 (5), 1983–2020.
- David, Joel M., Hugo A. Hopenhayn, and Venky Venkateswaran**, “Information, Misallocation, and Aggregate Productivity,” *Quarterly Journal of Economics*, 2016, 131 (2), 943–1005.
- David, Paul A.**, “The Dynamo and the Computer: An Historical Perspective on the Modern Productivity Paradox,” *American Economic Review*, 1990, 80 (2), 355–361.
- DeMarzo, Peter M., Dimitri Vayanos, and Jeffrey Zwiebel**, “Persuasion Bias, Social Influence, and Unidimensional Opinions,” *The Quarterly Journal of Economics*, 2003, 118 (3), 909–968.
- Enke, Benjamin and Florian Zimmermann**, “Correlation Neglect in Belief Formation,” *The Review of Economic Studies*, 2019, 86 (1), 313–332.
- Eyster, Erik and Matthew Rabin**, “Naive Herding in Rich-Information Settings,” *American Economic Journal: Microeconomics*, 2010, 2 (4), 221–243.
- Farboodi, Maryam and Laura Veldkamp**, “Long-Run Growth of Financial Data Technology,” *American Economic Review*, 2020, 110 (8), 2485–2523.
- Fedyk, Anastassia and James Hodson**, “When Can the Market Identify Old News?,” *Journal of Financial Economics*, 2023, 149 (1), 92–113.
- Financial Stability Board**, “The Financial Stability Implications of Artificial Intelligence,” Technical Report, Financial Stability Board November 2024.
- Girardi, Giulio, Kathleen W. Hanley, Stanislava Nikolova, Loriana Pelizzon, and Mila Getmansky Sherman**, “Portfolio Similarity and Asset Liquidation in the Insurance Industry,” *Journal of Financial Economics*, 2021, 142 (1), 69–96.
- Goldstein, Itay and Liyan Yang**, “Information Diversity and Complementarities in Trading and Information Acquisition,” *Journal of Finance*, 2015, 70 (4), 1723–1765.
- , **Hao Jiang, and David T. Ng**, “Investor Flows and Fragility in Corporate Bond Funds,” *Journal of Financial Economics*, 2017, 126 (3), 592–613.
- Gonzalez, Tanja Artiga, Teodor Dyakov, Justus Inhoffen, and Evert Wipplinger**, “Crowding of International Mutual Funds,” *Journal of Banking and Finance*, 2024, 164, 107202.
- Gopinath, Gita, Şebnem Kalemli-Özcan, Loukas Karabarbounis, and Carolina Villegas-Sanchez**, “Capital allocation and productivity in South Europe,” *The Quarterly Journal of Economics*, 2017, 132 (4), 1915–1967.

- Greenwood, Jeremy and Mehmet Yorukoglu**, “1974,” *Carnegie-Rochester Conference Series on Public Policy*, 1997, 46, 49–95.
- Greenwood, Robin, Augustin Landier, and David Thesmar**, “Vulnerable Banks,” *Journal of Financial Economics*, 2015, 115 (3), 471–485.
- Grossman, Sanford J. and Joseph E. Stiglitz**, “On the Impossibility of Informationally Efficient Markets,” *American Economic Review*, 1980, 70 (3), 393–408.
- Gu, Shihao, Bryan Kelly, and Dacheng Xiu**, “Empirical Asset Pricing via Machine Learning,” *The Review of Financial Studies*, 05 2020, 33 (5), 2223–2273.
- Guo, Hongye and Jessica A. Wachter**, “Correlation Neglect in Asset Prices,” Working Paper 35753, National Bureau of Economic Research September 2026.
- Hellwig, Christian and Laura Veldkamp**, “Knowing What Others Know: Coordination Motives in Information Acquisition,” *Review of Economic Studies*, 2009, 76 (1), 223–251.
- Hoberg, Gerard, Nitin Kumar, and Nagpurnanand Prabhala**, “Mutual Fund Competition, Managerial Skill, and Alpha Persistence,” *Review of Financial Studies*, 2018, 31 (5), 1896–1929.
- Kacperczyk, Marcin and Amit Seru**, “Fund manager use of public information: New evidence on managerial skills,” *The Journal of Finance*, 2007, 62 (2), 485–528.
- Koch, Andrew, Stefan Ruenzi, and Laura Starks**, “Commonality in Liquidity: A Demand-Side Explanation,” *Review of Financial Studies*, 2016, 29 (8), 1943–1974.
- Lakonishok, Josef, Andrei Shleifer, and Robert W. Vishny**, “The Impact of Institutional Trading on Stock Prices,” *Journal of Financial Economics*, 1992, 32 (1), 23–43.
- Lopez-Lira, Alejandro and Yuehua Tang**, “Can ChatGPT forecast stock price movements? Return predictability and large language models,” *Journal of Financial Economics*, 2026, 184, 104335.
- Manconi, Alberto, Massimo Massa, and Ayako Yasuda**, “The Role of Institutional Investors in Propagating the Crisis of 2007–2008,” *Journal of Financial Economics*, 2012, 104 (3), 491–518.
- Myatt, David P. and Chris Wallace**, “Endogenous Information Acquisition in Coordination Games,” *Review of Economic Studies*, 2012, 79 (1), 340–374.
- Nieuwerburgh, Stijn Van and Laura Veldkamp**, “Information Acquisition and Under-Diversification,” *Review of Economic Studies*, 2010, 77 (2), 779–805.
- Pavan, Alessandro**, “Attention, coordination, and bounded recall,” *Journal of Economic Theory*, 2025, 227, 106013.
- Peress, Joel and Daniel Schmidt**, “Noise traders incarnate: Describing a realistic noise trading process,” *Journal of Financial Markets*, 2021, 54, 100618.
- Pool, Veronika K., Noah Stoffman, and Scott E. Yonker**, “The People in Your Neighborhood: Social Interactions and Mutual Fund Portfolios,” *Journal of Finance*, 2015, 70 (6), 2679–2732.
- Rossi, Alberto G and Stephen Utkus**, “The diversification and welfare effects of robo-advising,” *Journal of Financial Economics*, 2024, 157, 103869.

- Schaal, Edouard and Mathieu Taschereau-Dumouchel**, “Coordinating business cycles,” *Journal of Monetary Economics*, 2025, 155, 103829. Informational Frictions in Macroeconomics (In honor of Robert E. Lucas, Jr.).
- Sheng, Jinfei, Zheng Sun, Baozhong Yang, and Alan L. Zhang**, “Generative AI and Asset Management,” *Review of Financial Studies*, 2026, p. hhag050.
- Sias, Richard W.**, “Institutional Herding,” *Review of Financial Studies*, 2004, 17 (1), 165–206.
- Solow, Robert M.**, “We’d Better Watch Out,” *The New York Times Book Review*, July 1987, p. 36. July 12.
- Veldkamp, Laura L.**, “Information Markets and the Comovement of Asset Prices,” *Review of Economic Studies*, 2006, 73 (3), 823–845.
- Verrecchia, Robert E.**, “Information Acquisition in a Noisy Rational Expectations Economy,” *Econometrica*, 1982, 50 (6), 1415–1430.
- Wagner, Wolf**, “Systemic Liquidation Risk and the Diversity-Diversification Trade-Off,” *Journal of Finance*, 2011, 66 (4), 1141–1175.
- Wahal, Sunil and Albert Yan Wang**, “Competition among Mutual Funds,” *Journal of Financial Economics*, 2011, 99 (1), 40–59.
- Wermers, Russ**, “Mutual Fund Herding and the Impact on Stock Prices,” *Journal of Finance*, 1999, 54 (2), 581–622.

A A Fixed-Price Portfolio Benchmark

This appendix isolates the portfolio implications of the information mechanism at given prices. Investors first acquire information and then choose budget-constrained portfolios. The exercise illustrates how common information can produce similar desired holdings; it does not impose financial-market clearing or provide an additional step needed to solve the macroeconomic model. We characterize portfolio choice, information acquisition, and aggregate desired positions, and then provide proofs and extensions.

A.1 Environment and Information Acquisition

The baseline model follows the portfolio-information-acquisition tradition in treating initial asset prices and the stochastic payoff process as exogenous (Van Nieuwerburgh and Veldkamp, 2010). This setting isolates how the quantity and composition of information affect portfolios. Prices convey no additional information in this benchmark.

Timing and assets. The economy has three dates, $t = 0, 1, 2$. There are $N \geq 2$ risky assets indexed by $j \in \{1, \dots, N\}$. Asset j has a stochastic date-2 payoff:

$$f_j \sim \mathcal{N}(\mu_j, \sigma_j^2), \quad (\text{A.1})$$

and payoffs are independent across assets. There is a continuum of investors indexed by $i \in [0, 1]$. Each investor has one unit of wealth at date 1 and takes the vector of strictly positive asset prices $\mathbf{p} = (p_1, \dots, p_N)'$ and the gross risk-free payoff r as given.

At date 0, investor i chooses how much information to acquire about each asset and how to divide this information between private research and an AI source. At date 1, the signals are observed, and the investor chooses a portfolio. Asset payoffs are realized at date 2.

Prior beliefs and signals. Investors share a common prior that coincides with the unconditional payoff distribution:

$$f_j \mid \mathcal{I}_{i0} \sim \mathcal{N}(\mu_j, (\lambda_j^0)^{-1}), \quad \lambda_j^0 = \sigma_j^{-2}. \quad (\text{A.2})$$

Investor i can acquire a private signal s_{ij}^P and an AI signal s_{ij}^A about asset j :

$$s_{ij}^P = f_j + u_{ij} + \epsilon_{ij}^P, \quad (\text{A.3})$$

$$s_{ij}^A = f_j + b_j + \epsilon_{ij}^A. \quad (\text{A.4})$$

The private component $u_{ij} \sim \mathcal{N}(0, \tau_u^{-1})$ is specific to investor i , whereas the AI model error $b_j \sim \mathcal{N}(0, \tau_b^{-1})$ is shared by all investors who use the AI signal. The remaining errors satisfy:

$$\epsilon_{ij}^P \sim \mathcal{N}\left(0, (x_{ij}^P)^{-1}\right), \quad \epsilon_{ij}^A \sim \mathcal{N}\left(0, (x_{ij}^A)^{-1}\right).$$

Fundamentals and signal errors are mutually independent across assets. The investor-specific errors are also independent across investors. Investors pay to increase x_{ij}^P and x_{ij}^A . The precisions τ_u and τ_b describe the irreducible errors in the two information technologies.

We define the effective precisions contributed by the private and AI signals as:

$$h_{ij}^P = \frac{\tau_u x_{ij}^P}{\tau_u + x_{ij}^P}, \quad h_{ij}^A = \frac{\tau_b x_{ij}^A}{\tau_b + x_{ij}^A}, \quad (\text{A.5})$$

and we can summarize the quantity and composition of information by:

$$h_{ij} = h_{ij}^P + h_{ij}^A, \quad \alpha_{ij} = \frac{h_{ij}^A}{h_{ij}}. \quad (\text{A.6})$$

Effective AI precision is $a_{ij} = \alpha_{ij} h_{ij}$. The feasible set satisfies:

$$h_{ij} > 0, \quad 0 < \alpha_{ij} < 1, \quad \alpha_{ij} h_{ij} < \tau_b, \quad (1 - \alpha_{ij}) h_{ij} < \tau_u. \quad (\text{A.7})$$

The first two inequalities describe an interior information structure. The last two follow from the upper bounds on effective precision generated by the irreducible signal errors.

Bayesian updating from $\{s_{ij}^P, s_{ij}^A\}$ gives the payoff posterior mean $\hat{\mu}_{ij}$ and precision $\hat{\lambda}_{ij}$:

$$\hat{\mu}_{ij} = \frac{\lambda_j^0 \mu_j + (1 - \alpha_{ij}) h_{ij} s_{ij}^P + \alpha_{ij} h_{ij} s_{ij}^A}{\lambda_j^0 + h_{ij}}, \quad \hat{\lambda}_{ij} = \lambda_j^0 + h_{ij}. \quad (\text{A.8})$$

For a single investor, h_{ij} determines the total information acquired. The composition variable α_{ij} determines how much of that information is exposed to the common AI model error.

The payoff posterior already accounts for uncertainty about b_j . An investor can also infer the common error from the two signals, but this inference provides no additional observation beyond those signals. Here the posterior distribution of f_j is sufficient for portfolio choice because payoffs depend only on fundamentals. Section 3 allows managers to learn from prices, so the covariance between their own AI signal and the price also enters their inference. Initial prices in the present section are fixed and supply no information.

Information costs. Raising the raw precisions $\{x_{ij}^A, x_{ij}^P\}$ reduces investor-specific signal noise and requires costly research. Write this cost in effective-information coordinates as

$$C(h_{ij}, \alpha_{ij}; c_A, c_P, \tau_b, \tau_u).$$

We assume that the cost is increasing in total effective precision and locally convex in the AI share. A useful specification, employed in the comparative-static analysis, is

$$C(h, \alpha; c_A, c_P, \tau_b, \tau_u) = c_A f\left(\frac{\tau_b \alpha h}{\tau_b - \alpha h}\right) + g\left(\frac{\tau_u (1 - \alpha) h}{\tau_u - (1 - \alpha) h}, c_P\right), \quad (\text{A.9})$$

where f and $g(\cdot, c_P)$ are nonnegative, strictly increasing, and convex functions, and $x^A = \frac{\tau_b \alpha h}{\tau_b - \alpha h}$ and $x^P = \frac{\tau_u (1 - \alpha) h}{\tau_u - (1 - \alpha) h}$ denote the raw signal precisions required to produce the desired effective precisions (Colombo et al., 2014). A decline in c_A lowers the price of AI information within a given technological frontier. An increase in τ_b improves the common-error precision and expands the upper bound on effective AI information.

Sequential optimization. After observing the signals, investor i chooses asset holdings $q_i = (q_{i1}, \dots, q_{iN})'$ to maximize mean-variance utility:

$$U_i = \mathbb{E}_i(W_i) - \frac{\rho}{2} \mathbb{V}_i(W_i),$$

where W_i denotes wealth at date $t = 2$, $\rho > 0$ measures risk aversion, and the expectations are taken conditional on the information set at date $t = 1$, after the signals are revealed. Conditional on posterior beliefs, the date-1 portfolio-choice problem can be written as:

$$\max_{q_i} \sum_{j=1}^N q_{ij} (\hat{\mu}_{ij} - r p_j) - \frac{\rho}{2} \sum_{j=1}^N q_{ij}^2 \hat{\lambda}_{ij}^{-1} \quad \text{subject to} \quad \sum_{j=1}^N p_j q_{ij} = 1. \quad (\text{A.10})$$

We abstract from short-sale constraints to keep the portfolio rule linear in posterior beliefs and permit a transparent characterization of portfolio similarity (Van Nieuwerburgh and Veldkamp, 2010). At date 0, the investor chooses $\{h_{ij}, \alpha_{ij}\}_{j=1}^N$ to maximize the expected value of the utility, $\mathbb{E}_0[U_i]$, net of information costs. We solve the portfolio problem first and then return to information acquisition.

A.2 Optimal Portfolios and Pairwise Similarity

For a given information structure, the portfolio problem (A.10) has the following solution.

Lemma A.1 (Optimal portfolio structure) *Suppose that asset payoffs are conditionally independent, posterior precisions are strictly positive, and short selling is unrestricted. Conditional on posterior beliefs, investor i 's optimal holding of asset j is*

$$q_{ij}^* = \frac{\hat{\lambda}_{ij} p_j}{B_i} + \frac{\hat{\lambda}_{ij}}{\rho} \left(m_{ij} - p_j \frac{A_i}{B_i} \right), \quad (\text{A.11})$$

where

$$m_{ij} \equiv \hat{\mu}_{ij} - r p_j, \quad A_i \equiv \sum_{\ell=1}^N \hat{\lambda}_{i\ell} p_{\ell} m_{i\ell}, \quad B_i \equiv \sum_{\ell=1}^N \hat{\lambda}_{i\ell} p_{\ell}^2.$$

Proof. See Appendix A.5. ■

The first term in (A.11) exhausts the budget using positions based on posterior precision. It assigns more weight to assets with less residual uncertainty, independently of signal realizations. The second term is an active position: the investor increases holdings when an asset's expected excess payoff rises relative to the portfolio-wide benchmark. Higher precision strengthens this response, while greater risk aversion dampens it.

To characterize the portfolio similarity between different asset managers, we first define the ex ante average portfolio of investor i as:

$$\bar{q}_i = \mathbb{E}_0[q_i^*], \quad (\text{A.12})$$

and the centered inner-product similarity between two distinct investors i and k can be expressed as:

$$\Omega_{ik} = (q_i^* - \bar{q}_i)'(q_k^* - \bar{q}_k). \quad (\text{A.13})$$

This measure removes average portfolio differences generated by asset prices, prior expected payoffs, and posterior precisions. It isolates the extent to which common signal realizations move the two portfolios in the same direction. Taking expectations at date $t = 0$ gives:

$$\mathbb{E}_0[(q_i^*)'(q_k^*)] = \bar{q}_i' \bar{q}_k + \mathbb{E}_0[\Omega_{ik}].$$

The following proposition characterizes the expected portfolio similarity given the information structure $\{(h_{ij}, \alpha_{ij})\}$:

Proposition A.1 (Common AI exposure and portfolio similarity) *Suppose that asset fundamentals and common AI model errors are independent across assets. Investor-specific signal errors are independent across investors. For two distinct investors i and k , given the information structure $\{(h_{ij}, \alpha_{ij})\}$,*

the expected portfolio similarity is given by:

$$E_0[\Omega_{ik}] = \frac{1}{\rho^2} \sum_{\ell=1}^N \omega_{ik,\ell} \left[h_{i\ell} h_{k\ell} \sigma_\ell^2 + \alpha_{i\ell} h_{i\ell} \alpha_{k\ell} h_{k\ell} \tau_b^{-1} \right], \quad (\text{A.14})$$

where $\omega_{ik,\ell} \geq 0$ depends on asset prices, posterior precisions, and the budget constraint. Its expression is given in Appendix A.6. Conditional on total information quantities and posterior precisions, expected portfolio similarity weakly increases with investors' joint exposure to each asset's common AI signal.

Proof. See Appendix A.6. ■

Proposition A.1 identifies two economically distinct sources of portfolio convergence. First, investors' active positions covary when their signals reveal the same underlying payoff. This component reflects common learning about fundamentals and is present even when research errors are investor-specific. Second, AI information introduces an additional source of comovement. A model error shared across users enters both investors' posterior beliefs and moves their portfolios in the same direction, although the error is unrelated to the asset's true payoff. Conditional on total effective information, the strength of this AI-specific component is proportional to the product of the investors' effective AI exposures, $\alpha_{ij} h_{ij} \alpha_{kj} h_{kj}$, and to the variance of the common model error, τ_b^{-1} . The weights $\omega_{ik,j}$ capture how asset prices, posterior precisions, and the budget constraint transmit a signal about asset j across the rest of each portfolio. Since these weights are nonnegative under the maintained independence assumptions, asset heterogeneity affects the size of the contribution from joint AI exposure.

Model robustness. Appendix A.8 studies two departures from the benchmark. With short-sale constraints, the portfolio rule is piecewise linear. For locally fixed active sets, common AI exposure continues to align portfolio responses, as shown in Appendix A.8.1. Appendix A.8.2 introduces multiple AI providers. Provider-specific errors diversify as the number of models grows, while an error component correlated across providers remains in the aggregate signal.

The symmetric benchmark. The general results above allow for heterogeneous assets and investors. We now impose symmetry across assets to obtain a clearer portfolio decomposition. The same specialization will also allow us to aggregate holdings across investors.

Assume

$$p_j = 1, \quad \mu_j = \mu, \quad \sigma_j^2 = \sigma^2, \quad \lambda_j^0 = \sigma^{-2} \quad \forall j. \quad (\text{A.15})$$

We focus on information choices that are symmetric across assets for each investor,

$$h_{ij} = h_i, \quad \alpha_{ij} = \alpha_i, \quad \hat{\lambda}_i = \sigma^{-2} + h_i. \quad (\text{A.16})$$

Let bars denote cross-asset averages for a given realization, for example:

$$\bar{f} = \frac{1}{N} \sum_{j=1}^N f_j, \quad \bar{b} = \frac{1}{N} \sum_{j=1}^N b_j, \quad \bar{u}_i = \frac{1}{N} \sum_{j=1}^N u_{ij}.$$

Applying Lemma A.1 gives:

$$q_{ij}^* = \frac{1}{N} + \frac{\hat{\lambda}_i}{\rho} (\hat{\mu}_{ij} - \bar{\mu}_i) = \frac{1}{N} + \frac{(1 - \alpha_i)h_i}{\rho} (s_{ij}^P - \bar{s}_i^P) + \frac{\alpha_i h_i}{\rho} (s_{ij}^A - \bar{s}_i^A). \quad (\text{A.17})$$

This is a decision rule in observed signals. Substituting the signal equations gives a decomposition in realized fundamentals and errors:

$$q_{ij}^* = \frac{1}{N} + \frac{h_i}{\rho} \left[(f_j - \bar{f}) + \alpha_i (b_j - \bar{b}) + \xi_{ij} \right], \quad (\text{A.18})$$

where

$$\xi_{ij} = (1 - \alpha_i)(u_{ij} - \bar{u}_i) + \alpha_i(\epsilon_{ij}^A - \bar{\epsilon}_i^A) + (1 - \alpha_i)(\epsilon_{ij}^P - \bar{\epsilon}_i^P). \quad (\text{A.19})$$

Equation (A.18) describes the sources of realized portfolio variation. The fundamental component, $(f_j - \bar{f})$, directs investment toward assets with higher realized payoffs. The AI component, $\alpha_i(b_j - \bar{b})$, loads on a model error shared across investors who acquire the AI signal, and greater reliance on AI information strengthens this channel. The final component contains investor-specific information and therefore creates dispersion across individual portfolios. A higher information precision h_i increases the response of portfolio choice to information, whereas greater risk aversion ρ dampens it.

In the symmetric-asset environment, the centered similarity measure reduces to

$$\Omega_{ik} = \sum_{j=1}^N \left(q_{ij}^* - \frac{1}{N} \right) \left(q_{kj}^* - \frac{1}{N} \right).$$

Proposition A.1 then implies:

$$\mathbb{E}_0[\Omega_{ik}] = \frac{N-1}{\rho^2} h_i h_k \left(\sigma^2 + \alpha_i \alpha_k \tau_b^{-1} \right). \quad (\text{A.20})$$

The AI-specific term is proportional to $(\alpha_i h_i)(\alpha_k h_k)$. The product structure motivates measuring joint alignment with AI-extractable information at the pair level.

A.3 Ex Ante Information Choice

We now return to date 0, before the signals are observed. Let $\mathcal{G}_i(\mathbf{h}_i)$ denote the expected gross value of the optimized portfolio. Bayesian posterior means satisfy

$$\mathbb{E}_0[\hat{\mu}_{ij}] = \mu_j, \quad \mathbb{V}_0(\hat{\mu}_{ij}) = \frac{1}{\lambda_j^0} - \frac{1}{\hat{\lambda}_{ij}}. \quad (\text{A.21})$$

The distribution of the investor's payoff posterior depends on total effective precision and is independent of the AI share at a fixed h_{ij} . Thus $\mathcal{G}_i$ depends on $\mathbf{h}_i$ alone. The investor's information problem is

$$\begin{aligned} \max_{\{h_{ij}, \alpha_{ij}\}_{j=1}^N} \quad & \mathcal{G}_i(\mathbf{h}_i) - \sum_{j=1}^N C(h_{ij}, \alpha_{ij}; c_A, c_P, \tau_b, \tau_u) \\ \text{subject to} \quad & \alpha_{ij} h_{ij} < \tau_b, \quad (1 - \alpha_{ij}) h_{ij} < \tau_u \quad \text{for every } j. \end{aligned} \quad (\text{A.22})$$

At an interior optimum,

$$C_\alpha(h_{ij}, \alpha_{ij}) = 0, \quad (\text{A.23})$$

$$C_h(h_{ij}, \alpha_{ij}) = \frac{\partial \mathcal{G}_i}{\partial h_{ij}}. \quad (\text{A.24})$$

Total information balances the value of a more precise posterior against its resource cost. Composition minimizes the cost of supplying that precision. The different cross-investor correlation of the two sources affects portfolio similarity, but has no separate private payoff consequence in this benchmark. Appendix A.7 gives the gross value and its derivatives.

An interior competitive information equilibrium in this baseline environment is a collection

$$\left\{ h_{ij}^*, \alpha_{ij}^*, q_{ij}^* \right\}_{i \in [0,1], j=1, \dots, N}$$

such that beliefs satisfy (A.8), portfolios satisfy Lemma A.1, and information choices maximize (A.22), with interior conditions (A.23) and (A.24). Prices remain exogenous. Section 3 endogenizes prices through financial-market clearing.

The symmetric information conditions also simplify. Differentiating the gross value with respect to a common increase in precision across assets, we can show that the interior information choice now satisfies:

$$C_h(h_i, \alpha_i) = \frac{N-1}{2N\rho} \sigma^2 + \frac{\rho}{2N^2(\sigma^{-2} + h_i)^2}, \quad (\text{A.25})$$

$$C_\alpha(h_i, \alpha_i) = 0. \quad (\text{A.26})$$

The first term on the right-hand side of (A.25) is the value of learning which assets have high realized payoffs, and the second reflects the value of reducing the conditional variance of a budget-constrained portfolio.

A.4 Aggregate Holdings and the Macroeconomic Connection

Consider aggregate holdings. Integrating (A.18) across investors and applying the law of large numbers to the investor-specific terms gives

$$Q_j \equiv \int_0^1 q_{ij}^* di = \frac{1}{N} + \frac{1}{\rho} \left[\left(\int_0^1 h_i di \right) (f_j - \bar{f}) + \left(\int_0^1 \alpha_i h_i di \right) (b_j - \bar{b}) \right]. \quad (\text{A.27})$$

Thus, investor-specific errors disappear in the aggregate, whereas fundamentals and the common AI model error survive aggregation.

Under an ex ante symmetric information-acquisition equilibrium,

$$h_i = h, \quad \alpha_i = \alpha \quad \forall i. \quad (\text{A.28})$$

Individual portfolios remain heterogeneous after signals are observed because ξ_{ij} is investor-specific. Aggregate holdings become:

$$Q_j = \frac{1}{N} + \frac{h}{\rho} \left[(f_j - \bar{f}) + \alpha(b_j - \bar{b}) \right]. \quad (\text{A.29})$$

Since $\sum_j Q_j = 1$, expected aggregate portfolio concentration is

$$\mathbb{E}_0 \left[\sum_{j=1}^N Q_j^2 \right] = \frac{1}{N} + \frac{N-1}{\rho^2} h^2 \left(\sigma^2 + \alpha^2 \tau_b^{-1} \right). \quad (\text{A.30})$$

Equations (A.20) and (A.30) describe the same common-information mechanism at two different levels of aggregation. Expected pairwise similarity, $\mathbb{E}_0[\Omega_{ik}]$, measures the covariance of active positions for a given investor pair. Expected aggregate concentration, $\mathbb{E}_0[\sum_{j=1}^N Q_j^2]$, measures how strongly common portfolio responses remain in aggregate holdings. Investor-specific research errors wash out from both objects in expectation, whereas variation in fundamentals and the common AI model error remain. At the pair level, the AI-specific component depends on the product of the two investors' effective AI exposures, $(\alpha_i h_i)(\alpha_k h_k)$. In the symmetric aggregate, it depends on the square of effective AI exposure, $(\alpha h)^2$. Pairwise similarity provides the model counterpart to the empirical evidence in Section 2. Aggregate concentration records the common demand response at given prices. Section 3 studies how this response enters prices when the financial market clears.

Connection to the motivating evidence. The model links common information to common portfolio adjustments. The shared-error term provides one source of the pairwise similarity documented in Section 2; common learning about fundamentals provides another. The incremental R^2 measure captures alignment with AI-extractable information, but does not directly estimate $\alpha_i h_i$, identify b_j , or separate these two sources of comovement. Likewise, a stronger model-generated score is not itself an estimate of signal precision. The evidence motivates the information channel. Sections 4 and 5 study the price and allocation consequences of the same information technology.

At fixed asset prices, common information appears in the quantities that investors demand. With a fixed net supply of financial claims, market clearing instead requires prices to adjust. The macroeconomic model uses this second setting and allows firms to learn from the resulting prices. The fixed supply of financial claims does not fix real investment, which firms choose after observing prices, and the common reliance on AI information could affect real investment through firms' learning and generate potential capital misallocation.

A.5 Proof of Lemma A.1

The first-order condition for the holding of asset j is

$$\hat{\mu}_{ij} - r p_j - \rho q_{ij} \hat{\lambda}_{ij}^{-1} - \zeta_i p_j = 0, \quad (\text{A.31})$$

which gives

$$q_{ij} = \frac{\hat{\lambda}_{ij}}{\rho} (m_{ij} - \zeta_i p_j). \quad (\text{A.32})$$

Substituting (A.32) into the budget constraint yields

$$1 = \frac{1}{\rho} \sum_{\ell=1}^N \hat{\lambda}_{i\ell} p_{\ell} (m_{i\ell} - \zeta_i p_{\ell}) = \frac{A_i - \zeta_i B_i}{\rho}.$$

Therefore,

$$\zeta_i = \frac{A_i - \rho}{B_i}. \quad (\text{A.33})$$

Substituting (A.33) into (A.32) gives

$$\begin{aligned} q_{ij}^* &= \frac{\hat{\lambda}_{ij}}{\rho} \left(m_{ij} - p_j \frac{A_i - \rho}{B_i} \right) \\ &= \frac{\hat{\lambda}_{ij} p_j}{B_i} + \frac{\hat{\lambda}_{ij}}{\rho} \left(m_{ij} - p_j \frac{A_i}{B_i} \right), \end{aligned}$$

which proves (A.11). The objective function is strictly concave because $\rho > 0$ and $\hat{\lambda}_{ij} > 0$ for every asset. The solution is therefore unique.

A.6 Proof of Proposition A.1

From Lemma A.1, the random component of the holding of asset j is

$$q_{ij}^* - \bar{q}_{ij} = \frac{\hat{\lambda}_{ij}}{\rho} \left[\hat{\mu}_{ij} - E_0(\hat{\mu}_{ij}) - \frac{p_j}{B_i} \sum_{\ell=1}^N \hat{\lambda}_{i\ell} p_{\ell} (\hat{\mu}_{i\ell} - E_0(\hat{\mu}_{i\ell})) \right]. \quad (\text{A.34})$$

Bayesian updating gives

$$\hat{\mu}_{i\ell} - E_0(\hat{\mu}_{i\ell}) = \frac{h_{i\ell}(f_{\ell} - \mu_{\ell}) + \alpha_{i\ell} h_{i\ell} b_{\ell} + e_{i\ell}}{\hat{\lambda}_{i\ell}}, \quad (\text{A.35})$$

where $e_{i\ell}$ collects investor-specific signal errors. Substituting (A.35) into (A.34) yields

$$q_{ij}^* - \bar{q}_{ij} = \frac{1}{\rho} \sum_{\ell=1}^N \chi_{ij\ell} [h_{i\ell}(f_{\ell} - \mu_{\ell}) + \alpha_{i\ell} h_{i\ell} b_{\ell} + e_{i\ell}], \quad (\text{A.36})$$

where

$$\chi_{ij\ell} = \begin{cases} 1 - \frac{\hat{\lambda}_{i\ell} p_{\ell}^2}{B_i}, & j = \ell, \\ -\frac{\hat{\lambda}_{ij} p_j p_{\ell}}{B_i}, & j \neq \ell. \end{cases} \quad (\text{A.37})$$

Substituting (A.36) into (A.13) gives

$$\begin{aligned} \mathbb{E}_0[\Omega_{ik}] &= \frac{1}{\rho^2} \sum_{j=1}^N \sum_{\ell=1}^N \sum_{n=1}^N \chi_{ij\ell} \chi_{kjn} \\ &\quad \times \mathbb{E}_0 [(h_{i\ell}(f_{\ell} - \mu_{\ell}) + \alpha_{i\ell} h_{i\ell} b_{\ell} + e_{i\ell}) \\ &\quad \times (h_{kn}(f_n - \mu_n) + \alpha_{kn} h_{kn} b_n + e_{kn})]. \end{aligned} \quad (\text{A.38})$$

Independence across assets implies that all terms with $\ell \neq n$ have zero expectation. Independence of investor-specific signal errors across investors implies that these errors do not contribute to the covariance between investors i and k . Therefore,

$$\mathbb{E}_0[\Omega_{ik}] = \frac{1}{\rho^2} \sum_{\ell=1}^N \left(\sum_{j=1}^N \chi_{ij\ell} \chi_{k\ell} \right) [h_{i\ell} h_{k\ell} \sigma_{\ell}^2 + \alpha_{i\ell} h_{i\ell} \alpha_{k\ell} h_{k\ell} \tau_b^{-1}]. \quad (\text{A.39})$$

Using (A.37), the portfolio-adjustment weight is:

$$\begin{aligned}\sum_{j=1}^N \chi_{ij\ell} \chi_{kj\ell} &= \left(1 - \frac{\hat{\lambda}_{i\ell} p_\ell^2}{B_i}\right) \left(1 - \frac{\hat{\lambda}_{k\ell} p_\ell^2}{B_k}\right) + \frac{p_\ell^2}{B_i B_k} \sum_{j \neq \ell} \hat{\lambda}_{ij} \hat{\lambda}_{kj} p_j^2. \\ \omega_{ik,\ell} &= \left(1 - \frac{\hat{\lambda}_{i\ell} p_\ell^2}{B_i}\right) \left(1 - \frac{\hat{\lambda}_{k\ell} p_\ell^2}{B_k}\right) + \frac{p_\ell^2}{B_i B_k} \sum_{j \neq \ell} \hat{\lambda}_{ij} \hat{\lambda}_{kj} p_j^2 \geq 0.\end{aligned}\quad (\text{A.40})$$

Each component of this expression is nonnegative because $B_i \geq \hat{\lambda}_{i\ell} p_\ell^2$ and $B_k \geq \hat{\lambda}_{k\ell} p_\ell^2$. Hence, $\omega_{ik,\ell} \geq 0$, and equations (A.14)–(A.40) follow.

A.7 Information Value in the Portfolio Benchmark

Define the prior expected excess return as:

$$\bar{m}_j = \mu_j - r p_j.$$

Conditional on posterior beliefs, substitution of the optimal portfolio into the objective gives the indirect utility:

$$U_i^*(\hat{\mu}_i, \hat{\Lambda}_i) = \frac{1}{2\rho} \sum_{j=1}^N \hat{\lambda}_{ij} (\hat{\mu}_{ij} - r p_j)^2 - \frac{1}{2\rho} \frac{\left[\sum_{j=1}^N \hat{\lambda}_{ij} p_j (\hat{\mu}_{ij} - r p_j) - \rho\right]^2}{B_i}. \quad (\text{A.41})$$

To see how the budget constraint enters this expression, write $m_{ij} = \hat{\mu}_{ij} - r p_j$ and substitute $q_{ij}^* = \hat{\lambda}_{ij} (m_{ij} - \zeta_i p_j) / \rho$ into the objective:

$$U_i^* = \frac{1}{\rho} \sum_j \hat{\lambda}_{ij} (m_{ij} - \zeta_i p_j) m_{ij} - \frac{1}{2\rho} \sum_j \hat{\lambda}_{ij} (m_{ij} - \zeta_i p_j)^2 = \frac{1}{2\rho} \left[\sum_j \hat{\lambda}_{ij} m_{ij}^2 - \zeta_i^2 B_i \right].$$

Using $\zeta_i = (A_i - \rho) / B_i$ gives (A.41). The second term imposes the budget constraint on the otherwise unconstrained portfolio value.

Using the posterior moments in (A.21) and independence across assets gives the ex ante value. Define:

$$S_i = \sum_{j=1}^N \hat{\lambda}_{ij} p_j \bar{m}_j - \rho, \quad T_i = \sum_{j=1}^N p_j^2 \left(\frac{\hat{\lambda}_{ij}^2}{\lambda_j^0} - \hat{\lambda}_{ij} \right). \quad (\text{A.42})$$

Chosen precisions, and hence B_i , are fixed before signals arrive. For the first term in (A.41),

$$\mathbb{E}_0 \left[\hat{\lambda}_{ij} (\hat{\mu}_{ij} - r p_j)^2 \right] = \hat{\lambda}_{ij} \left[\bar{m}_j^2 + \text{Var}_0(\hat{\mu}_{ij}) \right]$$

$$= \hat{\lambda}_{ij} \bar{m}_j^2 + \frac{\hat{\lambda}_{ij}}{\lambda_j^0} - 1.$$

For the budget term, the random variable $W_i = \sum_j \hat{\lambda}_{ij} p_j (\hat{\mu}_{ij} - r p_j) - \rho$ has mean S_i . Independence across assets gives

$$\text{Var}_0(W_i) = \sum_j \hat{\lambda}_{ij}^2 p_j^2 \left(\frac{1}{\lambda_j^0} - \frac{1}{\hat{\lambda}_{ij}} \right) = T_i.$$

Thus $\mathbb{E}_0[W_i^2] = S_i^2 + T_i$. Taking expectations in (A.41) yields the ex ante gross utility function:

$$\mathbb{E}_0[U_i^*] = \mathcal{G}_i(\mathbf{h}_i) = \frac{1}{2\rho} \left\{ \sum_{j=1}^N \left[\hat{\lambda}_{ij} \bar{m}_j^2 - 1 + \frac{\hat{\lambda}_{ij}}{\lambda_j^0} \right] - \frac{S_i^2 + T_i}{B_i} \right\}. \quad (\text{A.43})$$

For completeness, the first-order condition for total information can be written in closed form. Increasing h_{ij} changes only $\hat{\lambda}_{ij}$, with $\partial \hat{\lambda}_{ij} / \partial h_{ij} = 1$. The derivatives of the objects in the budget term are

$$\frac{\partial B_i}{\partial h_{ij}} = p_j^2, \quad \frac{\partial S_i}{\partial h_{ij}} = p_j \bar{m}_j, \quad \frac{\partial T_i}{\partial h_{ij}} = p_j^2 \left(\frac{2\hat{\lambda}_{ij}}{\lambda_j^0} - 1 \right).$$

Applying the quotient rule to $(S_i^2 + T_i)/B_i$ and differentiating the remaining terms of (A.43) gives:

$$C_h(h_{ij}, \alpha_{ij}) = \frac{1}{2\rho} \left[\bar{m}_j^2 + \frac{1}{\lambda_j^0} - \frac{R'_{ij} B_i - (S_i^2 + T_i) p_j^2}{B_i^2} \right], \quad (\text{A.44})$$

where

$$R'_{ij} = 2S_i p_j \bar{m}_j + p_j^2 \left(\frac{2\hat{\lambda}_{ij}}{\lambda_j^0} - 1 \right). \quad (\text{A.45})$$

The right-hand side of (A.24) is the marginal private value of increasing effective precision for asset j . The first two terms capture the direct value of a more precise posterior. The final fraction records how the budget constraint changes the value of information through the investor's remaining positions.

With symmetric assets and information, let $L_i = \sigma^{-2} + h_i$ and $\bar{m} = \mu - r$. Then

$$B_i = N L_i, \quad S_i = N L_i \bar{m} - \rho, \quad T_i = N(L_i^2 \sigma^2 - L_i).$$

In particular,

$$\frac{S_i^2 + T_i}{B_i} = N L_i \bar{m}^2 - 2\rho \bar{m} + \frac{\rho^2}{N L_i} + L_i \sigma^2 - 1.$$

Substitution into (A.43) cancels $NL_i\bar{m}^2$. Using $L_i\sigma^2 - 1 = h_i\sigma^2$ gives

$$\mathcal{G}_i(h_i) = \mu - r - \frac{\rho}{2N(\sigma^{-2} + h_i)} + \frac{N-1}{2\rho}h_i\sigma^2. \quad (\text{A.46})$$

Since the same precision is acquired for each of the N assets, total information cost is $NC(h_i, \alpha_i)$. The first-order conditions (A.25) and (A.26) are therefore

$$C_h(h_i, \alpha_i) = \frac{\mathcal{G}'_i(h_i)}{N} = \frac{\rho}{2N^2(\sigma^{-2} + h_i)^2} + \frac{N-1}{2\rho N}\sigma^2, \quad C_\alpha(h_i, \alpha_i) = 0.$$

A.8 Portfolio Extensions

This subsection considers two extensions of the baseline model. The first imposes short-selling constraints and shows that common AI exposure continues to align local portfolio responses. The second introduces multiple AI providers and identifies the residual cross-investor covariance in model errors that sustains the mechanism.

A.8.1 Short-Selling Constraint and Common AI Exposure

The baseline analysis allows unrestricted portfolio positions in order to obtain a globally linear portfolio rule. This appendix introduces the short selling constraint

$$q_{ij} \geq 0 \quad \text{for every investor } i \text{ and asset } j. \quad (\text{A.47})$$

The constraint makes portfolio choice piecewise linear because the set of assets held by an investor becomes endogenous. The common-information mechanism remains present. Within any region in which investors' active sets are unchanged, a common AI shock moves their portfolios in aligned directions, and the strength of this aligned response increases with their joint reliance on AI information. The constraint also creates an extensive-margin channel because a common signal can cause several investors to enter or exit the same asset at the same time.

Portfolio Choice with Short-Selling Constraint. Conditional on posterior beliefs, investor i solves

$$\begin{aligned} & \max_{\{q_{ij}\}_{j=1}^N} \sum_{j=1}^N q_{ij}m_{ij} - \frac{\rho}{2} \sum_{j=1}^N q_{ij}^2 \hat{\lambda}_{ij}^{-1} \\ & \text{subject to} \quad \sum_{j=1}^N p_j q_{ij} = 1, \quad q_{ij} \geq 0, \end{aligned} \quad (\text{A.48})$$

where

$$m_{ij} \equiv \hat{\mu}_{ij} - rp_j.$$

Prices and posterior precisions are strictly positive. The objective is strictly concave, and the feasible set is compact, so the optimal portfolio exists and is unique.

Lemma A.2 (Portfolio choice with short-selling constraints) *The unique solution to (A.48) is*

$$q_{ij}^* = \frac{\hat{\lambda}_{ij}}{\rho} [m_{ij} - \zeta_i p_j]_+, \quad [x]_+ \equiv \max\{x, 0\}, \quad (\text{A.49})$$

where the scalar ζ_i is uniquely determined by

$$\sum_{j=1}^N \frac{\hat{\lambda}_{ij} p_j}{\rho} [m_{ij} - \zeta_i p_j]_+ = 1. \quad (\text{A.50})$$

Define the active set

$$\mathcal{S}_i \equiv \{j : m_{ij} > \zeta_i p_j\}. \quad (\text{A.51})$$

For any candidate active set $\mathcal{S}_i$, let

$$A_i(\mathcal{S}_i) \equiv \sum_{j \in \mathcal{S}_i} \hat{\lambda}_{ij} p_j m_{ij}, \quad B_i(\mathcal{S}_i) \equiv \sum_{j \in \mathcal{S}_i} \hat{\lambda}_{ij} p_j^2. \quad (\text{A.52})$$

Then

$$\zeta_i = \frac{A_i(\mathcal{S}_i) - \rho}{B_i(\mathcal{S}_i)}, \quad (\text{A.53})$$

and the optimal holding can be written as

$$q_{ij}^* = \begin{cases} \frac{\hat{\lambda}_{ij} p_j}{B_i(\mathcal{S}_i)} + \frac{\hat{\lambda}_{ij}}{\rho} \left[m_{ij} - p_j \frac{A_i(\mathcal{S}_i)}{B_i(\mathcal{S}_i)} \right], & j \in \mathcal{S}_i, \\ 0, & j \notin \mathcal{S}_i. \end{cases} \quad (\text{A.54})$$

Proof. Let ζ_i be the multiplier on the budget constraint and let $v_{ij} \geq 0$ be the multiplier on $q_{ij} \geq 0$. The Lagrangian is

$$\mathcal{L}_i = \sum_{j=1}^N q_{ij} m_{ij} - \frac{\rho}{2} \sum_{j=1}^N q_{ij}^2 \hat{\lambda}_{ij}^{-1} + \zeta_i \left(1 - \sum_{j=1}^N p_j q_{ij} \right) + \sum_{j=1}^N v_{ij} q_{ij}.$$

The KKT conditions are

$$m_{ij} - \rho q_{ij} \hat{\lambda}_{ij}^{-1} - \zeta_i p_j + v_{ij} = 0, \quad (\text{A.55})$$

$$q_{ij} \geq 0, \quad v_{ij} \geq 0, \quad v_{ij} q_{ij} = 0. \quad (\text{A.56})$$

If $q_{ij} > 0$, complementary slackness implies $v_{ij} = 0$, and (A.55) gives

$$q_{ij} = \frac{\hat{\lambda}_{ij}}{\rho} (m_{ij} - \zeta_i p_j) > 0.$$

If $q_{ij} = 0$, then

$$v_{ij} = \zeta_i p_j - m_{ij} \geq 0,$$

which is equivalent to $m_{ij} \leq \zeta_i p_j$. Combining the two cases yields (A.49). Substitution into the budget constraint gives (A.50).

The left-hand side of (A.50) is continuous and weakly decreasing in ζ_i . It is strictly decreasing whenever at least one asset is active. It equals zero for sufficiently large ζ_i and diverges as $\zeta_i \rightarrow -\infty$. Hence, the budget equation has a unique solution.

For a fixed active set, the budget condition becomes

$$1 = \frac{1}{\rho} \sum_{j \in \mathcal{S}_i} \hat{\lambda}_{ij} p_j (m_{ij} - \zeta_i p_j) = \frac{A_i(\mathcal{S}_i) - \zeta_i B_i(\mathcal{S}_i)}{\rho}.$$

Solving for ζ_i gives (A.53). Substituting this expression into the interior rule for $j \in \mathcal{S}_i$ and rearranging gives (A.54). Strict concavity of the objective establishes uniqueness. ■

The long-only solution has the same benchmark-plus-active-tilt structure as the unconstrained portfolio rule. The benchmark and all portfolio-wide averages are now calculated over the endogenous active set. The positive-part operator makes the global solution nonlinear in posterior beliefs, while the rule remains linear within each active-set region.

The Symmetric Case. To connect the extension to the symmetric benchmark in the main text, suppose

$$p_j = 1, \quad \mu_j = \mu, \quad \lambda_j^0 = \sigma^{-2}, \quad h_{ij} = h_i, \quad \alpha_{ij} = \alpha_i \quad (\text{A.57})$$

for all assets j . The common posterior precision of investor i is

$$\hat{\lambda}_i = \sigma^{-2} + h_i.$$

Let $K_i \equiv |\mathcal{S}_i|$ and define the active-set average of a variable x_{ij} as

$$\bar{x}_{i, \mathcal{S}_i} \equiv \frac{1}{K_i} \sum_{j \in \mathcal{S}_i} x_{ij}.$$

Corollary A.1 (Symmetric long-only portfolio) Under (A.57), the cutoff and optimal holdings are

$$\zeta_i = \bar{m}_{i,\mathcal{S}_i} - \frac{\rho}{K_i \hat{\lambda}_i}, \quad (\text{A.58})$$

$$q_{ij}^* = \begin{cases} \frac{1}{K_i} + \frac{\hat{\lambda}_i}{\rho} (m_{ij} - \bar{m}_{i,\mathcal{S}_i}), & j \in \mathcal{S}_i, \\ 0, & j \notin \mathcal{S}_i. \end{cases} \quad (\text{A.59})$$

The active set satisfies

$$\begin{aligned} m_{ij} &> \bar{m}_{i,\mathcal{S}_i} - \frac{\rho}{K_i \hat{\lambda}_i}, & j \in \mathcal{S}_i, \\ m_{ij} &\leq \bar{m}_{i,\mathcal{S}_i} - \frac{\rho}{K_i \hat{\lambda}_i}, & j \notin \mathcal{S}_i. \end{aligned} \quad (\text{A.60})$$

Proof. Under symmetry,

$$A_i(\mathcal{S}_i) = \hat{\lambda}_i \sum_{j \in \mathcal{S}_i} m_{ij}, \quad B_i(\mathcal{S}_i) = K_i \hat{\lambda}_i.$$

Substitution into (A.53) gives (A.58). Substituting the cutoff into the positive holding rule yields

$$\begin{aligned} q_{ij}^* &= \frac{\hat{\lambda}_i}{\rho} \left(m_{ij} - \bar{m}_{i,\mathcal{S}_i} + \frac{\rho}{K_i \hat{\lambda}_i} \right) \\ &= \frac{1}{K_i} + \frac{\hat{\lambda}_i}{\rho} (m_{ij} - \bar{m}_{i,\mathcal{S}_i}) \end{aligned}$$

for $j \in \mathcal{S}_i$. The inequalities in (A.60) are the KKT conditions for active and inactive assets. ■

Define the effective AI exposure of investor i by

$$a_i \equiv \alpha_i h_i. \quad (\text{A.61})$$

The posterior mean satisfies

$$\hat{\mu}_{ij} = \frac{\sigma^{-2} \mu + h_i s_{ij}}{\hat{\lambda}_i},$$

where

$$s_{ij} = f_j + \alpha_i b_j + (1 - \alpha_i) u_{ij} + \alpha_i \epsilon_{ij}^A + (1 - \alpha_i) \epsilon_{ij}^P. \quad (\text{A.62})$$

The common prior mean and the risk-free payoff cancel when the posterior payoff is demeaned within the active set. Consequently, for $j \in \mathcal{S}_i$,

$$q_{ij}^* = \frac{1}{K_i} + \frac{h_i}{\rho} (f_j - \bar{f}_{\mathcal{S}_i}) + \frac{\alpha_i h_i}{\rho} (b_j - \bar{b}_{\mathcal{S}_i}) + \frac{h_i}{\rho} \zeta_{ij}^{\mathcal{S}_i}, \quad (\text{A.63})$$

where

$$\xi_{ij}^{\mathcal{S}_i} = (1 - \alpha_i) (u_{ij} - \bar{u}_{i,\mathcal{S}_i}) + \alpha_i (\epsilon_{ij}^A - \bar{\epsilon}_{i,\mathcal{S}_i}^A) + (1 - \alpha_i) (\epsilon_{ij}^P - \bar{\epsilon}_{i,\mathcal{S}_i}^P). \quad (\text{A.64})$$

Equation (A.63) displays the same three components as the unconstrained solution. Investors respond to asset fundamentals, to the common AI model error, and to investor-specific information. The active-set average replaces the full-market average.

Common AI Exposure and Portfolio Similarity. With a short-selling constraint, the active set varies with the realized shocks. Expected portfolio similarity therefore contains an extensive margin, through the distribution of active sets, and an intensive margin, through holdings conditional on those sets. Its unconditional expectation is:

$$\mathbb{E}_0[\Omega_{ik}] = \sum_{\mathcal{S}_i, \mathcal{S}_k} \Pr(\mathcal{S}_i, \mathcal{S}_k) \times \mathbb{E}_0[\Omega_{ik} \mid \mathcal{S}_i, \mathcal{S}_k], \quad (\text{A.65})$$

where $\Pr(\mathcal{S}_i, \mathcal{S}_k)$ captures the extensive margin and $\mathbb{E}_0[\Omega_{ik} \mid \mathcal{S}_i, \mathcal{S}_k]$ captures similarity conditional on a pair of active sets. For a common AI error b_ℓ , (A.63) implies

$$\frac{\partial q_{ij}^*}{\partial b_\ell} = \frac{\alpha_i h_i}{\rho} \mathbf{1}\{\ell \in \mathcal{S}_i\} \left[\mathbf{1}\{j = \ell\} - \frac{\mathbf{1}\{j \in \mathcal{S}_i\}}{K_i} \right]. \quad (\text{A.66})$$

A positive common signal about asset ℓ raises its portfolio weight and finances this increase through reductions in the other active positions.

Proposition A.2 (Common AI exposure with fixed active set) *Consider two distinct investors i and k at a state where their active sets $\mathcal{S}_i$ and $\mathcal{S}_k$ are locally stable. Let*

$$M_{ik} \equiv |\mathcal{S}_i \cap \mathcal{S}_k|.$$

For a common AI error b_ℓ , define the similarity of the two investors' local portfolio responses by

$$\mathcal{R}_{ik,\ell}^A \equiv \sum_{j=1}^N \frac{\partial q_{ij}^*}{\partial b_\ell} \frac{\partial q_{kj}^*}{\partial b_\ell}. \quad (\text{A.67})$$

If $\ell \notin \mathcal{S}_i \cap \mathcal{S}_k$, then $\mathcal{R}_{ik,\ell}^A = 0$. If $\ell \in \mathcal{S}_i \cap \mathcal{S}_k$, then

$$\mathcal{R}_{ik,\ell}^A = \frac{(\alpha_i h_i)(\alpha_k h_k)}{\rho^2} \frac{(K_i - 1)(K_k - 1) + (M_{ik} - 1)}{K_i K_k} \geq 0. \quad (\text{A.68})$$

The inequality is strict whenever both investors hold at least two assets.

Proof. If ℓ is inactive for either investor, (A.66) implies that one response vector is zero, so

$\mathcal{R}_{ik,\ell}^A = 0$. Suppose $\ell \in \mathcal{S}_i \cap \mathcal{S}_k$. Substituting (A.66) into (A.67) gives

$$\mathcal{R}_{ik,\ell}^A = \frac{(\alpha_i h_i)(\alpha_k h_k)}{\rho^2} \sum_{j=1}^N \left[\mathbf{1}\{j = \ell\} - \frac{\mathbf{1}\{j \in \mathcal{S}_i\}}{K_i} \right] \times \left[\mathbf{1}\{j = \ell\} - \frac{\mathbf{1}\{j \in \mathcal{S}_k\}}{K_k} \right].$$

Expanding the product and summing over assets yields

$$\mathcal{R}_{ik,\ell}^A = \frac{(\alpha_i h_i)(\alpha_k h_k)}{\rho^2} \left[1 - \frac{1}{K_i} - \frac{1}{K_k} + \frac{M_{ik}}{K_i K_k} \right].$$

The bracket can be rewritten as

$$\frac{K_i K_k - K_i - K_k + M_{ik}}{K_i K_k} = \frac{(K_i - 1)(K_k - 1) + (M_{ik} - 1)}{K_i K_k}.$$

Because $M_{ik} \geq 1$ whenever ℓ belongs to the intersection, the expression is nonnegative. It is strictly positive when $K_i \geq 2$ and $K_k \geq 2$. ■

Proposition A.2 shows that, conditional on locally fixed active sets, the alignment of portfolio responses increases with the product of effective AI exposures, $(\alpha_i h_i)(\alpha_k h_k)$. Changes in active sets add a second synchronization margin: a common AI shock can lead several investors to enter or exit the same asset. This extensive margin generally has no closed-form expression because the active sets are endogenous. The unconstrained baseline isolates the same common-error mechanism while preserving a globally linear portfolio rule.

A.8.2 Multiple AI Providers

The baseline uses one AI signal to isolate the common-information mechanism. Multiple providers can reduce this exposure when their model errors are sufficiently independent. Correlated errors leave a residual common component even when investors diversify across many models.

Suppose there are M AI service providers, indexed by $m \in \{1, \dots, M\}$. The signal that investor i obtains from provider m about asset j is:

$$s_{ij,m}^A = f_j + b_{j,m} + \epsilon_{ij,m}^A, \quad (\text{A.69})$$

where $b_{j,m}$ is provider m 's model error and $\epsilon_{ij,m}^A$ is an investor-specific error whose precision can be increased at a cost. Because model errors may be correlated across providers, write the signal vector as:

$$\mathbf{s}_{ij}^A = \mathbf{1}f_j + \mathbf{b}_j + \boldsymbol{\epsilon}_{ij}^A, \quad (\text{A.70})$$

where $\mathbf{b}_j \sim \mathcal{N}(0, \Sigma_b^{-1})$ and $\boldsymbol{\epsilon}_{ij}^A \sim \mathcal{N}(0, (\mathbf{X}_{ij}^A)^{-1})$. Define $\Omega_{ij} = \Sigma_b^{-1} + (\mathbf{X}_{ij}^A)^{-1}$. Bayesian updat-

ing then gives:

$$\hat{\lambda}_{ij} = \lambda_{ij}^0 + \mathbf{1}'_M \Omega_{ij}^{-1} \mathbf{1}_M + \frac{\tau_u x_{ij}^P}{\tau_u + x_{ij}^P}, \quad (\text{A.71})$$

$$\hat{\mu}_{ij} = \frac{\lambda_{ij}^0 \mu_{ij}^0 + \mathbf{1}'_M \Omega_{ij}^{-1} \mathbf{s}_{ij}^A + \frac{\tau_u x_{ij}^P}{\tau_u + x_{ij}^P} \mathbf{s}_{ij}^P}{\hat{\lambda}_{ij}}. \quad (\text{A.72})$$

Define:

$$\omega_{ij} = \frac{\Omega_{ij}^{-1} \mathbf{1}_M}{\mathbf{1}'_M \Omega_{ij}^{-1} \mathbf{1}_M}, \quad (\text{A.73})$$

This is investor i 's weight vector for asset j . Since $\omega'_{ij} \mathbf{1}_M = 1$, the provider signals can be aggregated into the sufficient signal:

$$\tilde{s}_{ij}^A = \omega'_{ij} \mathbf{s}_{ij}^A = f_j + \omega'_{ij} \mathbf{b}_j + \omega'_{ij} \epsilon_{ij}^A. \quad (\text{A.74})$$

The model-error component of this aggregate signal is investor specific because aggregation weights may differ:

$$\tilde{b}_{ij} = \omega'_{ij} \mathbf{b}_j. \quad (\text{A.75})$$

For two investors i and k , the relevant object for portfolio comovement is therefore the cross-investor covariance:

$$\text{Cov}(\tilde{b}_{ij}, \tilde{b}_{kj}) = \omega'_{ij} \Sigma_b^{-1} \omega_{kj}. \quad (\text{A.76})$$

Positive residual covariance preserves the common-error channel even when investors use several providers. For example, suppose $\mathbb{V}(b_{j,m}) = \tau_b^{-1}$ and $\text{Corr}(b_{j,m}, b_{j,n}) = \rho_b \in [0, 1]$ for every $m \neq n$. Under symmetric provider weights, $\omega_{ij} = M^{-1} \mathbf{1}_M$ for every investor, and $\Sigma_b^{-1} = \tau_b^{-1} [(1 - \rho_b) \mathbf{I}_M + \rho_b \mathbf{1}_M \mathbf{1}'_M]$. The variance of the common aggregate error is:

$$\mathbb{V}(\tilde{b}_{ij}) = \frac{\tau_b^{-1}}{M} + \frac{M-1}{M} \rho_b \tau_b^{-1}. \quad (\text{A.77})$$

Thus, as $M \rightarrow \infty$, $\mathbb{V}(\tilde{b}_{ij}) \rightarrow \rho_b \tau_b^{-1} > 0$. Diversification across providers eliminates independent model-specific noise but not the component shared across models, for example because providers use similar training data or architectures. Under common provider weights, the baseline analysis applies directly after replacing the single AI signal by $\tilde{s}_{ij}^A$. With heterogeneous weights, the exact formulas change, but the portfolio-similarity mechanism remains whenever $\omega'_{ij} \Sigma_b^{-1} \omega_{kj} > 0$.

B Macroeconomic Equilibrium and Comparative Statics

B.1 Proof of Lemma 1

Proof. Let $\mathbf{1} = (1, 1, 1)'$. Gaussian updating gives

$$V_{di}^\kappa = \left[\tau_d + \mathbf{1}'(R_i^\kappa)^{-1}\mathbf{1} \right]^{-1}$$

and

$$(\ell_{Pi}^\kappa, \ell_{Ai}^\kappa, \ell_{qi}^\kappa)' = V_{di}^\kappa (R_i^\kappa)^{-1} \mathbf{1}.$$

The private-signal block contributes h_{Pi} . The inverse of the lower 2×2 block is

$$\frac{1}{D_i^\kappa} \begin{pmatrix} v_q^\kappa & -c_{Aq}^\kappa \\ -c_{Aq}^\kappa & v_{Ai}^\kappa \end{pmatrix}.$$

Substitution gives (4.4)–(4.5). ■

B.2 Proof of Lemma 2

Proof. Let $\mu_i^\kappa \equiv \widehat{v}_i^\kappa - q$. The first-order condition is $\mu_i^\kappa - \rho m_i \Omega_i^\kappa = 0$, which gives (4.8). Substitution yields

$$m_i^* \mu_i^\kappa - \frac{\rho}{2} (m_i^*)^2 \Omega_i^\kappa = \frac{(\mu_i^\kappa)^2}{2\rho \Omega_i^\kappa}.$$

The square appears because favorable and unfavorable signals generate profitable long and short positions, respectively. ■

B.3 Proof of Proposition 1

Proof. Using the conjectured investment rule, the manager's conditional expected payoff is

$$\widehat{v}_i^\kappa = \delta y_N + \chi z + \mu_L + \chi \omega_s s + \delta \left(\ell_{Pi}^\kappa s_i^P + \ell_{Ai}^\kappa s_i^A + \ell_{qi}^\kappa \widetilde{q} \right) + g(k_0 + k_s s + k_q \widetilde{q}).$$

Subtracting $q = \beta_0 + \beta_s s + \beta_q \widetilde{q}$ gives

$$\begin{aligned} \widehat{v}_i^\kappa - q &= [\delta y_N + \chi z + \mu_L + g k_0 - \beta_0] \\ &\quad + [\chi \omega_s + g k_s - \beta_s] s + \delta \ell_{Pi}^\kappa s_i^P + \delta \ell_{Ai}^\kappa s_i^A \\ &\quad + [\delta \ell_{qi}^\kappa + g k_q - \beta_q] \widetilde{q}. \end{aligned}$$

The equilibrium price coefficients derived below satisfy

$$\beta_0 = \delta y_N + \chi z + \mu_L + gk_0, \quad \beta_s = \chi \omega_s + gk_s, \quad \beta_q = \delta \bar{\ell}_T^\kappa + gk_q.$$

Consequently,

$$\widehat{v}_i^\kappa - q = \gamma_{Pi} s_i^P + \gamma_{Ai} s_i^A + \gamma_{qi} \widetilde{q}.$$

Under the manager's subjective model, the expected excess gain satisfies:

$$\widehat{v}_i^\kappa - q = (\gamma_{Pi} + \gamma_{Ai} + \gamma_{qi})d + \sqrt{\kappa}(\gamma_{Ai} + \psi\gamma_{qi})b^c + \sqrt{1 - \kappa}\gamma_{Ai}b_i^I + \gamma_{Pi}v_i^P + \gamma_{Ai}\epsilon_i^A + \zeta\gamma_{qi}n.$$

Here $v_i^P = u_i + \epsilon_i^P$ has variance h_{Pi}^{-1} . The primitive shocks in this expression are subjectively independent and mean zero, so its variance is (4.10). Moreover, $\mathbb{E}_{0,i}^\kappa[\widehat{v}_i^\kappa - q] = 0$. Because Gaussian posterior variances depend on chosen precisions but not on signal realizations, Ω_i^κ is deterministic at date 0. Taking the date-0 expectation of (4.9) therefore gives (4.12). ■

B.4 Proof of Proposition 2

Proof. At a symmetric equilibrium, all managers have the same residual payoff variance Ω^κ . Integrating the holding rule and imposing market clearing gives

$$0 = \int_0^1 m_i di + n = \frac{\int_0^1 \widehat{v}_i^\kappa di - q}{\rho\Omega^\kappa} + n, \quad q = \int_0^1 \widehat{v}_i^\kappa di + \rho\Omega^\kappa n.$$

Aggregation uses the true signal process. Investor-specific errors wash out, so $\int_0^1 s_i^P di = d$ and $\int_0^1 s_i^A di = d + b$. Consequently,

$$\int_0^1 \widehat{d}_i^\kappa di = (\ell_P^\kappa + \ell_A^\kappa)d + \ell_A^\kappa b + \ell_q^\kappa \widetilde{q}.$$

Substitute this expression and the investment rule into the clearing price:

$$\begin{aligned} q &= \delta y_N + \chi z + \mu_L + gk_0 + (\chi \omega_s + gk_s)s \\ &\quad + \delta(\ell_P^\kappa + \ell_A^\kappa)d + \delta\ell_A^\kappa b + (\delta\ell_q^\kappa + gk_q)\widetilde{q} + \rho\Omega^\kappa n. \end{aligned}$$

Now use $\widetilde{q} = d + \psi b + \zeta n$ and compare with $q = \beta_0 + \beta_s s + \beta_q(d + \psi b + \zeta n)$. The three stochastic coefficients must satisfy

$$\begin{aligned} d : \quad & \beta_q = \delta(\ell_P^\kappa + \ell_A^\kappa + \ell_q^\kappa) + gk_q, \\ b : \quad & \beta_q \psi = \delta\ell_A^\kappa + (\delta\ell_q^\kappa + gk_q)\psi, \end{aligned}$$

$$n : \quad \beta_q \zeta = (\delta \ell_q^\kappa + g k_q) \zeta + \rho \Omega^\kappa.$$

The first equation gives (4.16). Subtracting ψ times the first equation from the second gives $\delta(\ell_P^\kappa + \ell_A^\kappa)\psi = \delta \ell_A^\kappa$. Subtracting ζ times the first equation from the third gives $\delta(\ell_P^\kappa + \ell_A^\kappa)\zeta = \rho \Omega^\kappa$. Thus

$$\psi = \frac{\ell_A^\kappa}{\ell_P^\kappa + \ell_A^\kappa}, \quad \zeta = \frac{\rho \Omega^\kappa}{\delta(\ell_P^\kappa + \ell_A^\kappa)}.$$

Matching the constant and public-signal terms gives $\beta_0 = \delta y_N + \chi z + \mu_L + g k_0$ and $\beta_s = \chi \omega_s + g k_s$. Finally, each manager chooses information while holding this aggregate price rule fixed. The two individual first-order conditions, evaluated at symmetric choices, supply the last two equations in (4.19). ■

B.5 Proof of Proposition 3

Proof. Treating aggregate output as known, as maintained in the large-economy equilibrium, expected profit conditional on $(s_j, \tilde{q}_j)$ is

$$Y_N^\delta Z^\chi M_j K_j^g - K_j.$$

Strict concavity follows from $0 < g < 1$. The first-order condition is

$$g Y_N^\delta Z^\chi M_j K_j^{g-1} = 1,$$

Rearranging gives $K_j^{1-g} = g Y_N^\delta Z^\chi M_j$, which proves (4.28). Taking logs and substituting (4.25) yields

$$\begin{aligned} k_j &= \Theta[\log g + \delta y_N + \chi z + \log M_j] \\ &= \Theta \left[\log g + \delta y_N + \chi z + \mu_L + \frac{1}{2} \Sigma^F \right] + \Theta \chi \omega_s s_j + \Theta \delta \omega_q^T \tilde{q}_j. \end{aligned}$$

The constant, s_j , and $\tilde{q}_j$ coefficients give (4.29)–(4.31). ■

B.6 Finite-Sample Allocation

For a finite cross section, define

$$K_N \equiv \frac{1}{N} \sum_{j=1}^N K_j, \quad D_N \equiv \frac{1}{N} \sum_{j=1}^N M_j^\Theta, \quad B_N \equiv \frac{1}{N} \sum_{j=1}^N X_j M_j^{\Theta-1}. \quad (\text{B.1})$$

Proposition B.1 (Finite-sample allocation index) *The investment rule implies*

$$K_j = K_N \frac{M_j^\Theta}{D_N}. \quad (\text{B.2})$$

Aggregate output admits the accounting representation

$$Y_N = Z \mathcal{A}_N^R K_N^\eta, \quad (\text{B.3})$$

where realized finite-sample allocative TFP is

$$\mathcal{A}_N^R = \frac{B_N^{1/\chi}}{D_N^\eta} = \frac{\left[N^{-1} \sum_{j=1}^N X_j M_j^{\Theta-1} \right]^{1/\chi}}{\left[N^{-1} \sum_{j=1}^N M_j^\Theta \right]^\eta}. \quad (\text{B.4})$$

Proof. Equation (4.28) has the form $K_j = C_N M_j^\Theta$. Averaging gives $K_N = C_N D_N$ and hence (B.2). Equations (3.2) and (3.4) imply

$$Y_N = Z \left[\frac{1}{N} \sum_{j=1}^N X_j K_j^g \right]^{1/\chi}.$$

Since $g^\Theta = \Theta - 1$,

$$K_j^g = K_N^g \frac{M_j^{\Theta-1}}{D_N^g}.$$

The common factor K_N^g / D_N^g can be taken outside the cross-sectional sum. Hence

$$\begin{aligned} Y_N &= Z \left[\frac{K_N^g}{D_N^g} \frac{1}{N} \sum_j X_j M_j^{\Theta-1} \right]^{1/\chi} \\ &= Z K_N^{g/\chi} \frac{B_N^{1/\chi}}{D_N^{g/\chi}} = Z K_N^\eta \frac{B_N^{1/\chi}}{D_N^\eta}, \end{aligned}$$

where the last equality uses $g/\chi = \eta$. This proves (B.3) and (B.4). ■

For finite N , Proposition B.1 is an exact accounting result conditional on the limiting atomistic investment rule. The equilibrium interpretation applies in the large- N limit, where aggregate output is known conditional on Z .

B.7 Proof of Proposition 4

Proof. The law of large numbers gives $D_N \rightarrow D \equiv \mathbb{E}[M^\Theta]$ and

$$B_N \rightarrow \mathbb{E}[XM^{\Theta-1}] = \mathbb{E}\left[\mathbb{E}[X \mid s, \tilde{q}]M^{\Theta-1}\right] = \mathbb{E}[M^\Theta] = D.$$

The iterated-expectations step uses the fact that $M^{\Theta-1}$ is measurable with respect to $(s, \tilde{q})$ and that $M = \mathbb{E}[X \mid s, \tilde{q}]$. Since $M > 0$, $D > 0$, and the continuous mapping theorem gives

$$\mathcal{A}_N^R \rightarrow \frac{D^{1/\chi}}{D^\eta} = D^{p_A},$$

where $p_A = \frac{1}{\chi} - \eta > 0$. This establishes (4.33). Next, let $\hat{L} \equiv \mathbb{E}[\log X \mid s, \tilde{q}]$. It is Gaussian with mean μ_L and variance $\Sigma_0 - \Sigma^F$, while

$$M = \exp(\hat{L} + \frac{1}{2}\Sigma^F).$$

Hence

$$\begin{aligned} \log \mathbb{E}[M^\Theta] &= \Theta\mu_L + \frac{\Theta}{2}\Sigma^F + \frac{\Theta^2}{2}(\Sigma_0 - \Sigma^F) \\ &= \Theta\mu_L + \frac{\Theta^2}{2}\Sigma_0 - \frac{\Theta(\Theta-1)}{2}\Sigma^F. \end{aligned}$$

Multiplying by p_A proves (4.35).

For the aggregate capital and output expressions in the main text, multiply the firm's first-order condition by K_j and take the cross-sectional limit:

$$K = gY^\delta Z^\chi \mathbb{E}[MK_j^\delta].$$

Investment is measurable with respect to the firm's information, so $\mathbb{E}[MK_j^\delta] = \mathbb{E}[XK_j^\delta]$. The final-good aggregator gives $\mathbb{E}[XK_j^\delta] = (Y/Z)^\chi$. Therefore,

$$K = gY^\delta Z^\chi (Y/Z)^\chi = gY,$$

where $\delta + \chi = 1$. Combining $K = gY$ with $Y = Z\mathcal{A}K^\eta$ gives $K^{1-\eta} = gZ\mathcal{A}$ and hence (4.36). ■

B.8 Proof of Lemma 3

Proof. We specialize the equilibrium system already obtained in (4.19). In terms of (h_i, α_i) , its information conditions are those in (4.13). Thus the system to simplify is

$$\begin{aligned}\psi &= \frac{\ell_A^\kappa}{\ell_P^\kappa + \ell_A^\kappa}, \\ \zeta &= \frac{\rho \Omega^\kappa}{\delta(\ell_P^\kappa + \ell_A^\kappa)}, \\ \left. \frac{\partial \mathcal{G}_i^\kappa}{\partial h_i} \right|_{(h_i, \alpha_i) = (h, \alpha)} &= \mathcal{C}_h^I(h, \alpha), \\ \left. \frac{\partial \mathcal{G}_i^\kappa}{\partial \alpha_i} \right|_{(h_i, \alpha_i) = (h, \alpha)} &= \mathcal{C}_\alpha^I(h, \alpha).\end{aligned}$$

The information derivatives hold the aggregate price rule and aggregate posterior loading fixed, as in the general equilibrium system.

Posterior beliefs. Substituting $\kappa = 0$ into (4.1) gives $c_{Aq}^0 = 0$ and $v_q^0 = \zeta^2 / \tau_n = 1/r$. The covariance matrix is therefore diagonal. The posterior formulas in Lemma 1 reduce to

$$\begin{aligned}V_{di}^0 &= \frac{1}{\tau_d + h_i + r}, \\ \hat{d}_i^0 &= V_{di}^0 [(1 - \alpha_i)h_i x_i^P + \alpha_i h_i x_i^A + r\tilde{q}], \\ \ell_{Pi}^0 &= V_{di}^0 (1 - \alpha_i)h_i, \quad \ell_{Ai}^0 = V_{di}^0 \alpha_i h_i, \quad \ell_{qi}^0 = V_{di}^0 r.\end{aligned}\tag{B.5}$$

The residual payoff variance in (4.7) becomes

$$\Omega_i^0 = \Omega_a + \delta^2 V_{di}^0.\tag{B.6}$$

Price equilibrium. At a symmetric information choice, $\ell_P^0 + \ell_A^0 = V_d^0 h$ and $\ell_A^0 = V_d^0 \alpha h$. Substituting these expressions into the first two equilibrium conditions gives

$$\psi = \alpha, \quad \zeta = \frac{\rho(\Omega_a + \delta^2 V_d^0)}{\delta V_d^0 h} = \frac{\rho[\Omega_a(\tau_d + h + r) + \delta^2]}{\delta h}.$$

Using $r = \tau_n / \zeta^2$ gives (5.4). Hence, for a given total precision, the price equations determine ψ from α and r from h .

Information quantity and composition. We now simplify the remaining two equilibrium conditions. Write $V_i = V_{di}^0$ and $\bar{\ell}_T = \bar{\ell}_T^0$ within this proof. In (4.10), setting $\kappa = 0$ combines the AI error terms using $1/\tau_b + 1/\pi_{Ai} = 1/(\alpha_i h_i)$. Together with the private signal error, they

contribute

$$\frac{\gamma_{Pi}^2}{(1 - \alpha_i)h_i} + \frac{\gamma_{Ai}^2}{\alpha_i h_i} = \delta^2 V_i^2 h_i.$$

Substituting the other posterior loadings into that same variance formula then yields

$$H_i^0 = \delta^2 \left\{ \frac{[V_i(h_i + r) - \bar{\ell}_T]^2}{\tau_d} + V_i^2 h_i + \frac{(V_i r - \bar{\ell}_T)^2}{r} \right\}. \quad (\text{B.7})$$

For a fixed aggregate price rule, both H_i^0 and Ω_i^0 depend on the manager's information choice only through h_i . Since $\mathcal{G}_i^0 = H_i^0 / (2\rho\Omega_i^0)$, it follows that $\partial \mathcal{G}_i^0 / \partial \alpha_i = 0$. The composition condition therefore reduces to

$$C_\alpha^I(h, \alpha) = 0.$$

This statement concerns the manager's subjective ex ante value. The realized posterior mean still depends on how the manager weights the two signals.

For the quantity condition, differentiate the individual objective with respect to h_i before imposing symmetry. Throughout this derivative, $(r, \bar{\ell}_T)$ remain fixed. At symmetry, write

$$V = \frac{1}{\tau_d + h + r}, \quad \bar{\ell}_T = V(h + r), \quad \Omega = \Omega_a + \delta^2 V.$$

Equation (B.7) then implies

$$H^0 = \delta^2 V^2 h \left(1 + \frac{h}{r} \right). \quad (\text{B.8})$$

Because $\partial V_i / \partial h_i = -V_i^2$, the derivatives of the three terms in braces in (B.7), evaluated at symmetry, give

$$\left. \frac{\partial H_i^0}{\partial h_i} \right|_{(h, \alpha)} = \delta^2 [0 + (V^2 - 2hV^3) + 2hV^3] = \delta^2 V^2, \quad \left. \frac{\partial \Omega_i^0}{\partial h_i} \right|_{(h, \alpha)} = -\delta^2 V^2.$$

The quotient rule therefore gives the marginal gross return to quantity:

$$\left. \frac{\partial \mathcal{G}_i^0}{\partial h_i} \right|_{(h, \alpha)} = \frac{\delta^2 V^2}{2\rho\Omega} \left(1 + \frac{H^0}{\Omega} \right). \quad (\text{B.9})$$

Only after taking this individual derivative do we substitute the equilibrium price precision $r(h)$. The resulting marginal return is $m(h)$, so the quantity condition is $C_h^I(h, \alpha) = m(h)$.

For completeness, under (3.13), the cost derivatives can be written using the marginal costs

of effective precision:

$$\text{MC}_P(h_P) \equiv c_P \frac{\tau_u^{\nu+1} h_P^{\nu-1}}{(\tau_u - h_P)^{\nu+1}}, \quad (\text{B.10})$$

$$\text{MC}_A(h_A; c_A, \tau_b) \equiv c_A \frac{\tau_b^{\nu+1} h_A^{\nu-1}}{(\tau_b - h_A)^{\nu+1}}. \quad (\text{B.11})$$

With $h_P = (1 - \alpha)h$ and $h_A = \alpha h$, the chain rule yields

$$\mathcal{C}_h^I = (1 - \alpha) \text{MC}_P + \alpha \text{MC}_A, \quad \mathcal{C}_\alpha^I = h(\text{MC}_A - \text{MC}_P).$$

Thus the conditions in (h, α) also imply $m(h) = \text{MC}_P = \text{MC}_A$. Since the two marginal costs are strictly increasing in their respective effective precisions, $\mathcal{C}_{\alpha\alpha}^I = h^2(\text{MC}'_P + \text{MC}'_A) > 0$. The interior composition condition therefore minimizes information cost at a given total precision.

Properties of the reduced system. For every $h > 0$, the right-hand side of (5.4) is positive at $r = 0$ and strictly decreasing in r , whereas the left-hand side increases from zero without bound. The positive solution exists and is unique. To establish the comparative statics, use the following abbreviations only in this proof:

$$T \equiv \tau_d + h + r(h), \quad D \equiv \Omega_a T + \delta^2. \quad (\text{B.12})$$

Implicit differentiation of the price equation gives

$$r'(h) = \frac{2r(D/h - \Omega_a)}{D + 2\Omega_a r} > 0, \quad (\text{B.13})$$

where $D/h - \Omega_a = [\Omega_a(\tau_d + r) + \delta^2]/h > 0$. Substituting $V = 1/T$, $\Omega = D/T$, and $r = \tau_n \delta^2 h^2 / (\rho^2 D^2)$ into (B.9) yields

$$m(h) = \frac{\delta^2}{2\rho} \left(\frac{1}{TD} + \frac{\delta^2 h}{T^2 D^2} \right) + \frac{\delta^2 \rho}{2\tau_n T^2} > 0. \quad (\text{B.14})$$

Finally, let $U = TD$. Since $T' = 1 + r' > 1$, $U' = T'(2\Omega_a T + \delta^2) > \delta^2$. Hence

$$\frac{d}{dh} \left(\frac{1}{U} + \frac{\delta^2 h}{U^2} \right) = \frac{\delta^2 - U'(1 + 2\delta^2 h/U)}{U^2} < 0.$$

The final term in (B.14) also decreases with h , which establishes $m'(h) < 0$. This last derivative compares marginal returns across symmetric price equilibria; the manager's first-order condition itself holds prices fixed. ■

B.9 Proof of Proposition 5

Proof. Throughout, $\kappa = 0$, $\nu > 1$, and all parameters other than the specified AI technology parameter remain fixed. We use the financial-information equilibrium reduction, under which $m(h) > 0$, $m'(h) < 0$, and $s'(h) < 0$. All derivatives below are equilibrium comparative statics. In particular, $m'(h)$ includes the price adjustment already incorporated in $m(h)$; the individual information first-order condition itself is derived holding the aggregate price rule fixed.

Information choices and the common local expansion. The first-order conditions for total precision and composition are

$$C_h^I(h, \alpha) = m(h), \quad C_\alpha^I(h, \alpha) = 0. \quad (\text{B.15})$$

With $h_P = (1 - \alpha)h$ and $h_A = \alpha h$, the chain rule gives

$$C_h^I = (1 - \alpha) \text{MC}_P(h_P) + \alpha \text{MC}_A(h_A), \quad C_\alpha^I = h [\text{MC}_A(h_A) - \text{MC}_P(h_P)],$$

where, suppressing technology parameters in the arguments,

$$\begin{aligned} \text{MC}_P(h_P) &= c_P h_P^{\nu-1} \left(1 - \frac{h_P}{\tau_u}\right)^{-(\nu+1)}, \\ \text{MC}_A(h_A) &= c_A h_A^{\nu-1} \left(1 - \frac{h_A}{\tau_b}\right)^{-(\nu+1)}. \end{aligned} \quad (\text{B.16})$$

Thus the nonlinear information system is equivalently

$$m(h) = \text{MC}_P(h - h_A), \quad m(h) = c_A h_A^{\nu-1} \left(1 - \frac{h_A}{\tau_b}\right)^{-(\nu+1)}. \quad (\text{B.17})$$

The interior domain is $0 < h - h_A < \tau_u$ and $0 < h_A < \tau_b$.

Recall that the no-AI equilibrium satisfies

$$m(h_0) = \text{MC}_P(h_0), \quad m_0 = m(h_0) > 0, \quad S_0 = s(h_0). \quad (\text{B.18})$$

Since $\text{MC}_P' > 0$ and $m' < 0$, the derivative of the first equation in (B.17) with respect to h is nonzero at $(h, h_A) = (h_0, 0)$. The implicit function theorem therefore determines h locally as a differentiable function of h_A . Differentiating that equation before taking any limit yields

$$m'(h) \frac{dh}{dh_A} = \text{MC}_P'(h_P) \left(\frac{dh}{dh_A} - 1 \right).$$

Consequently,

$$\frac{dh}{dh_A} = \frac{\text{MC}'_P(h_P)}{\text{MC}'_P(h_P) - m'(h)} \equiv \lambda(h_P, h) \in (0, 1), \quad \frac{dh_P}{dh_A} = \lambda - 1 < 0. \quad (\text{B.19})$$

We write λ for $\lambda(h_P, h)$ when its arguments are clear. Since $\alpha = h_A/h$, the exact composition response is

$$\frac{d\alpha}{dh_A} = \frac{1 - \alpha\lambda}{h}. \quad (\text{B.20})$$

As $h_A \rightarrow 0$, $\lambda \rightarrow \lambda_0$, where

$$\lambda_0 = \frac{\text{MC}'_P(h_0)}{\text{MC}'_P(h_0) - m'(h_0)} \in (0, 1). \quad (\text{B.21})$$

Taylor expansion now gives

$$\begin{aligned} h - h_0 &= \lambda_0 h_A + o(h_A), \\ h_P - h_0 &= -(1 - \lambda_0)h_A + o(h_A), \\ \alpha &= \frac{h_A}{h_0} + o(h_A), \quad m(h) = m_0 + m'(h_0)\lambda_0 h_A + o(h_A). \end{aligned} \quad (\text{B.22})$$

At $\kappa = 0$, the true price-error variance is $S_q^T = s(h) + \alpha^2/\tau_b$. Using $\alpha = h_A/h$ exactly,

$$S_q^T - S_0 = s'(h_0)\lambda_0 h_A + o(h_A) + \frac{h_A^2}{\tau_b h^2}.$$

Moreover,

$$\frac{h_A^2}{\tau_b h^2} - \frac{h_A^2}{\tau_b h_0^2} = \frac{h_A^2}{\tau_b} \left(\frac{1}{h^2} - \frac{1}{h_0^2} \right) = o(h_A).$$

The last equality follows because $h - h_0 = O(h_A)$ and $0 < h_A/\tau_b < 1$. This proves the common expansion (5.16) along either admissible path. The TFP formula also gives the exact differential

$$d \log \mathcal{A} = -\frac{p_A \Theta(\Theta - 1) \delta^2}{2(1 + \tau_d S_q^T)^2} dS_q^T, \quad \Gamma_0 = \frac{p_A \Theta(\Theta - 1) \delta^2}{2(1 + \tau_d S_0)^2} > 0. \quad (\text{B.23})$$

Part (i): fixed accuracy and high acquisition costs. Fix $\tau_b > 0$ and let $c_A \rightarrow \infty$. The private-information equation implies $h \geq h_0$: if $h < h_0$, then $m(h) > m_0$, whereas $\text{MC}_P(h - h_A) < \text{MC}_P(h_0) = m_0$. Hence $m(h) \leq m_0$. The AI equation in (B.17) then implies

$$c_A h_A^{v-1} \leq m(h) \leq m_0, \quad h_A \leq \left(\frac{m_0}{c_A} \right)^{1/(v-1)} \rightarrow 0.$$

The private-information equation therefore converges to the no-AI equation on the regular branch, so $h \rightarrow h_0$ and $m(h) \rightarrow m_0$. Since τ_b is fixed, $h_A/\tau_b \rightarrow 0$. The AI equation reduces to

$$c_A h_A^{\nu-1} = m_0[1 + o(1)],$$

and thus

$$h_A = \left(\frac{m_0}{c_A}\right)^{1/(\nu-1)} [1 + o(1)]. \quad (\text{B.24})$$

Substitution into (B.22) gives

$$\begin{aligned} h - h_0 &= \lambda_0 \left(\frac{m_0}{c_A}\right)^{1/(\nu-1)} + o\left(c_A^{-1/(\nu-1)}\right), \\ h_P - h_0 &= -(1 - \lambda_0) \left(\frac{m_0}{c_A}\right)^{1/(\nu-1)} + o\left(c_A^{-1/(\nu-1)}\right), \\ \alpha &= \frac{1}{h_0} \left(\frac{m_0}{c_A}\right)^{1/(\nu-1)} + o\left(c_A^{-1/(\nu-1)}\right). \end{aligned} \quad (\text{B.25})$$

In particular, $\alpha^2/\tau_b = O(h_A^2) = o(h_A)$, and

$$S_q^T - S_0 = s'(h_0) \lambda_0 \left(\frac{m_0}{c_A}\right)^{1/(\nu-1)} + o\left(c_A^{-1/(\nu-1)}\right). \quad (\text{B.26})$$

To establish the marginal effect, we differentiate the nonlinear system directly. This avoids imposing an unproved restriction on the derivative of the remainder in (B.26). The exact total differential of the logarithm of the AI equation is

$$\begin{aligned} \frac{m'(h)}{m(h)} dh &= \frac{dc_A}{c_A} + \left[\frac{\nu-1}{h_A} + \frac{\nu+1}{\tau_b - h_A} \right] dh_A \\ &\quad - \frac{(\nu+1)h_A}{\tau_b(\tau_b - h_A)} d\tau_b. \end{aligned} \quad (\text{B.27})$$

Combine this differential with $dh = \lambda dh_A$ from (B.19) and set $d\tau_b = 0$. We obtain

$$\frac{dh_A}{dc_A} = - \frac{1}{c_A \left[\frac{\nu-1}{h_A} + \frac{\nu+1}{\tau_b - h_A} - \frac{m'(h)}{m(h)} \lambda \right]} < 0. \quad (\text{B.28})$$

The bracket is strictly positive. Its leading term is $(\nu-1)/h_A$, because the remaining terms stay

bounded as $h_A \rightarrow 0$ at fixed τ_b . It follows that

$$\begin{aligned}\frac{dh_A}{dc_A} &= -\frac{h_A}{(\nu-1)c_A}[1+o(1)] \\ &= -\frac{m_0^{1/(\nu-1)}}{\nu-1}c_A^{-\nu/(\nu-1)}[1+o(1)].\end{aligned}\tag{B.29}$$

Using the exact quantity and composition responses, differentiation of S_q^T gives

$$\begin{aligned}\frac{dS_q^T}{dc_A} &= \left[s'(h)\lambda + \frac{2\alpha(1-\alpha\lambda)}{\tau_b h} \right] \frac{dh_A}{dc_A} \\ &= \frac{[-s'(h_0)]\lambda_0}{\nu-1} m_0^{1/(\nu-1)} c_A^{-\nu/(\nu-1)} + o\left(c_A^{-\nu/(\nu-1)}\right) > 0\end{aligned}\tag{B.30}$$

for sufficiently large c_A . Finally, (B.23) yields

$$\frac{d \log \mathcal{A}}{dc_A} = -\frac{\Gamma_0[-s'(h_0)]\lambda_0}{\nu-1} m_0^{1/(\nu-1)} c_A^{-\nu/(\nu-1)} + o\left(c_A^{-\nu/(\nu-1)}\right) < 0.\tag{B.31}$$

Thus a marginal decline in AI acquisition costs raises TFP near the high-cost boundary.

Part (ii): fixed acquisition costs and low model accuracy. Fix $c_A \in (0, \infty)$ and let $\tau_b \downarrow 0$. Since $0 < h_A < \tau_b$, we have $h_A \rightarrow 0$, and the private-information equation again implies $h \rightarrow h_0$ and $m(h) \rightarrow m_0$. The ratio h_A/τ_b need not be small. Rearranging the exact AI equation gives

$$\left(1 - \frac{h_A}{\tau_b}\right)^{\nu+1} = \frac{c_A}{m(h)} \tau_b^{\nu-1} \left(\frac{h_A}{\tau_b}\right)^{\nu-1}.\tag{B.32}$$

The right-hand side tends to zero because $\nu > 1$, $m(h) \rightarrow m_0 > 0$, and $0 < h_A/\tau_b < 1$. Hence $h_A/\tau_b \rightarrow 1$. Substituting this limit into (B.32) gives the rate at which effective precision approaches its upper bound:

$$1 - \frac{h_A}{\tau_b} = \left(\frac{c_A}{m_0}\right)^{1/(\nu+1)} \tau_b^{(\nu-1)/(\nu+1)} [1+o(1)].\tag{B.33}$$

Multiplying by τ_b yields

$$h_A = \tau_b - \left(\frac{c_A}{m_0}\right)^{1/(\nu+1)} \tau_b^{2\nu/(\nu+1)} [1+o(1)].\tag{B.34}$$

Because $1 < 2\nu/(\nu+1) < 2$, the correction is $o(\tau_b)$. The common expansion (B.22) therefore

implies

$$\begin{aligned} h - h_0 &= \lambda_0 \tau_b + o(\tau_b), \\ h_P - h_0 &= -(1 - \lambda_0) \tau_b + o(\tau_b), \\ \alpha &= \frac{\tau_b}{h_0} + o(\tau_b). \end{aligned} \tag{B.35}$$

The raw-precision choice explains why h_A/τ_b tends to one even though AI use vanishes. Since $h_A = \tau_b \pi_A / (\tau_b + \pi_A)$, its first-order condition is

$$m(h) \frac{\tau_b^2}{(\tau_b + \pi_A)^2} = c_A \pi_A^{\nu-1}. \tag{B.36}$$

If π_A/τ_b remained bounded along any sequence approaching zero accuracy, the left-hand side would stay bounded away from zero while the right-hand side would vanish. Thus $\pi_A/\tau_b \rightarrow \infty$. Using (B.33),

$$\pi_A = \frac{\tau_b h_A}{\tau_b - h_A} = \left(\frac{m_0}{c_A} \right)^{1/(\nu+1)} \tau_b^{2/(\nu+1)} [1 + o(1)] \rightarrow 0. \tag{B.37}$$

Raw acquisition precision decreases more slowly than model precision, so acquisition noise becomes small relative to the common model error. The convergence $h_A/\tau_b \rightarrow 1$ therefore concerns the utilization of the available effective precision, while $\alpha \rightarrow 0$. It relies on the vanishing marginal acquisition cost implied by $\nu > 1$, in addition to a fixed finite c_A .

To obtain the parameter derivatives, set $dc_A = 0$ in the exact differential (B.27) and use $dh = \lambda dh_A$. This gives

$$\frac{dh_A}{d\tau_b} = \frac{\frac{(\nu+1)h_A}{\tau_b(\tau_b - h_A)}}{\frac{\nu-1}{h_A} + \frac{\nu+1}{\tau_b - h_A} - \frac{m'(h)}{m(h)}\lambda} > 0. \tag{B.38}$$

Multiplying numerator and denominator by $(\tau_b - h_A)/(\nu+1)$,

$$\frac{dh_A}{d\tau_b} = \frac{h_A/\tau_b}{1 + \frac{\nu-1}{\nu+1} \frac{\tau_b - h_A}{h_A} - \frac{m'(h)}{m(h)} \frac{\lambda(\tau_b - h_A)}{\nu+1}} \rightarrow 1. \tag{B.39}$$

The numerator tends to one, the relative gap $(\tau_b - h_A)/h_A$ tends to zero, and $m'(h)/m(h)$ remains finite at h_0 . Equations (B.19) and (B.20) then give

$$\frac{dh}{d\tau_b} \rightarrow \lambda_0, \quad \frac{dh_P}{d\tau_b} \rightarrow -(1 - \lambda_0), \quad \frac{d\alpha}{d\tau_b} \rightarrow \frac{1}{h_0}. \tag{B.40}$$

These limits follow from the nonlinear system rather than differentiation of the asymptotic

remainder in (B.34).

The price-error derivative separates the quantity, exposure, and direct accuracy effects:

$$\frac{dS_q^T}{d\tau_b} = s'(h) \frac{dh}{d\tau_b} + \frac{2\alpha}{\tau_b} \frac{d\alpha}{d\tau_b} - \frac{\alpha^2}{\tau_b^2}. \quad (\text{B.41})$$

Since $\alpha/\tau_b \rightarrow 1/h_0$, the three terms converge respectively to

$$s'(h_0)\lambda_0, \quad \frac{2}{h_0^2}, \quad -\frac{1}{h_0^2}.$$

Consequently,

$$\lim_{\tau_b \downarrow 0} \frac{dS_q^T}{d\tau_b} = s'(h_0)\lambda_0 + \frac{1}{h_0^2}. \quad (\text{B.42})$$

The level expansion gives the same right derivative at the boundary:

$$\begin{aligned} S_q^T - S_0 &= s'(h_0)\lambda_0\tau_b + \frac{1}{\tau_b} \left(\frac{\tau_b}{h_0} + o(\tau_b) \right)^2 + o(\tau_b) \\ &= \left[s'(h_0)\lambda_0 + \frac{1}{h_0^2} \right] \tau_b + o(\tau_b). \end{aligned} \quad (\text{B.43})$$

Here the value at $\tau_b = 0$ denotes the continuous no-AI limit. Applying (B.23) establishes

$$\left. \frac{d \log \mathcal{A}}{d\tau_b} \right|_{0^+} = \lim_{\tau_b \downarrow 0} \frac{d \log \mathcal{A}}{d\tau_b} = \Gamma_0 \left[[-s'(h_0)]\lambda_0 - \frac{1}{h_0^2} \right]. \quad (\text{B.44})$$

The initial derivative is strictly negative if and only if $h_0^{-2} > [-s'(h_0)]\lambda_0$, and strictly positive under the reverse inequality. Continuity then gives the corresponding sign for sufficiently small positive τ_b . At equality, the first-order effect is zero and higher-order terms are required.

■

C General Information Costs

C.1 Proof of Proposition 6

Proof. All comparative statics are taken along the symmetric equilibrium branch. When obtaining an individual's information first-order condition, the aggregate price rule is held fixed. The functions $m(h)$ and $s(h)$ subsequently incorporate the equilibrium adjustment of that price rule. We keep the private-information cost and all parameters other than the specified AI pa-

parameter fixed.

Step 1. Express the general cost in effective-precision coordinates. The raw and effective AI precisions satisfy

$$h_A = \frac{\tau_b \pi_A}{\tau_b + \pi_A}, \quad \pi_A = \frac{\tau_b h_A}{\tau_b - h_A}, \quad 0 < h_A < \tau_b. \quad (\text{C.1})$$

Holding τ_b fixed gives

$$\frac{\partial h_A}{\partial \pi_A} = \left(\frac{\tau_b}{\tau_b + \pi_A} \right)^2, \quad \frac{\partial \pi_A}{\partial h_A} = \left(1 - \frac{h_A}{\tau_b} \right)^{-2}.$$

The marginal cost of effective AI precision is therefore

$$\text{MC}_A(h_A; c_A, \tau_b) = c_A f'(\pi_A) \left(1 - \frac{h_A}{\tau_b} \right)^{-2}. \quad (\text{C.2})$$

At $\kappa = 0$, information quantity and composition satisfy $\mathcal{C}_h^I = m(h)$ and $\mathcal{C}_\alpha^I = 0$ at an interior choice. Since $h_P = (1 - \alpha)h$ and $h_A = \alpha h$, the chain rule gives

$$\mathcal{C}_h^I = (1 - \alpha) \text{MC}_P + \alpha \text{MC}_A, \quad \mathcal{C}_\alpha^I = h(\text{MC}_A - \text{MC}_P).$$

The composition condition equates the two marginal costs; the quantity condition equates their common value to $m(h)$. Thus the nonlinear system is

$$m(h) = \text{MC}_P(h - h_A), \quad m(h) = \text{MC}_A(h_A; c_A, \tau_b). \quad (\text{C.3})$$

Equivalently, the AI condition in raw precision is

$$m(h) \left(\frac{\tau_b}{\tau_b + \pi_A} \right)^2 = c_A f'(\pi_A). \quad (\text{C.4})$$

Using (C.1) in this equality proves (6.6).

Step 2. Determine the response of private and total information to AI information. Recall the no-AI equilibrium and its marginal quantities:

$$m(h_0) = \text{MC}_P(h_0) = m_0 > 0, \quad S_0 = s(h_0),$$

$$\lambda_0 = \frac{\text{MC}'_P(h_0)}{\text{MC}'_P(h_0) - m'(h_0)} \in (0, 1), \quad \Gamma_0 = - \frac{\partial \log \mathcal{A}}{\partial S_q^T} \Big|_{S_q^T = S_0} > 0. \quad (\text{C.5})$$

The derivative with respect to h of the first equation in (C.3) is $m'(h) - \text{MC}'_P(h - h_A) < 0$. The implicit function theorem therefore determines h as a differentiable function of h_A near $(h_0, 0)$.

Differentiating that equation gives

$$m'(h) \frac{dh}{dh_A} = MC'_P(h_P) \left(\frac{dh}{dh_A} - 1 \right).$$

Solving for the derivative, and writing λ for this derivative within the proof, yields

$$\lambda \equiv \frac{dh}{dh_A} = \frac{MC'_P(h_P)}{MC'_P(h_P) - m'(h)} \in (0, 1), \quad \frac{dh_P}{dh_A} = \lambda - 1 < 0. \quad (\text{C.6})$$

Differentiating $\alpha = h_A/h$ then gives

$$\frac{d\alpha}{dh_A} = \frac{1 - \alpha\lambda}{h} > 0. \quad (\text{C.7})$$

Consequently, as $h_A \rightarrow 0$ along this branch,

$$\begin{aligned} h - h_0 &= \lambda_0 h_A + o(h_A), & h_P - h_0 &= -(1 - \lambda_0) h_A + o(h_A), \\ \alpha &= \frac{h_A}{h_0} + o(h_A), & m(h) &\rightarrow m_0, & \lambda &\rightarrow \lambda_0. \end{aligned} \quad (\text{C.8})$$

In particular, positive AI information raises h above h_0 on the branch, so $m(h) < m_0$.

Step 3. Locate the adoption boundary. At $\pi_A = 0$ and any $\tau_b > 0$, the marginal conversion from raw to effective precision is one. The marginal gross return to entering AI research is therefore m_0 , while the marginal cost is $c_A f'_+(0)$. It is also necessary for a positive solution of (C.3): convexity and (C.2) imply $MC_A \geq c_A f'_+(0)$, whereas Step 2 gives $m(h) < m_0$ when $h_A > 0$.

For later use, differentiate (C.2) at fixed c_A and τ_b :

$$\frac{\partial MC_A}{\partial h_A} = c_A f''(\pi_A) \left(1 - \frac{h_A}{\tau_b} \right)^{-4} + \frac{2c_A f'(\pi_A)}{\tau_b} \left(1 - \frac{h_A}{\tau_b} \right)^{-3} > 0. \quad (\text{C.9})$$

The first term is nonnegative and the second is strictly positive. Hence, along the private-information equation, $MC_A - m(h)$ is strictly increasing in h_A , with derivative $\partial MC_A / \partial h_A - m'(h)\lambda > 0$. Any root on this branch is unique.

First fix $\tau_b > 0$ and suppose $f'_+(0) = 0$. At an interior equilibrium, (C.2) implies

$$0 < f'(\pi_A) \leq \frac{m(h)}{c_A} \leq \frac{m_0}{c_A}.$$

As $c_A \rightarrow \infty$, the right-hand side vanishes. Since f' is nondecreasing and $f'(\pi) > 0$ for every $\pi > 0$, this forces $\pi_A \rightarrow 0$. It follows from (C.1) that $h_A \rightarrow 0$ and $h_A/\tau_b \rightarrow 0$. To verify that the branch exists near this boundary, fix any sufficiently small positive h_A in the domain of Step 2.

At that value, $MC_A - m(h) \rightarrow \infty$ as $c_A \rightarrow \infty$, while its value at $h_A = 0$ is $-m_0 < 0$. Continuity and strict monotonicity give a unique root between these two values for sufficiently large c_A .

Next suppose $f'_+(0) > 0$, keeping τ_b fixed. At an interior equilibrium,

$$m_0 \geq m(h) \geq c_A f'_+(0) \left(1 - \frac{h_A}{\tau_b}\right)^{-2}.$$

Rearranging gives the bound

$$0 < \frac{h_A}{\tau_b} \leq 1 - \sqrt{\frac{c_A f'_+(0)}{m_0}}. \quad (C.10)$$

The upper bound vanishes as $c_A \uparrow m_0 / f'_+(0)$, so $h_A \rightarrow 0$ and $h_A / \tau_b \rightarrow 0$ at this finite boundary. Existence just below the threshold again follows by continuity: $MC_A - m(h) < 0$ at zero, while at the threshold itself it is strictly positive at any sufficiently small positive h_A . Its derivative is strictly positive, so the nearby root is unique.

If $c_A f'_+(0) \geq m_0$, the no-AI choice satisfies the entry condition with the opposite weak inequality and there is no positive root on the branch. This is the no-adoption region described after the proposition; the proposition itself restricts attention to the strict interior side.

Step 4. Differentiate the nonlinear system with respect to the acquisition cost. At fixed h_A and τ_b , linearity in c_A gives $\partial MC_A / \partial c_A = MC_A / c_A = m(h) / c_A$ at equilibrium. Differentiating the AI equation in (C.3), using $dh = \lambda dh_A$, gives

$$m'(h) \lambda \frac{dh_A}{dc_A} = \frac{\partial MC_A}{\partial h_A} \frac{dh_A}{dc_A} + \frac{m(h)}{c_A}.$$

Solving,

$$\frac{dh_A}{dc_A} = -\frac{m(h) / c_A}{\partial MC_A / \partial h_A - m'(h) \lambda} < 0. \quad (C.11)$$

Together with Step 2, this also gives $dh / dc_A < 0$, $d\alpha / dc_A < 0$, and $dh_P / dc_A > 0$.

The true price-error variance is

$$S_q^T = s(h) + \frac{\alpha^2}{\tau_b} = s(h) + \frac{h_A^2}{\tau_b h^2}. \quad (C.12)$$

Its exact derivative at fixed τ_b is therefore

$$\frac{dS_q^T}{dc_A} = \left[s'(h) \lambda + \frac{2\alpha(1 - \alpha\lambda)}{\tau_b h} \right] \frac{dh_A}{dc_A}. \quad (C.13)$$

At either cost boundary in Step 3, the bracket tends to $s'(h_0) \lambda_0 < 0$. Since $dh_A / dc_A < 0$ at each interior point, $dS_q^T / dc_A > 0$ sufficiently close to the boundary.

Holding the production parameters fixed, the aggregate TFP formula implies the exact differential

$$d \log \mathcal{A} = -\Gamma_0 \left(\frac{1 + \tau_d S_0}{1 + \tau_d S_q^T} \right)^2 dS_q^T. \quad (\text{C.14})$$

The multiplying factor is strictly positive, proving (6.4). For comparison, the level expansion in this case is

$$S_q^T - S_0 = s'(h_0) \lambda_0 h_A + o(h_A), \quad \frac{\alpha^2}{\tau_b} = O(h_A^2).$$

The marginal conclusion follows from (C.11)–(C.13), not from differentiating this remainder.

Step 5. Obtain the low-accuracy limit of information acquisition. Fix $c_A > 0$ satisfying $c_A f'_+(0) < m_0$ and let $\tau_b \downarrow 0$. Since $0 < h_A < \tau_b$, we have $h_A \rightarrow 0$. Step 2 then gives $h \rightarrow h_0$ and $m(h) \rightarrow m_0$. The local interior branch exists for sufficiently small τ_b : $\text{MC}_A - m(h)$ is negative at zero, strictly increasing in h_A , and diverges to infinity as $h_A \uparrow \tau_b$. For small τ_b , the entire interval $0 \leq h_A < \tau_b$ lies in the private-information neighborhood constructed in Step 2.

We next show that $\pi_A \rightarrow 0$. Otherwise, there would be a sequence with $\pi_A \geq \epsilon$ for some fixed $\epsilon > 0$. Along that sequence, (C.4) would imply

$$0 < c_A f'(\epsilon) \leq c_A f'(\pi_A) \leq m_0 \left(\frac{\tau_b}{\tau_b + \epsilon} \right)^2 \rightarrow 0,$$

a contradiction. Continuity of the right derivative therefore gives $f'(\pi_A) \rightarrow f'_+(0)$. Taking limits in (6.6), with $0 < 1 - h_A/\tau_b < 1$, proves (6.7).

For the remaining proof only, abbreviate this limiting fraction by

$$\omega_0 \equiv 1 - \sqrt{\frac{c_A f'_+(0)}{m_0}} \in (0, 1]. \quad (\text{C.15})$$

Thus $h_A = \omega_0 \tau_b + o(\tau_b)$, and Step 2 gives

$$h - h_0 = \lambda_0 \omega_0 \tau_b + o(\tau_b), \quad h_P - h_0 = -(1 - \lambda_0) \omega_0 \tau_b + o(\tau_b), \quad \alpha = \frac{\omega_0}{h_0} \tau_b + o(\tau_b). \quad (\text{C.16})$$

If $f'_+(0) > 0$, then $0 < \omega_0 < 1$ and

$$\frac{\pi_A}{\tau_b} = \frac{h_A/\tau_b}{1 - h_A/\tau_b} \rightarrow \frac{\omega_0}{1 - \omega_0}. \quad (\text{C.17})$$

If $f'_+(0) = 0$, then $\omega_0 = 1$ and the same identity gives $\pi_A/\tau_b \rightarrow \infty$, even though $\pi_A \rightarrow 0$.

Substitution into (C.12) yields

$$S_q^T - S_0 = \left[s'(h_0)\lambda_0\omega_0 + \frac{\omega_0^2}{h_0^2} \right] \tau_b + o(\tau_b). \quad (\text{C.18})$$

This establishes the right derivative of the continuous extension at $\tau_b = 0$. The next two steps establish the limit of the derivatives at positive τ_b directly from the nonlinear system.

Step 6. Differentiate the nonlinear system with respect to model accuracy. When h_A is held fixed, the derivatives needed in (C.2) are

$$\frac{\partial \pi_A}{\partial \tau_b} = -\frac{h_A^2}{(\tau_b - h_A)^2}, \quad \frac{\partial}{\partial \tau_b} \left(1 - \frac{h_A}{\tau_b} \right) = \frac{h_A}{\tau_b^2}.$$

Applying the product and chain rules gives

$$-\frac{\partial \text{MC}_A}{\partial \tau_b} = c_A f''(\pi_A) \left(1 - \frac{h_A}{\tau_b} \right)^{-4} \left(\frac{h_A}{\tau_b} \right)^2 + \frac{2c_A f'(\pi_A)}{\tau_b} \left(1 - \frac{h_A}{\tau_b} \right)^{-3} \frac{h_A}{\tau_b} > 0. \quad (\text{C.19})$$

Now differentiate the equilibrium AI equation at fixed c_A :

$$m'(h)\lambda \frac{dh_A}{d\tau_b} = \frac{\partial \text{MC}_A}{\partial h_A} \frac{dh_A}{d\tau_b} + \frac{\partial \text{MC}_A}{\partial \tau_b}.$$

Hence the exact derivative is

$$\frac{dh_A}{d\tau_b} = \frac{-\partial \text{MC}_A / \partial \tau_b}{\partial \text{MC}_A / \partial h_A - m'(h)\lambda} > 0. \quad (\text{C.20})$$

Divide the numerator and denominator by $c_A f'(\pi_A) / [\tau_b(1 - h_A/\tau_b)^3]$.

Using $m(h) = c_A f'(\pi_A) / (1 - h_A/\tau_b)^2$ and the identity $\tau_b / (1 - h_A/\tau_b) = \tau_b + \pi_A$ gives the equivalent formula

$$\frac{dh_A}{d\tau_b} = \frac{h_A}{\tau_b} \frac{2 + \pi_A f''(\pi_A) / f'(\pi_A)}{2 + (\tau_b + \pi_A) f''(\pi_A) / f'(\pi_A) - \frac{m'(h)}{m(h)} \lambda \tau_b \left(1 - \frac{h_A}{\tau_b} \right)}. \quad (\text{C.21})$$

Step 7. Evaluate the derivative limit, treating the two initial marginal costs separately. First suppose $f'_+(0) > 0$. Step 5 implies $f'(\pi_A) \rightarrow f'_+(0) > 0$ and $\tau_b / \pi_A \rightarrow (1 - \omega_0) / \omega_0 < \infty$. The additional curvature condition gives

$$\frac{\pi_A f''(\pi_A)}{f'(\pi_A)} \rightarrow 0, \quad \frac{(\tau_b + \pi_A) f''(\pi_A)}{f'(\pi_A)} = \left(1 + \frac{\tau_b}{\pi_A} \right) \frac{\pi_A f''(\pi_A)}{f'(\pi_A)} \rightarrow 0.$$

Also, $m'(h)/m(h)$ and λ remain bounded, while $\tau_b(1 - h_A/\tau_b) \rightarrow 0$. Every term in (C.21) now has a limit, yielding $dh_A/d\tau_b \rightarrow \omega_0$.

Next suppose $f'_+(0) = 0$. We cannot assume that $\pi_A f''(\pi_A)/f'(\pi_A)$ has a finite limit. Instead, compare the two nonnegative terms in (C.9) with their counterparts in (C.19). Because $0 < h_A/\tau_b < 1$, these formulas imply

$$\left(\frac{h_A}{\tau_b}\right)^2 \leq \frac{-\partial \text{MC}_A / \partial \tau_b}{\partial \text{MC}_A / \partial h_A} \leq \frac{h_A}{\tau_b}. \quad (\text{C.22})$$

Both bounds tend to one. Moreover, the second term in (C.9), together with the equilibrium condition, gives

$$\frac{\partial \text{MC}_A}{\partial h_A} \geq \frac{2m(h)}{\tau_b(1 - h_A/\tau_b)} \rightarrow \infty.$$

Since $m'(h)\lambda$ is bounded,

$$\frac{\partial \text{MC}_A / \partial h_A}{\partial \text{MC}_A / \partial h_A - m'(h)\lambda} \rightarrow 1.$$

Multiplying this limit by the middle ratio in (C.22) and using (C.20) yields $dh_A/d\tau_b \rightarrow 1 = \omega_0$.

Thus in both cases the exact nonlinear system gives

$$\begin{aligned} \frac{dh_A}{d\tau_b} &\rightarrow \omega_0, & \frac{dh}{d\tau_b} &\rightarrow \lambda_0 \omega_0, \\ \frac{dh_P}{d\tau_b} &\rightarrow -(1 - \lambda_0) \omega_0, & \frac{d\alpha}{d\tau_b} &\rightarrow \frac{\omega_0}{h_0}. \end{aligned} \quad (\text{C.23})$$

The last three limits use (C.6)–(C.7). They are not inferred by differentiating the remainders in (C.16).

Step 8. Translate the information responses into price informativeness and TFP. Differentiating (C.12) at fixed c_A gives

$$\frac{dS_q^T}{d\tau_b} = s'(h) \frac{dh}{d\tau_b} + \frac{2\alpha}{\tau_b} \frac{d\alpha}{d\tau_b} - \frac{\alpha^2}{\tau_b^2}. \quad (\text{C.24})$$

The three terms represent the change in noise-trader contamination, the change in the common-error loading, and the direct reduction in model-error variance, respectively. By Steps 5 and 7, their limits are

$$s'(h_0) \lambda_0 \omega_0, \quad \frac{2\omega_0^2}{h_0^2}, \quad -\frac{\omega_0^2}{h_0^2}.$$

Consequently,

$$\lim_{\tau_b \downarrow 0} \frac{dS_q^T}{d\tau_b} = s'(h_0) \lambda_0 \omega_0 + \frac{\omega_0^2}{h_0^2}. \quad (\text{C.25})$$

This agrees with the right derivative obtained from (C.18). Since $S_q^T \rightarrow S_0$, the factor multiply-

ing dS_q^T in (C.14) tends to $-\Gamma_0$. It follows that

$$\left. \frac{d \log \mathcal{A}}{d\tau_b} \right|_{0+} = \lim_{\tau_b \downarrow 0} \frac{d \log \mathcal{A}}{d\tau_b} = \Gamma_0 \omega_0 \left[[-s'(h_0)] \lambda_0 - \frac{\omega_0}{h_0^2} \right]. \quad (\text{C.26})$$

Substituting (C.15) proves (6.5). Both Γ_0 and ω_0 are positive, so the expression is negative precisely under (6.8), and positive under the reverse strict inequality. The existence of this derivative limit gives the same strict sign at every sufficiently small positive τ_b . If the bracket is zero, these first-order results do not determine the sign. ■

C.2 Cost Derivatives and Equilibrium Responses

Step 1: Coordinate changes and individual optimality. Suppress fixed c_P where it is unnecessary. For the proof, distinguish the two representations of the same acquisition cost by writing

$$\mathcal{C}^I(a, p; c_A, t) = \mathcal{C} \left(\frac{1}{2} \log \left(1 + \frac{a+p}{\tau_d} \right), \frac{a}{a+p}; c_A, c_P, t \right), \quad a = h_A, \quad p = h_P, \quad t = \tau_b. \quad (\text{C.27})$$

The inverse transformation is

$$a = \alpha H(\mathcal{K}), \quad p = (1 - \alpha) H(\mathcal{K}), \quad H'(\mathcal{K}) = 2(\tau_d + H(\mathcal{K})). \quad (\text{C.28})$$

Thus the $\mathcal{C}_{h_A h_P}$ used in the main text is $\mathcal{C}_{ap}^I$; it is distinct from $\mathcal{C}_{\mathcal{K}\alpha}$.

In a unilateral information deviation, manager i holds the aggregate price rule fixed. Under $\kappa = 0$, write gross information value as $\mathcal{V}_i(h_i)$. The objective is $\mathcal{V}_i(H(\mathcal{K}_i)) - \mathcal{C}(\mathcal{K}_i, \alpha_i; c_A, c_P, t)$. At a symmetric equilibrium $\mathcal{V}'_i(h) = m(h)$, so differentiation with respect to quantity and composition gives

$$H'(\mathcal{K})m(h) - \mathcal{C}_{\mathcal{K}} = 0, \quad \mathcal{C}_{\alpha} = 0. \quad (\text{C.29})$$

A change in composition at fixed $\mathcal{K}_i$ leaves gross value unchanged. These are the main-text information conditions.

Applying the chain rule to (C.27) also gives

$$\begin{aligned} \mathcal{C}_a^I &= \frac{\mathcal{C}_{\mathcal{K}}}{H'(\mathcal{K})} + \frac{1-\alpha}{h} \mathcal{C}_{\alpha}, \\ \mathcal{C}_p^I &= \frac{\mathcal{C}_{\mathcal{K}}}{H'(\mathcal{K})} - \frac{\alpha}{h} \mathcal{C}_{\alpha}. \end{aligned} \quad (\text{C.30})$$

Hence the equilibrium conditions are equivalently

$$\mathcal{C}_a^I(a, p; c_A, t) = \mathcal{C}_p^I(a, p; c_A, t) = m(a + p), \quad \mathcal{C}_\alpha = h(\mathcal{C}_a^I - \mathcal{C}_p^I). \quad (\text{C.31})$$

The no-AI cost identity implies $\mathcal{C}^I(0, p; c_A, t) = G(p)$ and $\mathcal{C}_p^I(0, p; c_A, t) = G'(p)$.

Step 2: Differentiate the symmetric equilibrium system. Across symmetric equilibria, the marginal value changes with aggregate information, including its effect on prices. Its derivative is $m'(h)$; it need not equal the curvature of an individual's value at fixed prices. Let

$$B(\mathcal{K}) = H'(\mathcal{K})m(H(\mathcal{K})), \quad B'(\mathcal{K}) = H''(\mathcal{K})m(h) + [H'(\mathcal{K})]^2 m'(h). \quad (\text{C.32})$$

At $\kappa = 0$, B has no direct dependence on c_A or t . For either parameter ϑ , total differentiation of $\mathcal{C}_\mathcal{K} - B = 0$ and $\mathcal{C}_\alpha = 0$ yields

$$\begin{aligned} [\mathcal{C}_{\mathcal{K}\mathcal{K}} - B']\mathcal{K}_\vartheta + \mathcal{C}_{\mathcal{K}\alpha}\alpha_\vartheta &= -\mathcal{C}_{\mathcal{K}\vartheta}, \\ \mathcal{C}_{\mathcal{K}\alpha}\mathcal{K}_\vartheta + \mathcal{C}_{\alpha\alpha}\alpha_\vartheta &= -\mathcal{C}_{\alpha\vartheta}. \end{aligned} \quad (\text{C.33})$$

The coefficient matrix is

$$J = \begin{pmatrix} \mathcal{C}_{\mathcal{K}\mathcal{K}} - B' & \mathcal{C}_{\mathcal{K}\alpha} \\ \mathcal{C}_{\mathcal{K}\alpha} & \mathcal{C}_{\alpha\alpha} \end{pmatrix}. \quad (\text{C.34})$$

The main-text regularity conditions are equivalent to positive definiteness of J , which implies $\mathcal{C}_{\alpha\alpha} > 0$, $\det J > 0$, and

$$\mathcal{C}_{\mathcal{K}\mathcal{K}} - B' - \frac{\mathcal{C}_{\mathcal{K}\alpha}^2}{\mathcal{C}_{\alpha\alpha}} > 0. \quad (\text{C.35})$$

Solving the second line of (C.33) for α_ϑ gives (6.12). Substitution into the first line gives the quantity formula in the main-text footnote. Equivalently,

$$\begin{pmatrix} \mathcal{K}_\vartheta \\ \alpha_\vartheta \end{pmatrix} = -J^{-1} \begin{pmatrix} \mathcal{C}_{\mathcal{K}\vartheta} \\ \mathcal{C}_{\alpha\vartheta} \end{pmatrix}. \quad (\text{C.36})$$

Step 3: Cost changes and the relative quantity response. Write $\Delta = \det J > 0$. Expanding the inverse in (C.36), and using the definitions of U_c, V_c in the main text, gives

$$\mathcal{K}_{c_A} = -\frac{U_c}{\Delta}, \quad \alpha_{c_A} = -\frac{V_c}{\Delta}, \quad \frac{h_{c_A}}{\alpha_{c_A}} = \frac{H'(\mathcal{K})U_c}{V_c}. \quad (\text{C.37})$$

The last equality holds whenever $V_c \neq 0$. Under $V_c > 0$ and (6.13), $\alpha_{c_A} < 0$ and the last ratio has a strictly positive lower limit. Thus $h_{c_A} < 0$ sufficiently close to the boundary. The cost

conditions are restrictions on derivatives evaluated along the local branch; positive definiteness of J alone does not imply these signs. They do not require the entry cost to be infinite.

Step 4: Accuracy changes and direct TFP responses. The same inverse gives

$$\begin{aligned}\mathcal{K}_t &= \frac{\mathcal{C}_{\mathcal{K}\alpha}\mathcal{C}_{\alpha t} - \mathcal{C}_{\alpha\alpha}\mathcal{C}_{\mathcal{K}t}}{\Delta}, \\ \alpha_t &= \frac{\mathcal{C}_{\mathcal{K}\alpha}\mathcal{C}_{\mathcal{K}t} - [\mathcal{C}_{\mathcal{K}\mathcal{K}} - B']\mathcal{C}_{\alpha t}}{\Delta}.\end{aligned}\tag{C.38}$$

These are interior derivatives and do not by themselves establish their limits as $t \downarrow 0$.

Differentiating $S_q^T = s(H(\mathcal{K})) + \alpha^2/t$ and combining the result with (C.36) gives an exact expression in cost derivatives at every regular interior equilibrium:

$$\begin{aligned}\frac{d \log \mathcal{A}}{d\vartheta} &= \left(-\frac{\partial \log \mathcal{A}}{\partial S_q^T} \right) \left[\begin{pmatrix} s'(h)H'(\mathcal{K}) & 2\alpha/t \end{pmatrix} J^{-1} \begin{pmatrix} \mathcal{C}_{\mathcal{K}\vartheta} \\ \mathcal{C}_{\alpha\vartheta} \end{pmatrix} \right. \\ &\quad \left. + \mathbf{1}_{\{\vartheta=t\}} \frac{\alpha^2}{t^2} \right].\end{aligned}\tag{C.39}$$

The prefactor is positive. The final term is the direct variance reduction from greater accuracy, holding information choices fixed. For cost changes, the expression simplifies to

$$\frac{d \log \mathcal{A}}{dc_A} = \frac{-\partial \log \mathcal{A} / \partial S_q^T}{\Delta} \left[s'(h)H'(\mathcal{K})U_c + \frac{2\alpha}{t}V_c \right].\tag{C.40}$$

Thus the exact condition for a cost reduction to raise TFP is $[-s'(h)]H'(\mathcal{K})U_c > (2\alpha/t)V_c$.

C.3 Proof of Lemma 4

Proof. *Step 1: Locate the optimal utilization fraction.* Let $h(t)$ and $\alpha(t)$ denote the maintained equilibrium branch. Feasibility gives $0 < \alpha(t)/t < 1$. Fix any $q \in (0, q_0)$. Since $h(t) \rightarrow h_0$, the joint-limit condition (6.14) implies

$$\mathcal{C}_\alpha \left(\mathcal{K}(t), \frac{qt}{h(t)}; c_A, c_P, t \right) < 0$$

for all sufficiently small t . At the same total information, the equilibrium satisfies

$$\mathcal{C}_\alpha(\mathcal{K}(t), \alpha(t); c_A, c_P, t) = 0.$$

Strict convexity in composition therefore gives $\alpha(t) > qt/h(t)$, or $a(t)/t > q$. The comparison configuration is feasible for small t : its AI precision is below t , and its private precision converges to $h_0 \in (0, \tau_u)$.

If $q_0 < 1$, fix any $q \in (q_0, 1)$. The positive sign in (6.14) similarly implies $\alpha(t) < qt/h(t)$, so $a(t)/t < q$ for small t . Taking fixed comparison fractions arbitrarily close to q_0 on each side proves $a(t)/t \rightarrow q_0$. If $q_0 = 1$, the lower comparison holds for every $q < 1$, while $a(t)/t < 1$ by feasibility. It again follows that

$$\frac{a(t)}{t} \rightarrow q_0, \quad a(t) = q_0 t + o(t), \quad \frac{\alpha(t)}{t} \rightarrow \frac{q_0}{h_0}. \quad (\text{C.41})$$

This step uses cost signs at fixed comparison fractions. It does not assume a limit for the equilibrium ratio.

Step 2: Expand the marginal cost of private information. Put $p(t) = h(t) - a(t)$. Then $p(t) \rightarrow h_0$. At each $t > 0$, continuity of $\mathcal{C}_p^I$ at $a = 0$ and the mean value theorem imply that, for some $\tilde{a}(t) \in (0, a(t))$,

$$\begin{aligned} \mathcal{C}_p^I(a(t), p(t); c_A, t) - G'(p(t)) &= \mathcal{C}_{ap}^I(\tilde{a}(t), p(t); c_A, t) a(t) \\ &= [\gamma_0 + o(1)] a(t). \end{aligned} \quad (\text{C.42})$$

The last equality follows from (6.15), since $0 < \tilde{a}(t) < a(t) < t$ and $p(t) \rightarrow h_0$. The joint limit controls the entire interval used by the mean value theorem; convergence of the cross partial only at the equilibrium endpoint would not suffice.

Step 3: Determine the order of total information. By the private-information condition in (C.31),

$$m(h(t)) = G'(h(t) - a(t)) + [\gamma_0 + o(1)] a(t). \quad (\text{C.43})$$

Because G'' is continuous near h_0 , the mean value theorem gives $G'(h - a) - G'(h) = -[G''(h_0) + o(1)]a$. Also $m(h_0) - G'(h_0) = 0$, so the same theorem applied to $m - G'$ gives

$$[m'(h_0) - G''(h_0) + o(1)](h - h_0) = [\gamma_0 - G''(h_0) + o(1)]a. \quad (\text{C.44})$$

The coefficient on the left is bounded away from zero for small t . Dividing by this coefficient proves, rather than assumes, that $h - h_0 = O(a)$ and

$$\frac{h - h_0}{a} \rightarrow \frac{G''(h_0) - \gamma_0}{G''(h_0) - m'(h_0)}. \quad (\text{C.45})$$

This conclusion also holds when the limiting numerator is zero.

Step 4: Obtain the boundary responses. Multiplying (C.45) by $a/t \rightarrow q_0$ yields

$$\lim_{t \downarrow 0} \frac{h - h_0}{t} = q_0 \frac{G''(h_0) - \gamma_0}{G''(h_0) - m'(h_0)} = L_0. \quad (\text{C.46})$$

Thus $h = h_0 + L_0 t + o(t)$, proving the lemma. For completeness, $p = h_0 + (L_0 - q_0)t + o(t)$ and $\alpha = (q_0/h_0)t + o(t)$. These are boundary expansions. Their proof does not differentiate an $o(t)$ remainder or assert convergence of the interior derivatives. ■

C.4 Proof of Proposition 7

Proof. *Step 1: Cost reductions near the no-AI boundary.* Fix $t > 0$. By $V_c > 0$, (C.37), and (6.13), $\alpha_{c_A} < 0$ and h_{c_A}/α_{c_A} has a strictly positive lower limit. Hence the cost branch can locally be parameterized by α . Along it,

$$\frac{dS_q^T}{d\alpha} = s'(h) \frac{dh}{d\alpha} + \frac{2\alpha}{t}, \quad \frac{dh}{d\alpha} = \frac{h_{c_A}}{\alpha_{c_A}} = \frac{H'(\mathcal{K})U_c}{V_c}. \quad (\text{C.47})$$

There is a constant $R > 0$ such that $dh/d\alpha \geq R$ sufficiently close to the boundary. Continuity and $s'(h_0) < 0$ give a constant $\varepsilon > 0$ such that $s'(h) \leq -\varepsilon$ there. Consequently,

$$\frac{dS_q^T}{d\alpha} \leq -\varepsilon R + \frac{2\alpha}{t} < 0$$

for sufficiently small α . Multiplication by $\alpha_{c_A} < 0$ gives $dS_q^T/dc_A > 0$. Since $\partial \log \mathcal{A}/\partial S_q^T < 0$, it follows that $d \log \mathcal{A}/dc_A < 0$. This argument concerns the interior side of the boundary and is unchanged if the entry threshold is finite. If an interval of costs instead has zero AI acquisition, the maintained no-AI cost identity makes information and TFP independent of c_A on that interval.

Step 2: The order of the true price-error components. Fix $c_A > 0$ and let $t \downarrow 0$. The lemma gives $a = q_0 t + o(t)$ and $h = h_0 + L_0 t + o(t)$. Because $h_0 > 0$,

$$\alpha = \frac{a}{h} = \frac{q_0}{h_0} t + o(t), \quad \frac{\alpha^2}{t} = \frac{q_0^2}{h_0^2} t + o(t). \quad (\text{C.48})$$

Differentiability of s gives $s(h) - s(h_0) = s'(h_0)L_0 t + o(t)$. Adding the two components yields

$$S_q^T - s(h_0) = \left[s'(h_0)L_0 + \frac{q_0^2}{h_0^2} \right] t + o(t). \quad (\text{C.49})$$

This is the true price-error variance, including the full common model error despite managers'

correlation neglect.

Step 3: Convert the price-error change into TFP. Let $\mathcal{A}_0$ denote TFP in the no-AI economy. The existing TFP mapping is differentiable at $S_q^T = s(h_0)$, where $\partial \log \mathcal{A} / \partial S_q^T = -\Gamma_0 < 0$. Since (C.49) is $O(t)$,

$$\log \mathcal{A} - \log \mathcal{A}_0 = -\Gamma_0 \left[s'(h_0)L_0 + \frac{q_0^2}{h_0^2} \right] t + o(t). \quad (\text{C.50})$$

Dividing by t and taking the limit proves (6.18). The derivative at 0^+ in the proposition is this right boundary difference quotient. Substituting the lemma's formula for L_0 into (6.18) determines the sign of the boundary effect. It is negative when the common-error term exceeds the information gain, positive under the reverse strict inequality, and zero at equality. ■

Recovery of the separable-cost results. For the cost in Proposition 6, write

$$C^I(a, p; c_A, t) = c_A f\left(\frac{ta}{t-a}\right) + G(p), \quad x_A = \frac{ta}{t-a}. \quad (\text{C.51})$$

Its cross derivative is identically zero, so (6.15) holds with $\gamma_0 = 0$. For every fixed $q \in (0, 1)$, set $a = qt$ and $p = h - qt$. Then $x_A = tq/(1-q) \rightarrow 0$, and

$$\frac{C_\alpha}{h} = \frac{c_A f'(tq/(1-q))}{(1-q)^2} - G'(h - qt) \rightarrow \frac{c_A f'_+(0)}{(1-q)^2} - m_0 \quad (\text{C.52})$$

as $t \downarrow 0$ and $h \rightarrow h_0$. If $f'_+(0) > 0$ and $c_A f'_+(0) < m_0$, the limiting expression has the unique zero $q_0 = 1 - \sqrt{c_A f'_+(0)/m_0} \in (0, 1)$, with the signs required by (6.14). If $f'_+(0) = 0$, it equals $-m_0 < 0$ at every $q < 1$, giving $q_0 = 1$. The lemma consequently implies

$$L_0 = \frac{G''(h_0)}{G''(h_0) - m'(h_0)} q_0 = \lambda_0 q_0. \quad (\text{C.53})$$

This recovers the generalized initial-marginal-cost result, including the special case $q_0 = 1$.

To verify inclusion of the cost path as well, let $F(a; t) = f(ta/(t-a))$. The equilibrium residuals in effective coordinates are $c_A F_a - m(a+p)$ and $G'(p) - m(a+p)$. Their Jacobian is

$$M = \begin{pmatrix} c_A F_{aa} - m' & -m' \\ -m' & G'' - m' \end{pmatrix}.$$

Near the no-AI equilibrium of Proposition 6, $F_{aa} > 0$, $G'' > 0$, and $m' < 0$. Thus, for every

nonzero vector v , $v'Mv = c_A F_{aa} v_a^2 + G'' v_p^2 - m'(v_a + v_p)^2 > 0$. The coordinate Jacobian

$$T = \frac{\partial(a, p)}{\partial(\mathcal{K}, \alpha)} = \begin{pmatrix} \alpha H' & H \\ (1 - \alpha) H' & -H \end{pmatrix}$$

has determinant $-HH' \neq 0$. At a zero of the residuals the response matrix satisfies $J = T'MT$, and is therefore positive definite.

Differentiating the private condition $m(h) = G'(h - a)$ along the branch gives

$$\lambda = \frac{dh}{da} = \frac{G''(p)}{G''(p) - m'(h)} \in (0, 1), \quad \frac{da}{dc_A} = -\frac{F_a}{c_A F_{aa} - m'(h)\lambda} < 0.$$

Since $d\alpha/da = (1 - \alpha\lambda)/h > 0$, $\alpha_{c_A} < 0$ and

$$\frac{h_{c_A}}{\alpha_{c_A}} = \frac{h\lambda}{1 - \alpha\lambda} \longrightarrow h_0\lambda_0 > 0.$$

Equation (C.37) then verifies both restrictions, $V_c > 0$ and (6.13). When $f'_+(0) > 0$, this verification applies as $c_A \uparrow m_0/f'_+(0)$ from the active side. When $f'_+(0) = 0$, it applies as $c_A \rightarrow \infty$.

D Additional Empirical Results

This appendix reports descriptive statistics, common selling and downside outcomes, and robustness checks for the portfolio-convergence results. The exercises address persistence in portfolio similarity, the construction of the AI-reliance measure, the definition of portfolio adjustment, the threshold used to define joint GPT alignment, manager-specific time-varying shocks, and residual dependence across manager pairs.

Appendix Table D.1 reports descriptive statistics for the variables used in the main regressions. The AI-reliance row is drawn from the manager-quarter sample rather than the manager-pair panel, so each estimated manager-quarter value enters once.

Table D.1: Summary Statistics for Main Regression Variables

Variable	N	Mean	SD	Median
<i>Panel A: AI reliance and portfolio similarity</i>				
AI reliance	7,581	0.0691	0.0917	0.0360
Common AI reliance	890,394	0.0043	0.0110	0.0014
Average AI reliance	890,394	0.0653	0.0618	0.0406
Joint GPT alignment	890,394	0.0146	0.1199	0.0000
Portfolio overlap	890,394	0.3044	0.1741	0.2848
Cosine similarity	890,394	0.4204	0.2553	0.3940
<i>Panel B: Validation variables</i>				
Pair-weighted GPT signal strength	859,357	-0.0617	0.2192	-0.0680
Pair-weighted earnings-call length	859,357	0.0713	0.1784	0.0819
Pair-weighted analyst coverage	859,357	0.9911	0.3608	0.9928
<i>Panel C: Similarity and pair-level outcomes</i>				
Portfolio overlap	1,233,217	0.2456	0.1827	0.2105
Cosine similarity	1,233,217	0.3437	0.2641	0.2951
Future common selling, minimum denominator	1,233,217	0.1513	0.1451	0.1129
Future common selling, Jaccard denominator	1,234,800	0.0301	0.0331	0.0201
Next-quarter market downturn	1,233,217	0.2865	0.4521	0.0000
Total downside comovement $\times 10,000$	1,234,801	35.5393	106.9504	0.0000
Joint downside indicator	1,234,801	0.2567	0.4368	0.0000
Conditional downside comovement $\times 10,000$	298,963	137.5984	173.6902	33.6524
<i>Panel D: Pair-level controls</i>				
Average log portfolio value	1,233,217	24.4325	0.7752	24.3109
Absolute log portfolio value difference	1,233,217	1.1224	0.9923	0.8483
Average number of holdings	1,233,217	1504.3964	972.3907	1370.0000
Absolute difference in number of holdings	1,233,217	1486.3973	1252.5271	1161.0000
Average portfolio HHI	1,233,217	0.0279	0.0518	0.0149
Absolute portfolio HHI difference	1,233,217	0.0342	0.0975	0.0095

Notes: The AI-reliance row uses all valid top-300 manager-quarter estimates from 2018Q1 through 2024Q4. The remaining Panel A variables use the common estimation sample for the eight baseline regressions of AI reliance and portfolio similarity. Each Panel B moderator uses the sample for its corresponding validation regression. Panel C variables use the sample for the corresponding pair-outcome regression. Similarity and Panel D controls use the minimum-denominator common-selling sample. Pair-outcome samples require at least one common stock holding in quarter t . Interaction terms, fixed effects, absorbed same-type indicators, and sample-selection indicators are not reported. Means of binary variables are sample proportions.

D.1 Portfolio Similarity and Pair-Level Outcomes

We next ask whether current portfolio similarity provides an ex ante indicator of common selling and joint downside exposure, and whether this relationship strengthens during market downturns. Prior work shows that portfolio similarity predicts common trading and liquidation behavior (Pool et al., 2015; Girardi et al., 2021). We therefore relate quarter- t similarity to selling and downside outcomes in quarter $t + 1$, examining whether the portfolio convergence documented in Section 2 is followed by common adjustments. The analysis is restricted to manager pairs with at least one common stock holding in quarter t , which removes pairs whose current common exposure is mechanically zero while preserving zero realizations of the future outcome variables. Because common holdings create the opportunity for common sales, the regressions map portfolio exposure into subsequent liquidation behavior.

Common selling and downside comovement. We first consider whether higher portfolio similarity predicts more common selling. Our primary outcome follows the standard definition of Pool et al. (2015). Let

$$I_{i,s,t+1}^- = \mathbf{1}\{\text{Shares}_{i,s,t+1} < \text{Shares}_{i,s,t}\}, \quad (\text{D.1})$$

indicate whether manager i reduces its split-adjusted holdings of stock s between quarters t and $t + 1$, or whether manager i sells asset s in period $t + 1$. We define the common-selling fraction as (Pool et al., 2015):

$$\text{Common_Selling}_{i,k,t+1} = \frac{\sum_s \min\{I_{i,s,t+1}^-, I_{k,s,t+1}^-\}}{\min\left\{\sum_s I_{i,s,t+1}^-, \sum_s I_{k,s,t+1}^-\right\}}. \quad (\text{D.2})$$

The numerator counts stocks sold by both managers, while the denominator is the smaller number of stocks sold by either manager. The measure is defined when both managers sell at least one stock and ranges from zero to one. Because it treats each sold stock as one observation, it captures overlap in the direction of portfolio adjustment without allowing a few large dollar trades to dominate the outcome.

We complement the baseline measure with two value-weighted definitions. For stock s , manager i 's selling value between t and $t + 1$, $S_{i,s,t+1}$, is the positive decline in split-adjusted shares multiplied by the quarter- $t + 1$ stock price. If the share holding does not decline, $S_{i,s,t+1}$ is set to zero. Let $S_{i,t+1} = \sum_s S_{i,s,t+1}$ denote manager i 's total selling, and define the pair-level common selling value as:

$$C_{i,k,t+1} = \sum_s \min\{S_{i,s,t+1}, S_{k,s,t+1}\}, \quad (\text{D.3})$$

which measures dollar selling that can be matched stock by stock across the pair. Following Pool et al. (2015) and Girardi et al. (2021), we construct the value-based common-selling overlap

measure as:¹⁵

$$Common_Selling_{i,k,t+1}^{Val} = \frac{C_{i,k,t+1}}{\min\{S_{i,t+1}, S_{k,t+1}\}}. \quad (D.4)$$

$Common_Selling_{i,k,t+1}^{Val}$ measures the value of stocks that are commonly sold by the two asset managers as a fraction of the smaller aggregate selling within the manager pair. We also consider an alternative Jaccard-type common-selling measure:

$$Common_Selling_{i,k,t+1}^{Jac} = \frac{C_{i,k,t+1}}{\sum_s \max\{S_{i,s,t+1}, S_{k,s,t+1}\}}. \quad (D.5)$$

Its denominator is algebraically equal to $S_{i,t+1} + S_{k,t+1} - C_{i,k,t+1}$, the union of the pair's selling values. It is more conservative because unmatched sales by either manager enlarge the denominator. Together, the three measures distinguish overlap in sold stocks from overlap in the dollar magnitude of liquidation and show whether the results depend on the normalization of selling intensity.

We next examine whether portfolio similarity is associated with joint downside performance. Managers with similar portfolios are exposed to many of the same asset-level shocks, so their returns may experience common downside pressure. We first define the following unconditional measure of pair-level downside comovement:

$$Downside_{i,k,t+1} = \min\{R_{i,t+1}, 0\} \times \min\{R_{k,t+1}, 0\}, \quad (D.6)$$

which equals zero whenever at least one manager has a nonnegative return. We multiply this variable by 10,000 for readability. Because the unconditional downside product has a large mass at zero, its variation reflects both the incidence of negative returns for the two managers and the severity of losses conditional on both returns being negative.¹⁶ We therefore also estimate a two-part model that separates the occurrence of joint negative returns from the severity of losses conditional on that event. The first part is a linear probability model for $\mathbf{1}\{R_{i,t+1} < 0, R_{k,t+1} < 0\}$. The second part uses $Downside_{i,k,t+1}$ conditional on both returns being negative. This decomposition shows whether similarity primarily predicts the probability of entering a joint downside state or the severity of losses within that state.

To investigate the relationship between portfolio similarity and the pair-level outcomes, we estimate the following equation for each outcome $Y_{i,k,t+1}$:

$$Y_{i,k,t+1} = \beta \times Similarity_{i,k,t} + \theta' Z_{i,k,t} + \varphi_{i,k} + \psi_t + \epsilon_{i,k,t}. \quad (D.7)$$

¹⁵Girardi et al. (2021) measures common selling using the inner product of two asset managers' selling vectors. We use the sum of the smaller selling value for each stock and normalize it following Pool et al. (2015).

¹⁶In our sample, 92.5% of the observations in which both asset managers experience negative returns are concentrated in 7 quarters.

The control vector contains the pair average and absolute difference in log portfolio value, number of holdings, and portfolio HHI. Concentration is included here because it can directly affect portfolio return volatility and liquidation behavior. Pair fixed effects absorb persistent differences in manager matching, mandates, and investment style; quarter fixed effects absorb market-wide conditions common to all pairs. Standard errors are clustered by manager pair.

Table D.2 reports six complementary outcomes. Panels A–C show that current portfolio similarity predicts greater overlap in next-quarter sales under all three definitions. Under the baseline measure, a one-standard-deviation increase in portfolio overlap predicts 3.6 percentage points more overlap in sold stocks, while a one-standard-deviation increase in cosine similarity predicts 1.0 percentage point more. The corresponding estimates are 8.4 and 3.2 percentage points for the value-based normalized common-selling measure and 2.1 and 0.9 percentage points under the weighted-Jaccard normalization. The similar pattern across a stock-name measure and two value-weighted measures indicates that the association is not confined to a few large sales or to the scale of the smaller seller. More similar portfolios are followed by greater overlap in the assets subsequently sold.

Panels D–F show that similarity also predicts joint downside exposure. The unconditional downside coefficient is positive and significant for both overlap and cosine similarity. The two-part estimates locate the source of this association. A one-standard-deviation increase in overlap predicts a 0.62-percentage-point increase in the probability that both managers have negative returns, while a one-standard-deviation increase in cosine similarity predicts an increase of approximately 1.39 percentage points. Conditional on both returns being negative, the overlap coefficient remains positive but is not statistically distinguishable from zero; the cosine coefficient is positive and highly significant. Thus, current portfolio similarity is an *ex ante* indicator of joint losses. The evidence on conditional loss severity is stronger for cosine similarity than for overlap. These results connect portfolio convergence to the joint downside exposure that motivates the financial-fragility model.

Market downturns. Asset managers are more likely to sell during aggregate market downturns (Goldstein et al., 2017). When their portfolios are similar, those sales are more likely to involve the same assets, which can make common selling especially relevant for market fragility (Manconi et al., 2012). We ask whether the predictive relationship between similarity and common selling is stronger in such periods. We define $Market_Down_{t+1}$ as an indicator equal to one when the value-weighted market return is in the bottom 20% of sample quarters, and estimate:

$$\begin{aligned} Common_Selling_{i,k,t+1}^d = & \beta \times Similarity_{i,k,t} + \lambda \times Similarity_{i,k,t} \times Market_Down_{t+1} \\ & + \theta' Z_{i,k,t} + \varphi_{i,k} + \psi_t + \epsilon_{i,k,t}, \end{aligned} \quad (D.8)$$

Table D.2: Portfolio Similarity and Pair-Level Outcomes

Similarity measure	(1) Portfolio overlap	(2) Cosine similarity
<i>Panel A: Future common selling</i>		
Similarity	0.1981*** (41.53)	0.0372*** (20.84)
Observations	1,233,217	1,233,217
<i>Panel B: Value-based future common selling</i>		
Similarity	0.4573*** (100.97)	0.1225*** (67.82)
Observations	1,233,217	1,233,217
<i>Panel C: Jaccard-weighted future common selling</i>		
Similarity	0.1126*** (84.34)	0.0345*** (66.27)
Observations	1,234,800	1,234,800
<i>Panel D: Total future downside comovement</i>		
Similarity	4.1582*** (4.25)	5.5694*** (11.54)
Observations	1,234,801	1,234,801
<i>Panel E: Both future returns are negative</i>		
Similarity	0.0338*** (8.02)	0.0526*** (30.48)
Observations	1,234,801	1,234,801
<i>Panel F: Conditional downside comovement</i>		
Similarity	5.8675 (1.55)	14.9124*** (7.76)
Observations	298,963	298,963
Pair controls	Yes	Yes
Pair FE	Yes	Yes
Quarter FE	Yes	Yes
SE cluster	Pair	Pair

Notes: Similarity is measured in quarter t and all outcomes in quarter $t + 1$. Panel A uses the Pool SaleOverlap measure based on the fraction of sold stock names common to both managers. Panel B scales matched selling value by the smaller manager's total selling. Panel C uses the union of the pair's selling values. Panel D reports downside comovement in the full outcome sample. Panel E is a linear probability model for both managers having negative returns. Panel F conditions on both returns being negative. Downside comovement in Panels D and F is multiplied by 10,000. Robust t -statistics are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

where $Common_Selling_{i,k,t+1}^d$ denotes each of the three common-selling measures defined above, and λ captures the difference in the similarity–common-selling association between normal and downturn quarters. A positive λ means that similarity is a stronger predictor of common selling during market downturns.

Table D.3 reports positive and statistically significant estimates of λ under all three common-selling definitions. For the baseline measure, a one-standard-deviation increase in overlap predicts an additional 0.28 percentage points of common selling during market downturns, and a one-standard-deviation increase in cosine similarity predicts an additional 0.16 percentage points. For the value-based common-selling measure, the corresponding increments are 0.82 and 0.58 percentage points; under the weighted-Jaccard normalization, they are 0.18 and 0.12 percentage points. Portfolio similarity is therefore a stronger predictor of overlapping stock-level sales in market downturns, when common liquidation is more consequential.

D.2 Results without Controlling for Current Similarity

Table D.4 removes current portfolio similarity from the baseline specifications. The coefficients on common AI reliance are positive and statistically significant for both overlap and cosine similarity. They are also larger than the corresponding baseline estimates. This difference is consistent with the strong persistence in pair-level portfolio composition: current similarity absorbs much of the variation that would otherwise be attributed to common reliance. The main specifications retain current similarity because they provide a more demanding test of whether common reliance predicts incremental portfolio convergence in the next quarter.

Table D.3: Portfolio Similarity, Market Downturns, and Common Selling

Similarity measure	(1) Portfolio overlap	(2) Cosine similarity
<i>Panel A: Future common selling</i>		
Similarity	0.1941*** (40.52)	0.0355*** (19.51)
Similarity $\times$ market downturn	0.0153*** (9.93)	0.0062*** (6.10)
Observations	1,233,217	1,233,217
<i>Panel B: Value-based future common selling</i>		
Similarity	0.4457*** (97.77)	0.1165*** (63.48)
Similarity $\times$ market downturn	0.0450*** (29.44)	0.0221*** (22.50)
Observations	1,233,217	1,233,217
<i>Panel C: Jaccard-weighted future common selling</i>		
Similarity	0.1100*** (82.56)	0.0333*** (63.94)
Similarity $\times$ market downturn	0.0099*** (24.54)	0.0046*** (18.34)
Observations	1,234,800	1,234,800
Pair controls	Yes	Yes
Pair FE	Yes	Yes
Quarter FE	Yes	Yes
SE cluster	Pair	Pair

Notes: The market-downturn indicator equals one for the bottom 20% of quarterly value-weighted market returns and is absorbed by quarter fixed effects. Robust t -statistics are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Future portfolio similarity	(1) Overlap	(2) Cosine
<i>Panel A: Controlling for average AI reliance</i>		
Common AI reliance	0.0833*** (10.45)	0.1339*** (7.81)
Average AI reliance	-0.0367*** (-22.26)	-0.0070* (-1.74)
<i>Panel B: Joint GPT alignment</i>		
Joint GPT alignment	0.0044*** (13.61)	0.0107*** (12.07)
Average AI reliance	-0.0290*** (-21.60)	0.0028 (0.86)
<i>Panel C: Controlling for individual AI reliance</i>		
Common AI reliance	0.0838*** (10.51)	0.1343*** (7.84)
AI reliance, manager i	-0.0225*** (-20.16)	-0.0070*** (-2.58)
AI reliance, manager k	-0.0148*** (-15.05)	-0.0005 (-0.22)
Pair controls	Yes	Yes
Pair FE	Yes	Yes
Quarter FE	Yes	Yes
SE cluster	Pair	Pair
Observations	890,394	890,394

Table D.4: Common AI Reliance and Future Similarity without Current Similarity

D.3 Alternative Fit-Based Measures of AI Reliance

The baseline reliance measure is the increase in in-sample R^2 when the fourteen GPT scores are added to the public-information regression. Since the full specification nests the baseline specification, this increase is mechanically nonnegative. We therefore consider three alternative measures that treat model fit and the additional regressors differently.

The first measure is the fraction of residual variation from the baseline regression that is removed by the GPT scores:

$$\text{Partial } R_{i,t}^2 = \frac{RSS_{base,i,t} - RSS_{full,i,t}}{RSS_{base,i,t}}. \quad (\text{D.9})$$

This normalization measures the explanatory contribution of the GPT scores relative to the variation left unexplained by the baseline public-information variables. Table D.5 reports positive and statistically significant common-reliance coefficients.

The next two measures penalize the full model for the additional GPT regressors. Let $\bar{R}_{base,i,t}^2$ and $\bar{R}_{full,i,t}^2$ denote adjusted R^2 from the two specifications. We define

$$\text{Adjusted } R^2 \text{ gain}_{i,t} = \max \left\{ \bar{R}_{full,i,t}^2 - \bar{R}_{base,i,t}^2, 0 \right\}, \quad (\text{D.10})$$

$$\text{Adjusted partial } R^2 \text{ gain}_{i,t} = \max \left\{ \frac{\bar{R}_{full,i,t}^2 - \bar{R}_{base,i,t}^2}{1 - \bar{R}_{base,i,t}^2}, 0 \right\}. \quad (\text{D.11})$$

The truncation at zero preserves the interpretation of reliance as nonnegative exposure to information extracted by the GPT scores. Unlike the raw R^2 gain, these measures assign a positive value only when the improvement in fit is large enough to offset the degrees-of-freedom penalty.

Table D.6 shows that the estimated coefficients remain positive, but the overlap estimates become less precise. Common adjusted- R^2 gain predicts future overlap at the 10 percent level, while common adjusted partial- R^2 gain has a similar coefficient but falls just outside conventional significance levels. The cosine estimates remain statistically significant under both adjustments. These results indicate that the baseline association is not solely a consequence of the mechanical increase in raw in-sample R^2 , although the precision of the overlap result depends on how strongly the additional GPT regressors are penalized.

Future portfolio similarity	(1) Overlap	(2) Cosine
<i>Panel A: Controlling for average partial R^2</i>		
Common partial R^2	0.0171*** (5.25)	0.0390*** (5.75)
Average partial R^2	-0.0063*** (-8.73)	0.0078*** (3.97)
Current similarity	0.7957*** (371.16)	0.6760*** (223.62)
<i>Panel B: Joint GPT alignment</i>		
Joint GPT alignment	0.0019*** (9.35)	0.0104*** (13.37)
Average partial R^2	-0.0050*** (-8.45)	0.0071*** (4.46)
Current similarity	0.7957*** (371.19)	0.6760*** (223.83)
<i>Panel C: Controlling for individual partial R^2</i>		
Common partial R^2	0.0172*** (5.28)	0.0392*** (5.77)
Partial R^2 , manager i	-0.0046*** (-9.78)	0.0017 (1.35)
Partial R^2 , manager k	-0.0018*** (-4.06)	0.0058*** (4.74)
Current similarity	0.7957*** (371.16)	0.6760*** (223.62)
Pair controls	Yes	Yes
Pair FE	Yes	Yes
Quarter FE	Yes	Yes
SE cluster	Pair	Pair
Observations	890,394	890,394

Table D.5: Partial- R^2 AI Reliance and Future Portfolio Similarity

	(1)	(2)
Future portfolio similarity	Overlap	Cosine
<i>Panel A: AI reliance based on adjusted-R^2 gain</i>		
Common adjusted- R^2 gain	0.0146* (1.73)	0.0453** (2.57)
Average adjusted- R^2 gain	-0.0068*** (-9.22)	0.0109*** (5.57)
Current similarity	0.7957*** (371.40)	0.6761*** (223.65)
Observations	890,394	890,394
<i>Panel B: AI reliance based on adjusted partial-R^2 gain</i>		
Common adjusted partial- R^2 gain	0.0130 (1.64)	0.0452*** (2.71)
Average adjusted partial- R^2 gain	-0.0068*** (-9.70)	0.0082*** (4.39)
Current similarity	0.7957*** (371.40)	0.6761*** (223.64)
Observations	890,394	890,394
Pair controls	Yes	Yes
Pair FE	Yes	Yes
Quarter FE	Yes	Yes
SE cluster	Pair	Pair

Table D.6: Degrees-of-Freedom-Adjusted AI Reliance and Future Portfolio Similarity

D.4 Alternative Portfolio-Adjustment Measures

The baseline manager-level regression uses changes in portfolio weights. Such changes may reflect active trading, but they can also arise mechanically from price movements. Table D.7 reconstructs AI reliance using two alternative dependent variables. Active rebalancing removes passive price drift from portfolio-weight changes, while the split-adjusted percentage change in shares measures trading quantities more directly. Under both definitions, common AI reliance positively predicts future overlap and cosine similarity. The portfolio-convergence result therefore does not depend on treating passive changes in portfolio weights as active information-based adjustments.

Future portfolio similarity	(1) Overlap	(2) Cosine
<i>Panel A: AI reliance based on active rebalancing</i>		
Common AI reliance	0.0104** (2.47)	0.0279*** (3.34)
Average AI reliance	-0.0053*** (-6.87)	0.0042** (2.02)
Current similarity	0.7957*** (371.20)	0.6760*** (223.63)
Observations	890,394	890,394
<i>Panel B: AI reliance based on split-adjusted share changes</i>		
Common AI reliance	0.0132*** (4.03)	0.0269*** (3.87)
Average AI reliance	-0.0063*** (-9.82)	0.0039** (2.17)
Current similarity	0.7958*** (371.58)	0.6760*** (223.61)
Observations	889,769	889,769
Pair controls	Yes	Yes
Pair FE	Yes	Yes
Quarter FE	Yes	Yes
SE cluster	Pair	Pair

Table D.7: Alternative AI Reliance Measures and Future Portfolio Similarity

D.5 Alternative Joint-Alignment Thresholds

The baseline joint-alignment indicator equals one when the fourteen GPT coefficients are jointly significant at the 5 percent level for each manager in the pair. Table D.8 varies this threshold. Requiring significance at the 1 percent level produces positive and statistically significant esti-

mates for both similarity measures, as does the baseline 5 percent threshold. When the threshold is relaxed to 10 percent, the estimates become small and statistically insignificant. The attenuation under the more permissive rule suggests that the association is concentrated among managers whose portfolio adjustments display stronger alignment with the GPT-extracted signals. A 10 percent rule includes more manager-quarter estimates with weaker evidence of alignment and therefore produces a noisier indicator.

Future portfolio similarity	(1) Overlap	(2) Cosine
<i>Panel A: Joint GPT alignment at the 1% threshold</i>		
Joint GPT alignment	0.0028*** (8.63)	0.0142*** (10.98)
Average AI reliance	-0.0070*** (-10.15)	0.0089*** (4.80)
Current similarity	0.7956*** (370.97)	0.6760*** (223.70)
Observations	890,394	890,394
<i>Panel B: Joint GPT alignment at the 5% threshold</i>		
Joint GPT alignment	0.0020*** (9.86)	0.0105*** (13.47)
Average AI reliance	-0.0076*** (-10.86)	0.0059*** (3.19)
Current similarity	0.7956*** (371.06)	0.6760*** (223.84)
Observations	890,394	890,394
<i>Panel C: Joint GPT alignment at the 10% threshold</i>		
Joint GPT alignment	-0.0002 (-1.25)	0.0008 (1.47)
Average AI reliance	-0.0059*** (-8.27)	0.0123*** (6.47)
Current similarity	0.7957*** (371.16)	0.6760*** (223.65)
Observations	890,394	890,394
Pair controls	Yes	Yes
Pair FE	Yes	Yes
Quarter FE	Yes	Yes
SE cluster	Pair	Pair

Table D.8: Alternative Joint-Alignment Thresholds and Future Portfolio Similarity

D.6 Manager-by-Quarter Fixed Effects

Table D.9 adds separate fixed effects for each manager in each quarter. These fixed effects absorb additive manager-specific shocks, including changes in fund flows, mandates, or risk constraints that may affect every pair containing the same institution. They also absorb each manager's own time-varying reliance measure. The coefficient on common reliance is therefore identified from the pair-level interaction in the two managers' reliance after accounting for these additive shocks.

The continuous common-reliance coefficient remains positive and statistically significant for future overlap. For cosine similarity, however, it becomes negative and is significant only at the 10 percent level. The joint-alignment indicator remains positive and statistically significant for both outcomes. The two measures therefore behave differently under saturated manager-specific time controls: the joint-alignment result is stable, whereas the sign of the continuous cosine estimate is sensitive to these controls.

	(1) Overlap	(2) Cosine
<i>Panel A: Continuous common AI reliance</i>		
Common AI reliance	0.0111*** (2.67)	-0.0142* (-1.67)
Current similarity	0.7609*** (290.50)	0.6268*** (102.70)
Observations	890,338	890,338
<i>Panel B: Joint GPT alignment</i>		
Joint GPT alignment	0.0008*** (4.68)	0.0026*** (4.05)
Current similarity	0.7609*** (290.52)	0.6268*** (102.70)
Observations	890,338	890,338
Pair controls	Yes	Yes
Pair FE	Yes	Yes
Manager $i \times$ quarter FE	Yes	Yes
Manager $k \times$ quarter FE	Yes	Yes
SE cluster	Pair	Pair

Table D.9: Common AI Reliance and Future Portfolio Similarity with Manager-by-Quarter Fixed Effects

D.7 Two-Manager Clustered Standard Errors

The baseline regressions cluster standard errors at the manager-pair level. This covariance estimator allows arbitrary serial correlation and heteroskedasticity within a pair, which is the most immediate dependence structure in a panel of persistent pair-level outcomes. Residuals may also be correlated across different pairs that contain the same institution. Manager-wide fund flows, mandate changes, risk constraints, and trading-capacity shocks are possible sources of such dependence.

To assess the sensitivity of the empirical analysis to shared-manager dependence, Tables D.10 and D.11 cluster by the manager identities assigned to positions i and k . This estimator broadens the covariance structure beyond pair clustering, although it is not a fully permutation-invariant dyadic covariance estimator.

The two clustering dimensions overlap extensively in the dense manager-pair panel. In several multiway-clustered regressions, especially the validation specifications with many interactions, the estimated covariance matrix is not positive semidefinite. Following Cameron et al. (2011), the calculations replace its negative eigenvalues with zero. This correction leaves the coefficient estimates unchanged but makes the reported standard errors dependent on the finite-sample covariance adjustment.

Table D.10 reports the baseline coefficients under two-manager clustering. The continuous common-reliance coefficients remain statistically significant at the 5 percent level for both overlap and cosine similarity. The joint-alignment coefficient remains positive, with significance at the 10 percent level for overlap and at the 1 percent level for cosine similarity. The estimated standard errors are larger than those obtained from pair clustering.

Table D.11 applies the same covariance specification to the validation regressions. The post-GPT interaction remains positive for both outcomes but is statistically significant only for cosine similarity. The triple interactions with GPT signal strength and earnings-call length also retain their positive signs, but most are estimated imprecisely. The cosine estimate for GPT signal strength is significant at the 10 percent level. The analyst-coverage interaction is negative and insignificant under this covariance estimator. The validation results thus have the predicted directional pattern in most specifications, but their precision is not uniform once the standard errors allow for shared-manager dependence.

	(1) Overlap	(2) Cosine
Future portfolio similarity		
<i>Panel A: Controlling for average AI reliance</i>		
Common AI reliance	0.0271** (2.41)	0.0627** (2.17)
Average AI reliance	-0.0097 (-1.62)	0.0052 (0.35)
Current similarity	0.7957*** (44.40)	0.6760*** (24.47)
<i>Panel B: Controlling for individual AI reliance</i>		
Common AI reliance	0.0272** (2.43)	0.0628** (2.17)
AI reliance, manager i	-0.0057 (-1.49)	0.0018 (0.18)
AI reliance, manager k	-0.0041 (-1.02)	0.0033 (0.35)
Current similarity	0.7956*** (44.40)	0.6760*** (24.47)
<i>Panel C: Joint GPT alignment</i>		
Joint GPT alignment	0.0020* (1.84)	0.0105*** (2.62)
Average AI reliance	-0.0076 (-1.43)	0.0059 (0.47)
Current similarity	0.7956*** (44.39)	0.6760*** (24.50)
Pair controls	Yes	Yes
Pair FE	Yes	Yes
Quarter FE	Yes	Yes
SE cluster	Manager i , manager k	Manager i , manager k
Observations	890,394	890,394

Table D.10: Baseline Results with Two-Manager Clustered Standard Errors

Table D.11: Validation with Two-Manager Clustered Standard Errors

	(1) Post-GPT	(2) GPT strength	(3) Call length	(4) Analyst coverage
<i>Panel A: Future portfolio overlap</i>				
Common reliance	0.0192** (2.02)	0.0125 (1.36)	0.0148 (1.58)	0.0040 (0.23)
Common reliance $\times$ Post-GPT	0.0202 (0.97)	0.0252 (1.19)	0.0207 (0.99)	0.0574 (0.99)
Common reliance $\times M^X \times$ Post-GPT		0.0807 (1.53)	0.0773 (0.86)	-0.0135 (-0.35)
Current similarity	0.7948*** (44.39)	0.7902*** (43.14)	0.7907*** (43.42)	0.7872*** (42.29)
<i>Panel B: Future cosine similarity</i>				
Common reliance	0.0180 (0.66)	-0.0020 (-0.07)	0.0121 (0.43)	-0.0745* (-1.67)
Common reliance $\times$ Post-GPT	0.1575** (2.57)	0.1762*** (2.86)	0.1704*** (2.92)	0.4174** (2.02)
Common reliance $\times M^X \times$ Post-GPT		0.2626* (1.67)	0.2386 (0.92)	-0.1610 (-1.22)
Current similarity	0.6736*** (24.41)	0.6700*** (24.34)	0.6705*** (24.39)	0.6644*** (23.59)
All lower-order interactions	Yes	Yes	Yes	Yes
Pair controls	Yes	Yes	Yes	Yes
Pair FE	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes
SE cluster	Manager i , manager k	Manager i , manager k	Manager i , manager k	Manager i , manager k
Observations	890,394	859,357	859,357	859,357

Notes: Each panel and column is a separate regression. The covariance estimator clusters by both manager identities. All lower-order terms and average-reliance interactions are included. Robust t -statistics are in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

D.8 Leave-One-Quarter-Out Robustness Checks

The post-GPT sample contains eight predictor quarters. To assess whether any single quarter disproportionately influences the validation estimates, we reestimate all four specifications for both dependent variables after omitting one predictor quarter at a time. Figure D.1 reports the portfolio-overlap estimates, and Figure D.2 reports the cosine-similarity estimates. The post-GPT and GPT-signal-strength coefficients retain their predicted signs under every omission for both similarity measures. The call-length estimates also remain positive, although precision declines under several post-GPT omissions. The analyst-coverage estimates are stable for cosine similarity but weaker and more variable for portfolio overlap. Overall, no single predictor quarter accounts for the principal validation patterns. This exercise measures the influence of individual quarters; it does not expand the time-series support of the post-GPT sample.

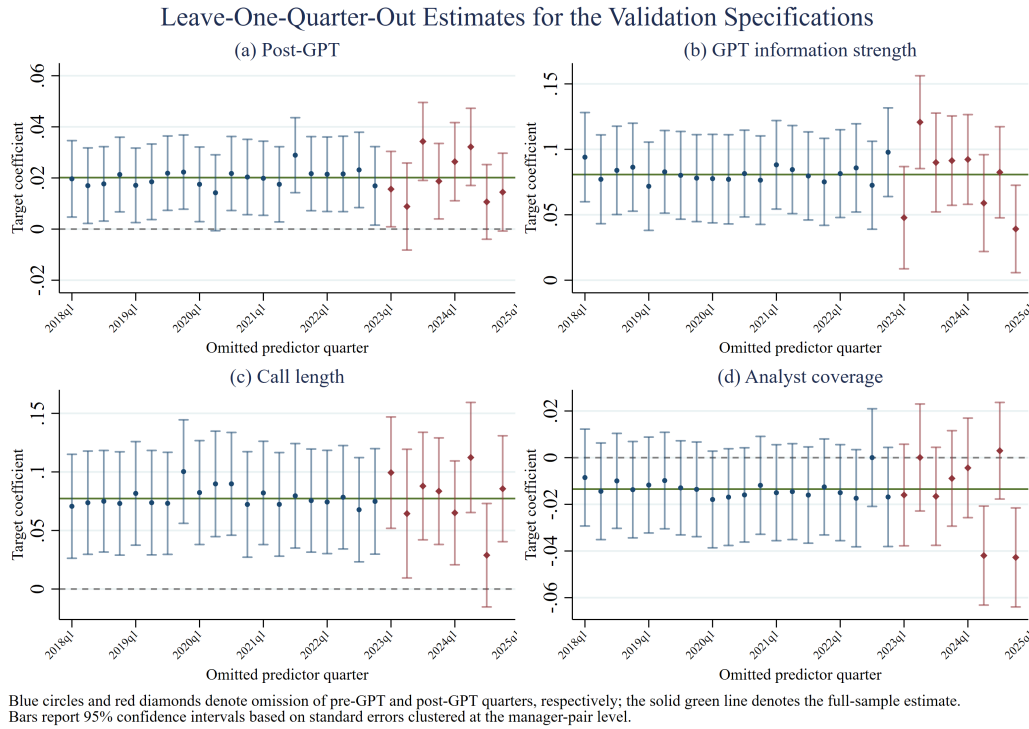


Figure D.1: Leave-One-Quarter-Out Estimates for the Validation Specifications: Portfolio Overlap

Notes: This figure shows the “leave-one-quarter-out” results for the portfolio-overlap validation regressions. We sequentially omit the predictor quarter indicated on the horizontal axis and plot the coefficients studied in the main text with 95% confidence intervals.

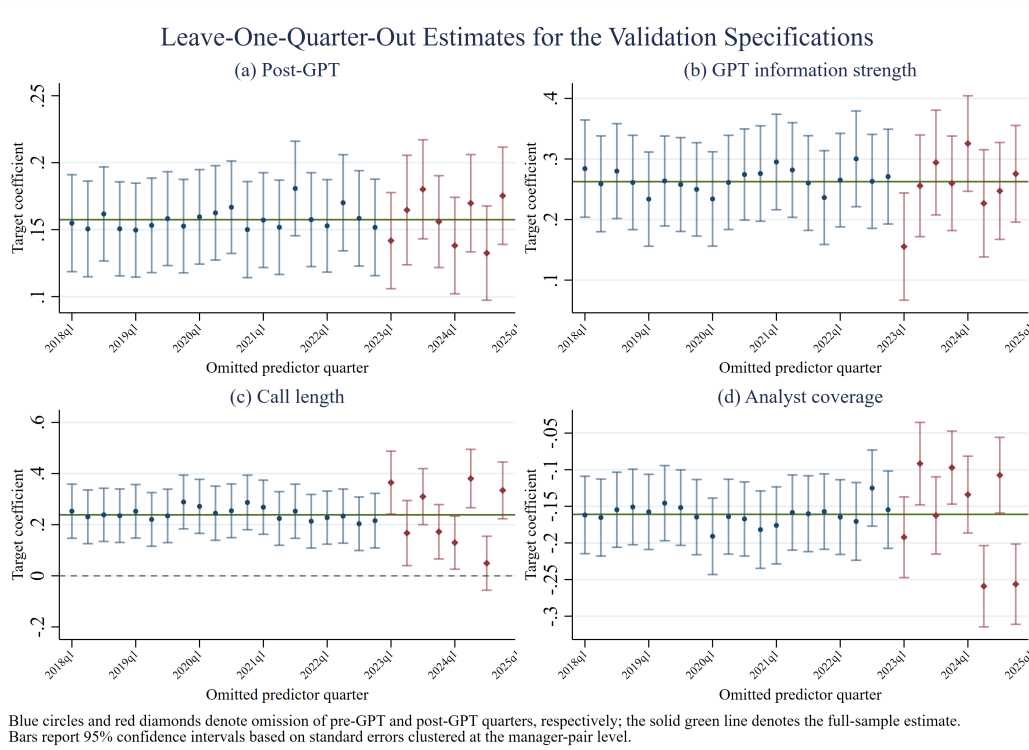


Figure D.2: Leave-One-Quarter-Out Estimates for the Validation Specifications: Cosine Similarity

Notes: This figure shows the “leave-one-quarter-out” results for the cosine-similarity validation regressions. We sequentially omit the predictor quarter indicated on the horizontal axis and plot the coefficients studied in the main text with 95% confidence intervals.